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BIOLOGY PAST QUESTIONS 2020 WITH SOLUTIONS

1. The cell as the basic unit of life consists of

- **A.** cell wall and vacuole
- **B.** nucleus and cell wall
- **C.** cytoplasm and nucleus
- **D.** cytoplasm and vacuole

Correct Option: Answer is C

Use the following Classes of Arthropoda to answer the question below

I. Crustacea II. Insecta III. Arachinda IV. Chilopoda

2. Which members of the classes live mainly in an aquatic habitat?

- **A.** I
- **B.** II
- **C.** III
- **D.** IV

• **Correct Option: Answer is A**

Use the following Classes of Arthropoda to answer the question below

I. Crustacea II. Insecta III. Arachinda IV. Chilopoda

3. Which of the classes is characterised by the possession of two pairs of antennae?

- **A.** I
- **B.** II
- **C.** III
- **D.** IV

• **Correct Option: Answer is A**

4. The organism with spiral chloroplasts and nucleus suspended by cytoplasmic strands is

- **A.** Volvox
- **B.** Spirogyra
- **C.** Paramecium
- **D.** Euglena

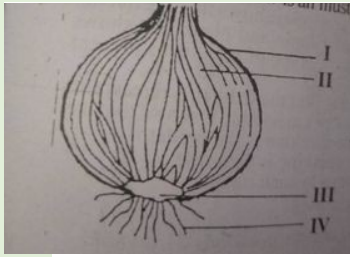
• **Correct Option: Answer is B**

5. The network of double membrane that conveys materials through the cytoplasm is the

- **A.** plasma membrane
- **B.** nuclear membrane
- **C.** mitochondrion
- **D.** endoplasmic reticulum

• **Correct Option: Answer is D**

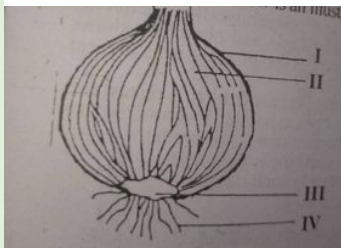
The diagram below is an illustration of the longitudinal section of a plant organ. Study to answer the question below.



6. Food is stored in the part labeled

- **A.** I
- **B.** II
- **C.** III
- **D.** IV
- **Correct Option: Answer is B**

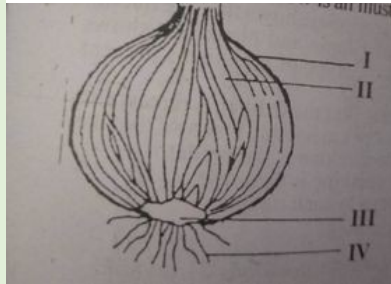
The diagram below is an illustration of the longitudinal section of a plant organ. Study it and answer the question below;



7. The plant is a

- **A.** stem tuber
- **B.** bulb
- **C.** stolon
- **D.** runner
- **Correct Option: Answer is B**

The diagram below is an illustration of the longitudinal section of a plant organ. Study it and answer the question below;



8. The part labeled I is the

- **A.** apical bud
- **B.** scale leaf
- **C.** adventitious root
- **D.** fleshy leaf
- **Correct Option: Answer is B**

9. Which of the following materials is not a living semi-permeable membrane?

- **A.** pig's bladder
- **B.** Unripe pawpaw fruit
- **C.** Yam tuber
- **D.** Sheet of cellophane
- **Correct Option: Answer is D**

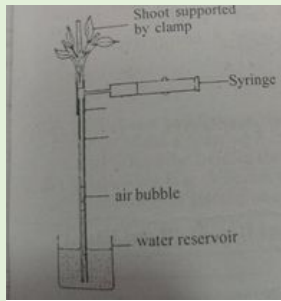
10. In an experiment, mould and yeast cells were transferred into an environment with low oxygen concentration. After a few days, the mould dies while the yeast cells did not. Which of the following statements best explains the following observation.

- **A.** The yeast cells carried out photosynthesis while the mould did not
- **B.** Photosynthesis does not take place in the absence of oxygen
- **C.** Respiration can take place in yeast cells in the absence of oxygen
- **D.** Respiration does not occur in the mould
- **Correct Option: Answer is C**

11. Which of the following tissues does not provide support in flowering plants?

- **A.** Collenchyma
- **B.** Parenchyma
- **C.** Xylem
- **D.** Phloem
- **Correct Option: Answer is B**
- **Explanation:**
- Parenchyma is a type of tissue consists of cells that carry out an essential function. In botany (plant biology), parenchyma is the **simple permanent ground tissues** that form the bulk of the plant tissues, such as the soft part of leaves, fruit pulp, and other plant organs

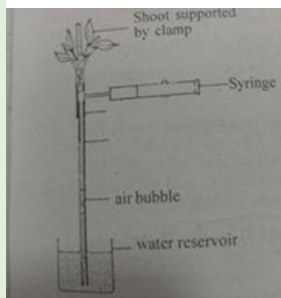
The diagram below is an illustration of an experimental set-up. Study it and answer the question below;



12. The set-up directly measure

- **A.** transpiration of water by the shoot
- **B.** evaporation of water from the leaves
- **C.** absorption of water by the shoot
- **D.** loss of mineral salts from the leaves
- **Correct Option: Answer is A**

The diagram below is an illustration of an experimental set-up. Study it and answer the question below;



13. The set-up can measure comparatively the rate of

- **A.** evaporation from leaves on a single shoot under different experimental conditions
- **B.** salt uptake by shoots from different plants
- **C.** transpiration of a single shoot of a plant under different experimental conditions
- **D.** water uptake by roots of different plants
- **Correct Option: Answer is C**

14. The respiratory organ of a cockroach

- **A.** air sac
- **B.** trachea
- **C.** lung book
- **D.** lung
- **Correct Option: Answer is C**

Explanation:

Book lung, **form of respiratory organ found in certain air-breathing arachnid arthropods** (scorpions and some spiders). Each book lung consists of a series of thin plates that are highly vascular (i.e., richly supplied with blood) and are arranged in relation to each other like the pages of a book.

15. The excretory product of some reptiles, birds and insects is

- **A.** urea
- **B.** urine
- **C.** ammonia
- **D.** uric acid
- **Correct Option: Answer is D**
- **Explanation:**

Birds, reptiles and desert dwelling animals excrete uric acid as a semi-solid material in their gut excreta, by a complicated, high energy-demanding process.

16. The part of the mammalian kidney that stores urine is the

- **A.** capsule
- **B.** medulla
- **C.** pelvis
- **D.** bladder
- **Correct Option: Answer is D**

17. The properties of endocrine system include the following except

- **A.** secretion of hormones
- **B.** transportation by blood to target organ
- **C.** having specific effect
- **D.** release of secretion into ducts
- **Correct Option: Answer is D**

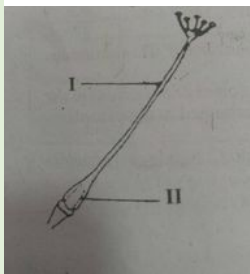
18. The part of the central nervous system that controls unconscious actions in humans is the

- **A.** cerebellum
- **B.** cerebrum
- **C.** spinal cord
- **D.** optic nerves
- **Correct Option: Answer is A**

19. What happens when the ciliary muscles of the eye contract? The

- **A.** lens become more concave
- **B.** lens become more convex
- **C.** lens gets a longer focal length
- **D.** suspensory ligament becomes tight
- **Correct Option: Answer is C**

The diagram below is an illustration of a part of a flower. Study it and answer the question below;



20. The function of the part labeled I is

- **A.** receiving the pollen grain
- **B.** passage for the male gamete to the ovary
- **C.** germination of the pollen grain
- **D.** site for double fertilization in the plant
- **Correct Option: Answer is B**

The diagram below is an illustration of a part of a flower. Study it and answer the question below.

21. The part labeled II is the

- **A.** unfused stigma

- **B.** fused style
- **C.** fused ovaries
- **D.** unfused anther
- **Correct Option: Answer is C**

22. The reagent used in testing for carbon (IV) oxide is

- **A.** copper sulphate solution
- **B.** lime water
- **C.** hydrochloric acid
- **D.** sodium hydroxide solution
- **Correct Option: Answer is B**

Explanation:

Lime is the most widely used reagent in water treatment applications. It is supplied in two forms: ... slaked (or hydrated) lime: **Ca(OH)₂**.

23. Oxygen comes out of the stomata during photosynthesis through the process known as

- **A.** diffusion
- **B.** transpiration pull
- **C.** osmosis
- **D.** active transport
- **Correct Option: Answer is A**

24. The first stable product of photosynthesis is

- **A.** starch
- **B.** fructose
- **C.** glucose
- **D.** sucrose
- **Correct Option: Answer is C**

25. An example of a trace elements is

- **A.** potassium
- **B.** calcium
- **C.** magnesium
- **D.** copper
- **Correct Option: Answer is D**

26. Which of the following statements about a mixture of a protein-digesting enzyme and starch solution would be correct? The protein-digesting enzyme

- **A.** has no effect on the starch solution
- **B.** leads to the production of amino acids
- **C.** leads to the production of glucose
- **D.** digest the starch
- **Correct Option: Answer is A**

27. The mutualism type of relationship is different from a parasitic relationship because in mutualism

- **A.** both organisms involved benefit
- **B.** none of the organisms benefits or harm each other
- **C.** both organisms harm each other
- **D.** only one of the organisms is harmed
- **Correct Option: Answer is A**

28. The table below shows the number of some organisms in habitats W and Y. Study it to answer the question below;

Organism	Number in the habitat	
	habitat W	habitat Y
Plankton	126	0
Antelope	0	51
Water flea	10	0
Tilapia	23	0
Lion	0	6
Frog	6	0
Grass	0	250

What type of habitat is W?

- **A.** Rainforest
- **B.** Pond
- **C.** Desert
- **D.** Ocean
- **Correct Option: Answer is B**

29. The table below shows the number of some organisms in habitats W and Y. Study it to answer the question below;

Organism	Number in the	
	habitat	

	habitat W	habitat Y
Plankton	126	0
Antelope	0	51
Water flea	10	0
Tilapia	23	0
Lion	0	6
Frog	6	0
Grass	0	250

Which of the following statements about habitat W is correct? The

- **A.** absence of lions and antelopes shows that the habitat is terrestrial
- **B.** presence of Tilapia and planktons shows that the habitat is not aquatic
- **C.** type of organisms present indicate that habitat is aquatic
- **D.** absence of grasses indicates that the habitat is terrestrial
- **Correct Option: Answer is C**

The table below shows the number of some organisms in habitats W and Y. Study it to answer the question below;

Organism	Number in the	
	habitat W	habitat Y
Plankton	126	0
Antelope	0	51
Water flea	10	0
Tilapia	23	0
Lion	0	6
Frog	6	0
Grass	0	250

30. The number of lions and antelopes in habitat Y shows that the lion

- **A.** is the prey while the antelope is the predator
- **B.** is the predator while the antelope is the prey
- **C.** and the antelope are preys to each other
- **D.** and the antelopes are predators
- **Correct Option: Answer is B**

The diagram below is an illustration of an ecological instrument. Study it and answer the question below;



31. A disadvantage of the abiotic factor measured by the instrument is that it

- **A.** is an agent of pollination
- **B.** is necessary for germination
- **C.** leads to flooding when in excess
- **D.** is used for irrigation
- **Correct Option: Answer is B**

The diagram below is an illustration of an ecological instrument. Study it and answer the question below;

32. When the instrument is in use, it is usually

- **A.** placed slightly above soil level
- **B.** placed on a table
- **C.** suspended in air
- **D.** suspended to moving water
- **Correct Option: Answer is A**

33. Soil with the finest texture is

- **A.** silt
- **B.** clay
- **C.** sand
- **D.** gravel

Correct Option: Answer is A
Explanation:

Silt is a solid, dust-like sediment that water, ice, and wind transport and deposit. Silt is made up of **rock**

and mineral particles that are larger than clay but smaller than sand. Individual silt particles are so small that they are difficult to see.

34. The position occupied by an organism in food chain is the

- **A.** energy level
- **B.** niche
- **C.** trophic level
- **D.** biomass
- **Correct Option: Answer is C**

35. The depletion of the ozone layer will result in the earth surface receiving more

- **A.** gamma rays
- **B.** infra-red rays
- **C.** ultraviolet rays
- **D.** X-rays
- **Correct Option: Answer is C**

Use the list of insects below to answer the question below;

I. Cotton stainer II. Honeybee III. Termite
IV. Weevil

36. The insects whose activities are both beneficial and harmful to humans are;

- **A.** I and II
- **B.** II and III
- **C.** II and IV
- **D.** III and IV

Correct Option: Answer is B

Use the list of insects below to answer the question below;

I. Cotton stainer II. Honeybee III. Termite
IV. Weevil

37. Which of the insects destroys grains?

- **A.** I
- **B.** II
- **C.** III
- **D.** IV
- **Correct Option: Answer is D**

38. Conservation of natural resources does not

- **A.** maintain a balanced ecosystem
- **B.** preserve the beauty of nature
- **C.** attract tourists
- **D.** threaten the survival of species
- **Correct Option: Answer is D**

39. A company was prohibited from producing bags made from natural leopard skin. This is an attempt to conserve

- **A.** land
- **B.** water
- **C.** wildlife
- **D.** minerals
- **Correct Option: Answer is A**

40. Which of the following substances is not a conservable natural resource?

- **A.** Mineral
- **B.** Air
- **C.** Soil
- **D.** Water
- **Correct Option: Answer is C**

41. A child that can receive blood donation from anybody belongs to the blood group

- **A.** O
- **B.** A
- **C.** B
- **D.** AB
- **Correct Option: Answer is D**
- **Explanation:**

If your blood type is:	You can give to:	You can receive from:
AB Positive	AB+ Only	All Blood Types
O Negative	All Blood Types	O-
A Negative	A-, A+, AB-, AB+	A-, O-
B Negative	B-, B+, AB-, AB+	B-, O-

42. Variation which exhibits a wide range from one extreme to the other is

- **A.** phenotypic variation
- **B.** discontinuous variation
- **C.** continuous variation
- **D.** genotypic variation
- **Correct Option: Answer is C**

43. Measurements of height and weight of students in a class show

- **A.** discontinuous variation
- **B.** continuous variation
- **C.** shortness is more prevalent
- **D.** fatness is less prevalent
- **Correct Option: Answer is B**

44. Which of the following statements about chromosomes is correct?

- **A.** All the chromosomes of a species are the same shape
- **B.** The number present in a specie is constant
- **C.** They are neatly arranged in a cytoplasm
- **D.** They bear ribosomes on their outer membranes
- **Correct Option: Answer is B**

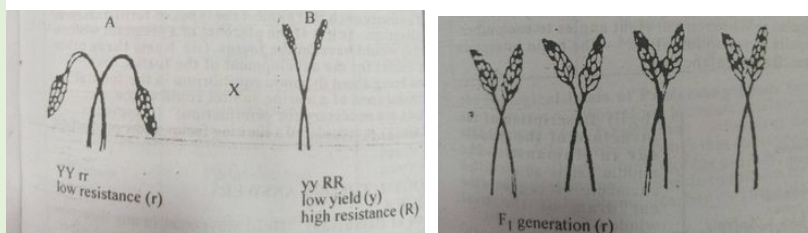
45. Which of the following diseases can be inherited?

- **A.** Pneumonia
- **B.** Whooping cough
- **C.** Sickle cell anemia
- **D.** Malaria
- **Correct Option: Answer is C**

Explanation:

Sickle cell anemia is **inherited in an autosomal recessive pattern**, which means that both copies of the gene in each cell have mutations. The parents of an individual with an autosomal recessive condition each carry one copy of the mutated gene, but they typically do not show signs and symptoms of the condition.

The diagram below is an illustration of a cross between plants A and B of the same species. Study it and answer the question below;



46. If the F1 generation are plants with high yield and resistance, the genotype of the F1 generation plants would be

- **A.** YYRR

- **B.** YyRr
- **C.** yyRr
- **D.** yyrr
- **Correct Option: Answer is A**

The diagram below is an illustrating of a cross between plants ! and B of the same species. Study it to answer the question below;

47. The process that gave rise to the F1 generation is

- **A.** self fertilization
- **B.** cross fertilization
- **C.** out-breeding
- **D.** test cross
- **Correct Option: Answer is B**

48. Replication of DNA molecules is catalysed by an enzyme called

- **A.** polymerase
- **B.** ptyalin
- **C.** pepsin
- **D.** amylase
- **Correct Option: Answer is A**

49. Who proposed the theory of evolution by natural selection

- **A.** Darwin
- **B.** Lamark
- **C.** Aristotle
- **D.** Linnaeus
- **Correct Option: Answer is A**

50. Which of the following statements best explains the reason why termites swarm at night?

- **A.** Light is not necessary for swarming
- **B.** They avoid day-flying birds
- **C.** Light destroys their wings
- **D.** They can only see in the dark
- **Correct Option: Answer is A**

BIOLOGY PAST QUESTIONS THEORY 2020 WITH SOLUTIONS

(a) Explain briefly how the human ear carries out its function of balancing.

(b) (i) make a drawing 8 cm - 10 cm long of the structure of the human ear and label fully.

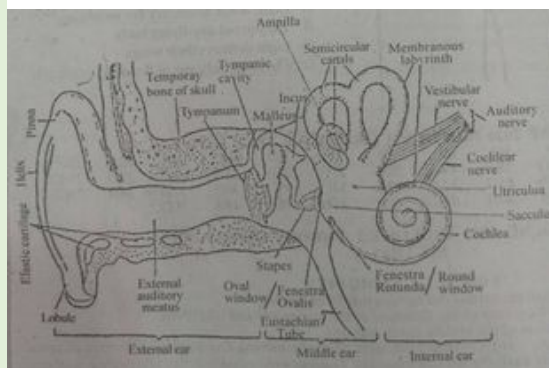
(ii) Describe briefly the structure of the middle ear in humans.

Solution & Explanation:

(a) How the human ear carries out its function of balancing;

- Head movement in any direction affect the fluid/endolymph, In the semi-circular canals, Which are at right angles to each other, The movement forces the sensory/hair cells in the ampulla to set up the impulses, The brain interprets the impulses and relays the impulses to the body muscles for balancing.

(b)(i) Diagram of the Human Ear;



(b)(ii) Description of the structure of the middle ear; The middle ear is an air filled chamber, It connects the ear drum of the oval window fenestra ovalis. It contains three

bones/hammer/malleus, anvil/incus and stapes/stirrup/ear ossicles. The malleus is attached to the tympanic membrane/tympanum/eardrum. Each of the bones has direct contact with the one next to it, the eustachian tube is formed at the base of the middle ear . The eustachian tube links the ear to the throat/pharynx.

2, (a)(i) What is a deficiency disease?

(ii) Complete the table below by naming five nutrient deficiency diseases in humans and stating one remedy for each of the diseases.

(b) Outline a chemical test for (i) Starch in a tuber of yam
(ii) Glucose in orange fruit

Nutrient deficiency disease	Remedy

Solution & Explanation:

(a)(i) Deficiency Disease; This is a disease caused by lack of or insufficient or inadequate supply of a specific vitamin or other nutrients/food substances.

(a)(ii)

Nutrient deficiency diseases	Remedy
Goitre/Goiter	Consumption of iodized salt/sea food
Rickets	Consumption of a diet rich in calcium/phosphorous/vitamin D
Anaemia/Anemia	Eating a diet rich in iron
Kwashiorkor	Diet of infant should contain protein
Scurvy	Consumption of foods/fruits rich in Vitamin C
Pellagra	Addition of niacin/vitamin K to diet
Night blindness	Consumption of food rich in Vitamin A
Beriberi	Consumption of food rich in vitamin B
Stunted growth/cretinism	Consumption of food rich in iodine
Marasmus	Consumption of protein and carbohydrate

(b) Procedure of Chemical Test for;

(i) Starch in a tuber of yam; The tuber of yam is peeled and sliced, a few drops of iodine solution is added to the peeled yam, blue-black colour formed indicates presence of starch.

(ii) Glucose in orange fruit; Orange fruit is cut and its juice is extracted into a beaker/test tube, few drops of Benedict's solution/Fehling's solutions A and B is added to equal volume of orange juice, add heat gently, a red/orange/brick-red precipitate shows the presence of glucose. For Benedict's solution when heated shows red/yellow precipitate confirming presence of glucose

3(a) Complete the table below;

Disease	Causative organism	Vector	Control
---------	--------------------	--------	---------

Cholera			
	Trypanosome		
		Female Anopheles mosquito	
Ringworm		None	

(b) Explain briefly the biological basis of preserving foods using each of the following methods;

(i) salting (ii) irradiation (iii) smoking

Solution & Explanation:

(a)

Disease	Causative Organism	Vector	Control
Cholera	Vibrio Cholerae	Housefly/ Cockroach	Good sanitation/good personal hygiene/ any other method of good sanitation/inoculation/vaccination
Trypanosomiasis/ Sleeping sickness	Trypanosome	Tsetsefly/ Glossina	Destroying habitat of vector/clearing the vegetation, killing vector/eradicate flies.
Malaria	Plasmodium sp	Female anopheles	Proper sanitation/usin

		mosquito	g spraying recommended insecticide/fumigation, clearing of bushes/drainages of breeding sites/sleeping under nets.
Ringworm	Tinea Sp/Fungi/Microsporum	None	Good personal hygiene, avoid contact with infected persons/material, use of drugs/griseofulvin

(b) Biological Basis of Preserving Foods by;

(i) Salting; Removes water/moisture from food by osmosis therefore prevents micro-organisms from growing on food by dehydrating and killing them.

(ii) Irradiation; Involves the exposure of food to high energy gamma rays/x-rays radiation to destroy enzymes that cause spoilage, delay ripening of fruits, sprouting of vegetables therefore allowing long storage periods.

(iii) Smoking; Reduces the water content of food thereby creating unfavourable conditions for microbial growth and survival also called curing. Smoking contains phenols/carbon that kills microbes on the surface of food.

4. (a) Explain briefly; (i) independent assortment of genes (ii) the reason blood group O in humans can only exist in homozygous form while blood groups A and B can exist both in homozygous and heterozygous forms.

(b) Complete the table below by naming the classes of vertebrates in their evolutionary trend and giving one example each of the classes.

Classes of vertebrate	One example

Solution & Explanation:

(a)(i) Independent Assortment of Genes; is the random selection of genes during meiosis to produce gametes, where the genes separate/segregate so that only one of a pair is found in any one gamete. In the dihybrid inheritance in heterozygote the pairs of alleles are found on different pairs of chromosome, Only four types of gametes are produced in approximately equal numbers.

(ii) Blood grouping in humans; Is determined by three alternative genes/alleles A, B and O. A person can only possess two alleles of the genes on a pair of homologous chromosomes. Gene O is recessive to both genes A and B which are dominant/co-dominant and possess antigen A and B in the red blood cells hence it is possible to have blood group A as AA (homozygous) or AO (heterozygous) and blood group B as BB or BO, Blood group O can only exist genotypically as OO and does not possess antigen A

or B in the red blood cells or in the homozygous recessive state.

(b) Classes of Vertebrates in their Evolutionary Trend;

Class of Vertebrates	One Example
Pisces/Osteichthyes/ Chondrichthyes	Tilapia; Tuna, Salmon, Shark, Dogfish, Catfish, Skate, Ray
Amphibia	Frog/Toad
Reptilia	Snake, Crocodile, Tortoise, Turtle, Lizard, Wall Gecko
Aves	Dove, Pigeon, Parrot, Weaver Bird, Hawk, Eagles, Domestic Fowl
Mammalia	Human, Monkeys, Rat, Rab

5(a) Explain briefly the following types of fertilization in animals; (i) external fertilization (ii) internal fertilization

(b) Name two groups of animal each that exhibit the types of fertilization in (a)

(c)(i) If the placenta in a pregnant woman is detached from the uterine wall, give three effect this will have on the foetus

(ii) Name three other features in the uterus of a pregnant woman that are useful for the development of the foetus

(d) Explain briefly how the activities of organisms bring about dynamic equilibrium in the habitat.

(e) State four problems that organisms in the intertidal zone of a marine habitat could encounter.

(f) Explain briefly the reasons that the following factors are necessary for germination; (i) moisture (ii) viable seeds

(g) Explain briefly the reason light energy is considered a limiting factor in the production of foods by autotrophs.

Solution & Explanation:

(a)(i) External Fertilization; Involves the release of eggs by the female to the outside environment where a male sheds its sperms on them for fertilization outside the body of the female.

(ii) Internal Fertilization; Involves the release of eggs inside the body of a female males deposit sperms inside the female where fertilization occurs.

(b)(i) Groups of Animal that exhibit external fertilization ;- Fish/pisces, amphibians

(ii) Groups of Animal that exhibit internal fertilization;- Reptiles, birds/aves, mammals

(c)(i) Effect of Detached Placenta on a Foetus;

The exchange of dissolved food substances/oxygen/antibodies between a foetus and the mother will cease, removal of waste products from foetus will cease production of progesterone/hormone will cease might lead to the death of foetus/miscarriage/abortion/still birth. Certain molecules may pass into the body of foetus.

(ii) Other Features Present in the Uterus of a Pregnant Woman; Umbilical cord, amnion, amniotic fluid, allantois, Chorion

(d) Dynamic Equilibrium; Nature does not encourage any population exceeding the carrying capacity of the habitat, living organisms play a role in the regulation of the abundance of organisms/population in the habitat. Interaction among various populations lead to competition which may result in the elimination of one species or its restriction to a non-competitive position, Plants sometimes compete by producing toxic chemicals that inhibit neighboring species, there also exists predator-prey relationship/association between animals all these forces act together to achieve dynamic equilibrium in the habitat.

(e) Problem that Organisms in Intertidal Zones of Marine Habitats could Encounter; Tendency of being carried away by waves drying up/desiccation/exposure to air, varied osmotic balance oxygen deficiency, low light intensity for producers, reduced visibility in animals, fluctuation of water temperature..

(f) Reasons the following factors are necessary for germination;

(i) Moisture;- A seed imbibes/absorbs moisture causing the seed to swell and the testa becomes more permeable to oxygen and carbon (IV) oxide. This swelling will rupture the testa before germination starts, the moisture/water activates the enzymes for metabolism leading to germination

(ii) Viable Seed;- Has a living/healthy embryo, intact healthy cotyledon/endosperm, whole without boring from insect.

(g) Reason Light energy is Considered a Limited Factor in the Production of Food by Autotrophs; Light energy is obtained from the sun which is an important factor for

photosynthesis/photolysis of water that is a process of manufacturing/producing food by autotrophs at optimum temperature/sunlight/solar/light energy green plants/autotrophs will manufacture food.

BIOLOGY PAST QUESTIONS 2019 WITH SOLUTIONS

- 1.** Which of the following structures will NOT be found in the nucleus of a cell?
- **A.** DNA
 - **B.** Nucleolus
 - **C.** Lysosome

- **D. Centriole**

• **Correct Option: Answer is C**

A lysosome is a **membrane-bound cell organelle that contains digestive enzymes**. ... They break down excess or worn-out cell parts. They may be used to destroy invading viruses and bacteria.

2. An example of organ level of organization is _____

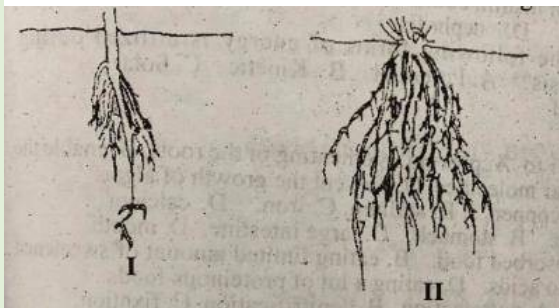
- **A. bird**
- **B. kidney**
- **C. spermatozoon**
- **D. xylem**

• **Correct Option: Answer is B**

3. A structural similarity between *Paramecium* and *Amoeba* is the presence of _____

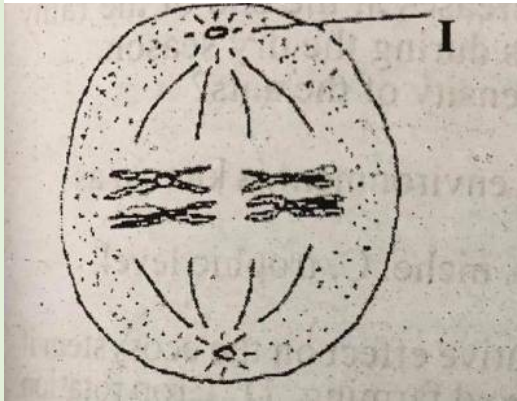
- **A. one food vacuole**
- **B. two contractile vacuoles**
- **C. two nuclei**
- **D. one gullet**

• **Correct Option: Answer is A**



4. The classes of plants the root systems in diagrams I and II above represent respectively are _____

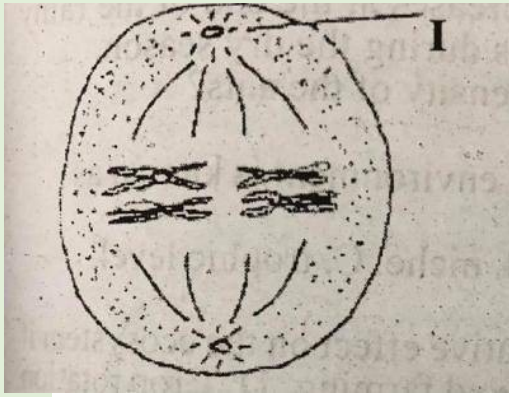
- **A.** Dicotyledoneae and Monocotyledoneae
- **B.** Monocotyledoneae and Dicotyledoneae
- **C.** Dicotyledoneae and Dicotyledoncae
- **D.** Monocotyledoneae and Monocotyledoneae
- **Correct Option: Answer is B**



- 5.** The diagram above is an illustration of a stage in a biological process. Study it and answer this question

What stage of cell division is illustrated in the diagram?

- **A.** Prophase
- **B.** Metaphase
- **C.** Anaphase
- **D.** Telophase
- **Correct Option: Answer is B**

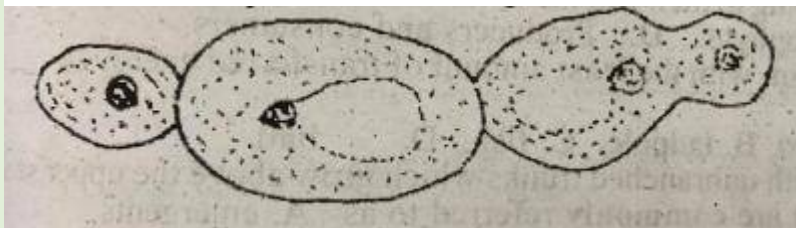


6. The part labelled I is the _____

- **A.** centromere
- **B.** spindle fibre
- **C.** centriole
- **D.** chiasmata
- **Correct Option: Answer is A**

7. The rate of diffusion of molecules between two media will be higher if _____

- **A.** the difference in concentration of the two media is low
- **B.** the difference in concentration of the two media is negligible
- **C.** the difference in concentration of the two media is high
- **D.** there is no difference in the concentration of the two media
- **Correct Option: Answer is D**



8. Anaerobic respiration in the organism illustrated in the diagram above produces carbon dioxide and _____

- **A.** ethanol
- **B.** water
- **C.** oxygen
- **D.** glucose
- **Correct Option: Answer is A**

9. Underground storage stems which grow horizontally in the soil are _____

- **A.** bulbs
- **B.** runners
- **C.** rhizomes
- **D.** corms
- **Correct Option: Answer is C**

Rhizome, also called creeping rootstalk, **horizontal underground plant stem** capable of producing the shoot and root systems of a new plant.

10. Muscle fatigue in the body of an athlete is due to _____

- **A.** low pH
- **B.** high oxygen content
- **C.** accumulation of lactic acid
- **D.** accumulation of carbonic acid
- **Correct Option: Answer is C**

11. The heart can beat continuously without being fatigued because it is composed of _____

- **A.** skeletal muscles
- **B.** smooth muscles

- C. cardiac muscles
- D. renal muscles
- **Correct Option: Answer is C**

12. The products of tissue respiration are_____

- A. glucose, oxygen, and water
- B. oxygen, water, and energy
- C. glucose, carbon dioxide, and energy
- D. water, carbon dioxide, and energy
- **Correct Option: Answer is D**

13. Excretory products in plants responsible for the discoloration of flowers are called_____

- A. alkaloids
- B. tannins
- C. anthocyanins
- D. resins
- **Correct Option: Answer is D**

Resins are typically viscous substances that convert into rigid polymers through a curing process.

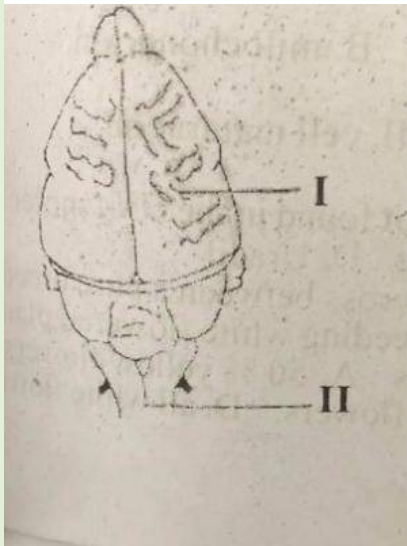
14. The process of maintaining a steady internal environment is known as_____

- A. osmoregulation
- B. equilibration
- C. homeostasis
- D. plasmolysis
- **Correct Option: Answer is C**

Homeostasis refers to **the capacity of the body to maintain the stability of diverse internal variables**, such as temperature, acidity, and water level, in the face of constant environmental disturbance.

15. Which of the following statements is not true about hormones?

- **A.** Their responses are voluntary
- **B.** They may affect more than one target organ
- **C.** They are chemical messengers in animals
- **D.** They are transported by the blood to the target organs
- **Correct Option: Answer is B**



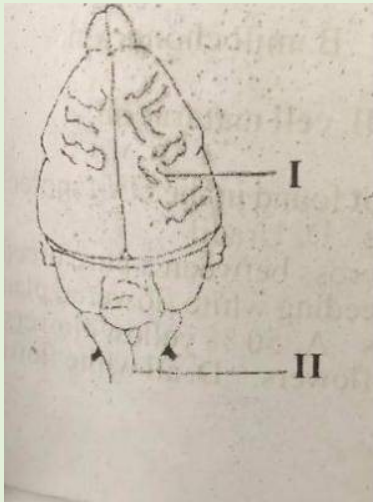
The diagram above is an illustration of a human brain. Study it and answer this question

16. The part labelled I is the _____

- **A.** olfactory lobe
- **B.** cerebellum
- **C.** cerebrum
- **D.** medulla oblongata

• **Correct Option: Answer is C**

Cerebrum, **the largest and uppermost portion of the brain.** The cerebrum consists of the cerebral hemispheres and accounts for two-thirds of the total weight of the brain



17. The function of the part labelled II is that it controls_____

- **A.** posture and balance of the body
- **B.** all voluntary actions
- **C.** temperature and appetite
- **D.** many involuntary actions

• **Correct Option: Answer is D**

The medulla oblongata plays a critical role in **transmitting signals between the spinal cord and the higher parts** of the brain and in controlling autonomic activities, such as heartbeat and respiration.

18. To focus on a distant object, the ciliary muscle of the eye_____

- **A.** contracts and the eye lens gets thicker
- **B.** relaxes and the eye lens gets thinner
- **C.** contracts and the eye lens gets thinner
- **D.** relaxes and the eye lens gets thicker

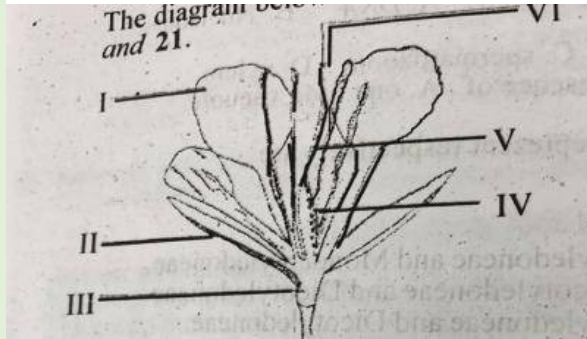
• **Correct Option: Answer is D**

19. The main function of the amniotic fluid is for_____

- **A.** respiration in the foetus

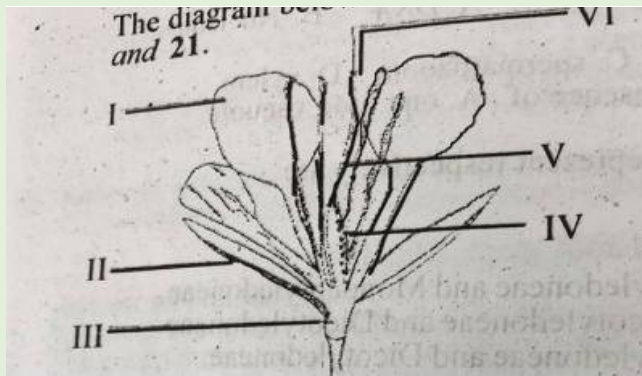
- **B.** protection of the foetus from shock
- **C.** Lubricating the birth canal foetus
- **D.** nourishing the foetus
- **Correct Option: Answer is B**

Amniotic fluid is responsible for: **Protecting the fetus:** The fluid cushions the baby from outside pressures, acting as a shock absorber. **Temperature control:** The fluid insulates the baby, keeping it warm and maintaining a regular temperature.



The diagram above is an illustration of the longitudinal section of a flower. Study it and answer this question

- 20.** Which part of the flower is responsible for the protection of young flower buds?
- **A.** I
 - **B.** II
 - **C.** III
 - **D.** IV
 - **Correct Option: Answer is B**



21. The essential parts of the flower are labelled_____

- **A.** I, II and III
- **B.** II, III and IV
- **C.** III, V and VI
- **D.** IV, V and VI
- **Correct Option: Answer is D**

22. Deficiency in insulin could lead to_____

- **A.** cirrhosis
- **B.** hepatitis
- **C.** diabetes
- **D.** nephritis
- **Correct Option: Answer is C**

23. Which of the following forms of energy is utilized during photosynthesis?

- **A.** Potential
- **B.** Kinetic
- **C.** Solar
- **D.** Chemical
- **Correct Option: Answer is C**

24. Light is excluded from a water culture experiment in order to_____

- **A.** prevent overheating of the roots
- **B.** enable the roots to grow straight down
- **C.** prevent ionization of water molecules
- **D.** prevent the growth of algae
- **Correct Option: Answer is C**

25. The major mineral present in the shell of Molluscs is_____

- **A.** Copper
- **B.** Sodium
- **C.** Iron
- **D.** Calcium
- **Correct Option: Answer is D**

26. The digestion of proteins starts in the_____

- **A.** small intestine
- **B.** stomach
- **C.** large intestine
- **D.** mouth
- **Correct Option: Answer is B**

27. Tooth decay is usually caused by_____

- **A.** undigested and absorbed food
- **B.** eating a limited amount of sweetened food.
- **C.** the bacteria in the mouth that metabolize sugar to acids
- **D.** eating a lot of proteinous foods

• **Correct Option: Answer is C**

28. The conversion of ammonium salts into nitrates is called_____

- **A.** putrefaction
- **B.** denitrification
- **C.** fixation
- **D.** nitrification

• **Correct Option: Answer is D**

29. The salinity of a brackish habitat_____

- **A.** increases immediately after rainfall
- **B.** increases at the end of the rain
- **C.** decreases with an increase in the number of organisms
- **D.** increases during the dry season

• **Correct Option: Answer is D**

30. If 90 ants are found in a field with a total area of 100 m. what is the population density of the ants?

- **A.** 0.09
- **B.** 0.90
- **C.** 9.00
- **D.** 90.00

• **Correct Option: Answer is B**

31. The complex relationship between the members of a community and their physical environment is known as_____

- **A.** commensalism

- **B.** symbiosis
- **C.** saprophytism
- **D.** ecology
- **Correct Option: Answer is D**

32. The specific role of a species within its environment is known as its _____

- **A.** habitat
- **B.** niche
- **C.** trophic level
- **D.** biotic community

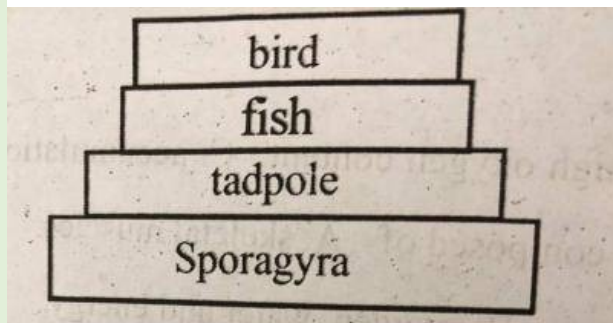
Correct Option: Answer is B

33. Which of the following soil fertility improvement methods will have the most negative effect on the ecosystem if not controlled?

- **A.** Application of N-P-K fertilizer
- **B.** Shifting cultivation
- **C.** Mixed farming
- **D.** Crop rotation
- **Correct Option: Answer is A**

34. Organisms in an ecosystem are usually grouped according to their trophic level into _____

- **A.** carnivores and omnivores,
- **B.** consumers and parasites
- **C.** producers, and saprophytes
- **D.** Producers and consumers
- **Correct Option: Answer is D**
-



35. In the pyramid of energy illustrated above, the organism with the least amount of transferred energy is _____

- **A.** Spirogyra
- **B.** Tadpole
- **C.** Fish
- **D.** Bird
- **Correct Option: Answer is D**

36. Tall trees with unbranched trunks which grow above the upper storey of the forest are commonly referred to as _____

- **A.** Emergents
- **B.** Herbs
- **C.** Lianes
- **D.** Shrubs
- **Correct Option: Answer is A**

37. Maize grains are best stored against insect attacks by _____

- **A.** keeping in a silo
- **B.** storing in the room

- **C.** leaving them on their cobs
- **D.** covering with leaves
- **Correct Option: Answer is A**

38.

- **A.** air
- **B.** soil
- **C.** water
- **D.** noise
- **Correct Option: Answer is C**

39. Which of the following resources is renewable?

- **A.** Petroleum
- **B.** Coal
- **C.** Iron
- **D.** Trees
- **Correct Option: Answer is C**

40. Which of the following roles is not performed by a government agency responsible for forest maintenance?

- **A.** Regulating the harvest of timber and poaching
- **B.** Creating, protecting and managing the forest reserves
- **C.** Tracking the movement of timber, wood, and wildlife
- **D.** Regulating mining activities in the forest
- **Correct Option: Answer is D**

41. Which of the following factors is not a source of variation in living organisms?

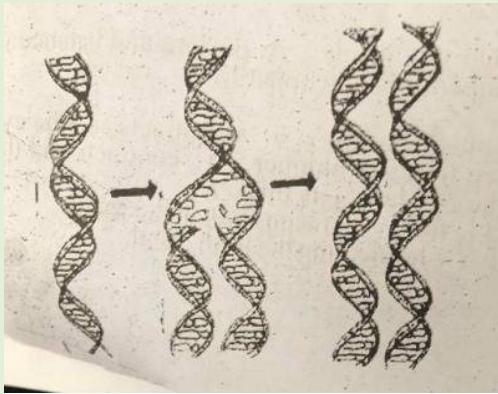
- **A.** Environment

- **B.** Inheritance
- **C.** Mitosis
- **D.** Meiosis
- **Correct Option: Answer is C**

42. Individuals with blood group AB can donate blood to individuals with blood group(s) _____

- **A.** A and B
- **B.** A and O
- **C.** AB only
- **D.** O only
- **Correct Option: Answer is A**

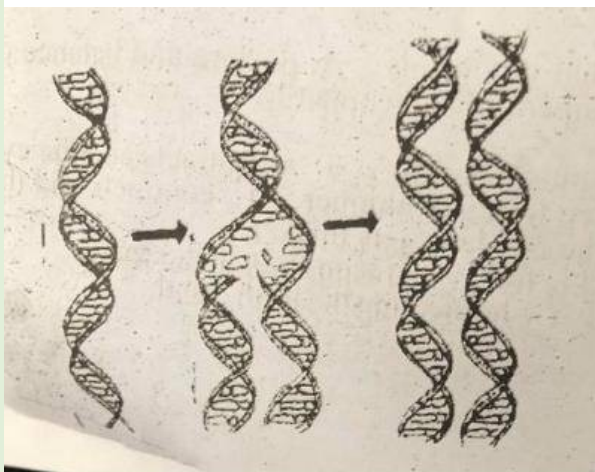
If your blood type is:	You can give to:	You can receive from:
AB Positive	AB+ Only	All Blood Types
O Negative	All Blood Types	O-
A Negative	A-, A+, AB-, AB+	A-, O-
B Negative	B-, B+, AB-, AB+	B-, O-



The diagram above is an illustration of a process occurring in a living animal cell. Study it and answer this question

43. The process occurs in the _____

- **A.** Nucleus
- **B.** Mitochondrion
- **C.** Golgi body
- **D.** Cytoplasm
- **Correct Option: Answer is A**



44. The process is called _____

- **A.** replication
- **B.** cell maturation
- **C.** crossing over

- **D.** mutation
- **Correct Option: Answer is A**

45. Which of the following nucleotides is not found in the DNA molecules?

- **A.** Adenine
- **B.** Cytosine
- **C.** Guanine
- **D.** Uracil
- **Correct Option: Answer is D**

46. In a complete dominance monohybrid cross between a pure breeding yellow-flowered plant Y and a pure breeding white-flowered plant y, the result of the first filial generation is _____

- **A.** 50 % yellow flowers
- **B.** 75% yellow flowers
- **C.** all yellow flowers
- **D.** all white flowers
- **Correct Option: Answer is C**

47. A man whose blood group is heterozygous **B** is married to a woman whose blood up is also heterozygous **B**

Which of the following statements is correct about the blood group of their four offspring?

- **A.** Three of them belong to blood group B
- **B.** Three of them belong to blood group O
- **C.** None of them belongs to blood group O
- **D.** Two of them belong to blood group B
- **Correct Option: Answer is A**

48. The cross between **RrTt** and **rrtt** where **R** is a gene for red colour and **T** for tallness will result in_____

- **A.** all the offspring being tall with red fruits
- **B.** 25 % tall with red fruits
- **C.** 50 % tall with red fruits
- **D.** 75 % tall with red fruits
- **Correct Option: Answer is C**

49. The yellow and black stripes on the body of wasps is an example of_____

- **A.** mimicry
- **B.** courtship behaviour
- **C.** camouflage
- **D.** warning colouration
- **Correct Option: Answer is D**

50. The biological importance of the dance by the worker honeybee is that it_____

- **A.** acts as a signal for mating
- **B.** serves as a warning signal
- **C.** signals the availability of food
- **D.** serves as a means of identifying members of the caste
- **Correct Option: Answer is C**

BIOLOGY PAST QUESTIONS THEORY 2019 WITH SOLUTIONS

51. a. Name five vertebrae in mammals with their corresponding locations. [10 marks]

b. State one function each of the following parts in a dicotyledonous leaf:

- (i) palisade;
- (ii) vascular bundle;
- (iii) epidermis.

[3 marks]

c. State one function each of the following parts in the stem of a flowering plant:

- (i) sieve tube;
- (ii) cortex;
- (iii) pith.

d. In a tabular form, state:

- (i) two external structural differences between the stem and root of a maize seedling. [2 marks]
- (ii) two internal structural differences between the stem and root of a maize seedling. [2 marks]

Solution & Explanation:

a. Vertebrae with corresponding location

Vertebrae	Location
Cervical	Neck
Thoracic	Chest/Thorax
Lumbar	Abdomen/Upper abdomen/loin
Sacral	Hip/lower abdomen
Caudal	Tail/coccyx

b. Function of:

(i) Palisade

- Manufactures food/carries out photosynthesis/site of photosynthesis;
- Contains chloroplasts for photosynthesis;

(ii) Vascular bundle

- Contains xylem for transport of water/mineral salts/phloem for translocation of manufactured organic substances/hormones;
- Provides mechanical support;

(iii) Epidermis

- It keeps the leaf's shape;
- Provides mechanical support;

c. Function of

(i) Sieve tube

- Translocates/transportes manufactured foods to other parts of the plant

(ii) Cortex

- The cells of cortex store starch;
- Outer cortex cells of green stems contain chloroplasts

and make food by photosynthesis/chlorophyllus cortex cells manufacture food;

(iii) Epidermis

- Cells of pith act as packing/food storage tissues;
- Supports the stem;

d. Structural differences

(i) External structural differences between stem and root of a maize seedling

Stem of a maize seedling	Root of a maize seedling
Fibrous root absent	fibrous root present
Chlorophyllus/greenish	non chlorophyllus/non-greenish
Leaves are present	leaves are absent
Thicker in size	thinner in size
Coleoptiles/shoot sheath present	coleoptiles/shoot sheath absent
Coleorhizae/radicle sheath absent	coleorhizae/radicle sheath present.

(ii) Internal structural differences between stem and root of a maize seedling

Stem of a maize seedling	Root of a maize seedling
Vascular bundles are scattered	vascular bundles are arranged/in a ring form
More vascular bundles	less vascular bundles

Cortex is narrower	cortex is wider
Pith is absent	pith is present
Endodermis absent	endodermis present
Phloem and xylem do not alternate	phloem alternates with xylem
Piliferous layer absent	piliferous layer present

52. a. State **three** features of the small intestine that increase the rate of absorption of digested food. [3 marks]

b. Explain **briefly** what happens to the glucose formed in a plant during photosynthesis. [3 marks]

c. Explain **briefly** the mode of feeding in each of the following organisms:

(i) Euglena; [4 marks]

(ii) Spirogyra. [4 marks]

d. State **three** characteristics of an Estuarine habitat. [3 marks]

e. State **three** differences between an aquatic habitat and a terrestrial habitat. [3 marks]

Solution & Explanation:

a. Features of the small intestine that increase the rate of absorption of digested food

- Ridges/furrows on inner linings;
- Small finger-like projections/villi;
- Minute projections/microvilli on the villi;

- Thin epithelial lining of the villi;
- Highly vascularised/well supplied with blood;
- Intestine is long/it has a large surface area.

b. Fate of glucose formed in a plant during photosynthesis

- Glucose combines with Nitrogen/Sulphur/Phosphorus to form proteins for growth;
- Glucose is stored as starch;
- Glucose is oxidised to form carbon dioxide/water/energy in respiration;
- Glucose is also converted into lipids/cellulose which is stored.

c. Mode of feeding of

(i) Euglena

- Exhibits both autotrophic and heterotrophic modes of nutrition;
- In autotrophic mode of feeding, the chlorophyll in its stellate chloroplast absorbs sunlight energy;
- And the dissolved carbon dioxide in the pond water diffuses into the cell of Euglena;
- Inside the chloroplast, the absorbed solar energy/sunlight is used in the synthesis of glucose from water and carbon dioxide;
- Glucose is converted into starch and stored in the paramylum granules;
- In heterotrophic mode of feeding, it takes in organic matter/diatoms e.t.c.;
- Through the gullet;
- And digests it in the reservoir.

(i) Spirogyra

- Manufactures its own food through photosynthesis;

- Its spiral chloroplast/chlorophyll absorbs the sunlight energy;
- It absorbs carbon dioxide from water/its environment to manufacture food;
- The manufactured food is used;
- while the rest/excess is stored in its pyrenoid in the form of starch;

d. Characteristics of an Estuarine Habitat

- It is shallow;
- It has a high level of nutrients;
- Salinity/pH/O₂ level fluctuates;
- Turbidity is high during rainy season;
- It is affected by tides;
- Species diversity is low;
- Muddy/soft soil/substratum

d. Differences between an aquatic and terrestrial habitat

Aquatic habitat	Terrestrial habitat
Temperature is lower	temperature is higher
Water is the medium for growth/reproduction	soil is a medium for growth
Presence of dissolved oxygen	presence of atmospheric oxygen
Organisms are not affected by humidity	organisms are affected by humidity
Organisms are affected by tide/waves/water/current	organisms are not affected by tide/waves/water current

Organisms are not affected by wind action	organisms are affected by wind action
Light affects the distribution of organisms more	light affects the distribution of organisms less
Water is abundant	water is not abundant

53. a. State **five** :

(i) challenges that make conservation of resources difficult; [5 marks]

(ii) ways by which the Government of a country ensures conservation of natural resources. [5 marks]

b. What is an endangered species? [3 marks]

c. A child with blood group O was born to a mother of blood group A.

(i) What is the blood group genotype of the child? [1 marks]

(ii) What are the possible blood group genotypes of the father? [3 marks]

d. State **three** reasons why genetic variation is important to plants. [3 marks]

Solution & Explanation:

a. (i) Challenges that make the conservation of resources difficult:

- Limited land area for development leads to the rapid destruction of the natural forest;
- Over grazing/over fishing;
- Poaching/ killing of endangered species;

- Bush burning/poor farming methods;
- Over cultivation of land;
- High demand for timber that has resulted in the felling of trees/indiscriminate cutting of trees for firewood;
- Damming of rivers for hydro-electricity and irrigation has resulted in the alteration of the ecosystem;
- Lack of awareness of the populace and policy makers of the need for conservation/lack of education;
- Improper enforcement of existing laws/poor leadership;
- Forest fire;
- Inadequate funding in conservation;
- Lack of international/local cooperation on conservation;
- Illegal/unlawful exploitation of minerals.

(ii) Ways by which Government ensures the conservation of resources:

- Creation of parks/gardens/wildlife/zoo/reserves;
- Putting in place a chemical and toxic waste dump programmes to avoid water pollution;
- Enactment of conservation laws;
- Enforcement of conservation laws;
- Creating awareness/education through the media;
- Establishment of conservation fund;
- Training of conservation personnel/employing experts to manage natural resources;
- Establishing local and international cooperation;
- Aforestation/planting of trees.

b. Endangered species

- Species whose population has become so small; that it can completely be wiped out/become extinct; if not properly managed.

- c. (i) Genotype of child: OO
(ii) Genotype of Father: OO; AO; BO;
- d. Importance of genetic variation to plants
- It enables plants to spread to colonies/new areas;
 - It may enable plants to survive harsh/adverse conditions/environments;
 - It may enable plants to be resistant to diseases;
 - It gives rise to varieties of desirable plant species;
 - It makes plants produce better fruits/yield;

54. A homozygous tall, red flower plant was crossed with a homozygous dwarf, white flower plant. The F1 were selfed and 160 F2 plants were obtained.

- a.(i) Name the type of genetic cross. [1 marks]
(ii) State **one** reason for the answer in 4(a)(i).

[1 marks]

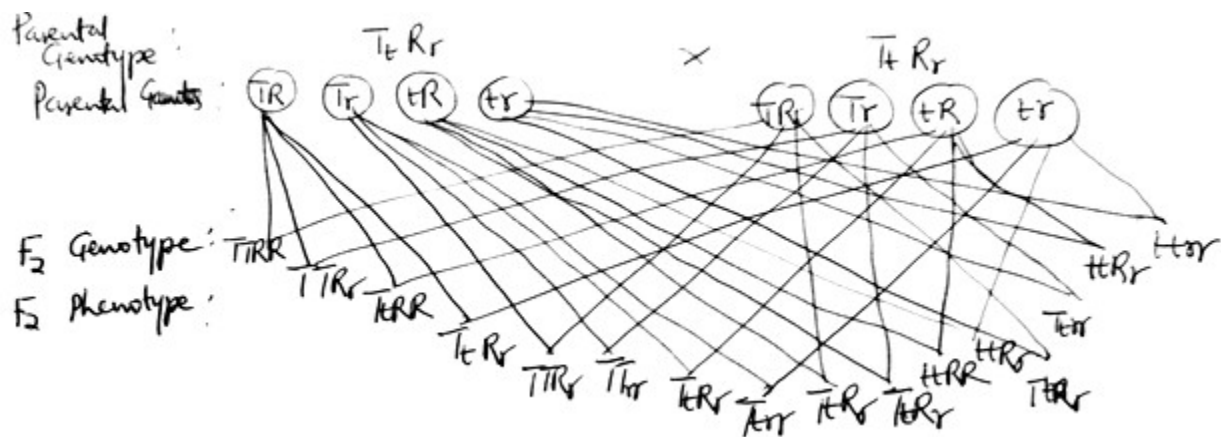
- b.(i) What is evolution? [2 marks]
(ii) With the aid of a genetic diagram, state the phenotypic ratio of F2 generation. [13 marks]
(iii) Calculate the number of tall and white flowers that would be obtained in the **F2** generation. [3 marks]

Solution & Explanation:

- a. (i) Type of genetic cross: Dihybrid
(ii) Reason for answer: Two characters are involved in the cross/involves two characters separable in inheritance.

- b. (i) Phenotype of F1: Tall and Red.
(ii) Genetic Diagram
Let T represent the gene for Tallness;

t represent the gene for Dwarfism;
 R represent the gene for Red;
 r represent the gene for White.



OR

Parental genotype TtRr X TtRr

X	TR	Tr	tR	tr
TR	TTRR Tall Red	TTRr Tall Red	TtRR Tall Red	TtRr Tall Red
Tr	TTRr Tall Red	TTrr Tall White	TtRr Tall Red	Tttr Tall White
tR	TtRR Tall Red	TtRr Tall Red	ttRR Dwarf Red	ttRr Dwarf Red
tr	TtRr Tall Red	Tttr Tall White	ttRr Dwarf Red	tttr Dwarf White

Crosses of parental genotypes

Tall and Red - 9

Tall and White - 3

Dwarf and Red - 3

Dwarf and White - 1

Parental genotype

Parental gametes (TR, Tr, tR, tr) shown

Correct Crosses of parental gamete

Phenotypic ratio 9:3:3:1

(iii) Calculation of Tall and White flowers in F₂

$$(3/16) \times (160/1) = 30$$

55. a. (i) Complete the table below with the respiratory surface of **each** of the listed organisms.

organism	respiratory surface
Tadpole (2days old)	
Cockroach	
Domestic fowl	
Talium	
Earth Worm	
Amoeba	

(ii) State **five** characteristic features of respiratory surfaces in organisms. [5 marks]

b. Make a drawing, 4 cm – 6 cm long of the respiratory organ in Tilapia and label fully. [9 marks]

Solution & Explanation:

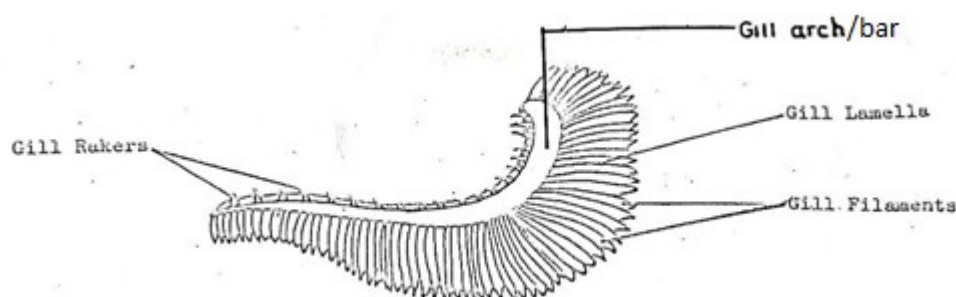
a. (i) Table

Organism	Respiratory
Tadpole (2 days old)	Gill (External)
Cockroach	Trachea/tracheoles
Domestic fowl	Lung/lungs
Talinum	Stoma/stomata
Earthworm	(moist) Skin surface/body surface
Amoeba	Cell membrane/plasmalemma/body surface

(ii) Characteristic Features of respiratory surfaces

- Large surface area;
- Thin-walled/thin body surface;
- Moist surface;
- Well-ventilated surface;
- Permeable surface;
- Highly vascularised/network of capillaries present.

b. Diagram of respiratory organ/gill of Tilapia



- 56.** a. List **five** parts of the alimentary canal of an earthworm. [5 marks]
b. State **six** ways by which water could be polluted by agricultural. [6 marks]
c. Describe briefly the life cycle of a housefly. [5 marks]
d. Make a diagram, 6 cm – 8 cm long of the hypogeal germination in a mature maize seedling and label fully. [8 marks]

Solution & Explanation:

- 57.** The table below shows the percentage composition of fat and protein in six different meat types. Study it and answer questions 2(a) to 2(d).

Meat Type	Fat (%)	Protein (%)
A	07.2	21.3
B	25.3	10.6
C	20.0	22.5
D	03.1	28.2
E	12.6	17.3
F	13.2	14.3

- a. (i) Which **two** of the meat types may be recommended for an obese patient? [2 marks]
(ii) State **one** reason for the answer in 2(a)(i). [1 mark]

- b. (i) Which **two** of the meat types would provide the most energy? [2 marks]
(ii) State **two** reasons for the answer in 2(b)(i). [2 marks]
- c. (i) Which **three** of the meat types could be recommended for a child suffering from kwashiorkor? [3 marks]
(ii) State **one** reason for the answer in 2(c)(i) [1 mark]
- d. Which of the meat types would most likely be suitable for:
(i) an active teenager; [1 mark]
(ii) a 70-year old human? [1 mark]
- e. Which other class of food provides energy? [1 mark]
- f. State **three** uses of fat in the human body. [3 marks]
- g. Describe briefly the procedure for testing for fat in a meat sample using a piece of white paper. [3 marks]

Solution & Explanation:

a. (i) Meat type that may be recommended for an obese patient

A, D

(ii) Reason for meat types

Meat types A and D have lower fat content compared to the other meat types.

b. (i) Meat type that would provide most energy

B, C

(ii) Reason

They have the highest fat composition/content; when oxidised would release much energy.

c. (i) Meat type that may be recommended for a child suffering from kwashiorkor

A, C, D and E

(ii) Reason

A child suffering from kwashiorkor is deficient in/lacks adequate amount of protein; needs more protein; and meat types A, C, D and E have high protein contents.

d. (i) Meat type most suitable for an active teenager

A, B, C, D, E, F/all meat types.

(ii) Meat type most suitable for a 70-year old human

A, D

e. Other class of food that provides energy

Carbohydrate

f. Uses of fat in the human body

- provides insulation;
- generates heat;
- provides energy;
- protects vital organs;
- forms part of the cell membrane/formation of phospholipids;
- smoothens the skin;
- forms solvent for some fat soluble vitamins/vitamin A,D,E,K.

g. Procedure of testing for fat in a meat sample using white paper

- The meat sample is rubbed against piece of (white) paper;
- and allowed to dry;
- a translucent spot on the paper;
- confirms the presence of fat in the meat sample.

58. a. Classify the following biological associations under the headings in the table below:

- (i) Remora and shark;
- (ii) Lichen;
- (iii) Cattle and white Egret;
- (iv) Tapeworm in the gut of humans;
- (v) Flowers and Honeybees;
- (vi) Mistletoe and Cacao plant.

Parasitism	Mutualism	Commensalism

[6 marks]

b. (i) State **three** adaptive features of parasites. [6 marks]

(ii) State **two** effects of parasites on their hosts. [2 marks]

c. (i) What are saprophytes? [2 marks]

(ii) Give **four** examples of saprophytes. [4 marks]

Solution & Explanation:9

Parasitism	Mutualism	Commensalism
Tapeworm in the gut of humans	Lichen	Remora and shark
Mistletoe and Cacao plant	Cattle and white Egret	
	Flowers and	

b. (i) Adaptive features of parasites

- possession of specialised organs/haustoria/suckers/hooks; for attachment to the surface of the body of the host;
- specialised piercing/sucking mouthparts; for sucking fluids/blood/plant juices;
- hermaphrodite/ability to reproduce rapidly/easily in large numbers/large quantity of eggs are produced; to propagate their lineage/species;
- large surface area; for absorption of digested food;
- possession of special features/structures/cuticle; to withstand harsh environment/conditions/being digested.

(ii) Effects of parasites on their host

Absorption of nutrients from the body of the host can:

- lead to stunted growth;
- lead to low resistance to diseases in the host;
- lead to loss of blood;
- lead to loss of weight;
- lead to septicaemia;
- cause diseases/injuries/harm;
- lead to death;
- de-value the worth of animal skin/hides/ reduce the value of farm animals.

c. (i) Saprophytes

are organisms that feed; on dead/decaying/decomposing organic matter.

(ii) Examples of saprophytes

- Fungi/Bacteria/
- Mushroom
- Azotobacter

- Rhizopus;
- Clostridium
- Mucor
- Streptomyces
- Toadstool
- Mold/mould
- Penicillium

59. Use the information below to answer questions (a)(i) and (a)(ii).

Hb A represents normal haemoglobin,
Hb S represents sickled haemoglobin.

- a. A female heterozygote for sickle cell married a sickler. With the aid of a genetic diagram, determine the:
 - (i) possible genotypes of their offspring; [8 marks]
 - (ii) phenotypic ratio of the offspring. [2 marks]
- b. Explain **briefly** the reason why a Rhesus negative woman married to a Rhesus positive man might lose her second pregnancy. [5 marks]
- c. Name **two** examples of features in animals that support the theory of use and disuse of body parts as used by Lamarck. [2 marks]
- d. List **three** structures in mammals that are vestigial. [3 marks]

Solution & Explanation: 10

(ii) Phenotypic ratio

2:2 or 1:1/Ratio of sickler to non-sicklers.

b. Reasons why a rhesus negative woman married to a rhesus positive man might lose her second pregnancy

- Rhesus negative mother has no antigen on the surface of her Red Blood Cells/RBC;
- the husband has antigen on his RBC;
- during her first pregnancy, the child carries positive Rhesus factor;
- the Rhesus factor enters the mother's blood stream;
- due to rupture of the placenta, blood seeps into the mother's blood;
- causing it to develop anti-Rhesus antibodies;
- causing agglutination in small proportion of foetal blood cells;
- in the second pregnancy, the child might be Rhesus positive;
- the child carrying Rhesus positive suffers massive destruction of red blood cells/erythroblastosis fetalis;
- which might result into miscarriage of the second child.

c. Examples of features in animals that support the theory of use and disuse of body parts as used by Lamarck

- long neck in Giraffe;
- smaller wings in Penguins;
- lost legs of snakes;
- long legs of Heron;
- webbed feet of ducks/frogs/geese.

d. Vestigial organs in mammals

- appendix in humans;
- pelvic and leg bones in whale;
- reduced coccyx in man/caudal vertebrae;
- pyramidalis nasi (extrinsic) muscle/muscles of the forehead;

- posterior auricular muscles to move the ear/pinna;
- erector muscle;
- nipples in males;

60. a. Explain **briefly** the importance of the following factors and organisms in the nitrogen cycle

- (i) *lightning*; [3 marks]
- (ii) *Nitrosomonas*; [2 marks]
- (iii) *Azotobacter*. [2 marks]

b. Name the excretory organ in:

- (i) insects: [1 mark]
- (ii) earthworm.

c. Define

- (i) hepatitis; [4 marks]
- (ii) kidney stones [4marks]

d. (i) State **three** effects of lack of sense receptors in the skin to humans. [3 marks]

(ii) List **three** layers of the epidermis in the skin of humans. [3 marks]

e. (i) What is **metamorphosis**? [2 marks]

(ii) State the type of metamorphosis exhibited by **each** of the following insects listed in the table below.

Insect	Type of metamorphosis
Grasshopper	
Cockroach	
Butterfly	
Mosquito	

Solution & Explanation:

a. Importance of the following in the Nitrogen Cycle

(i) Lightning

- When lightning occurs, nitrogen reacts with oxygen in air to form nitrous oxide/oxides of nitrogen;
- which is further oxidised to form nitrogen (IV) oxide /nitrogen dioxide;
- which dissolves in rain water to form nitric and nitrous acids;
- (that fall to the ground) when it reaches the soil combines with bases/mineral salts to form nitrates;
- which the plants absorb.

(ii) Nitrosomonas

- Is a nitrifying bacterium in the soil;
- that converts ammonia to nitrites in the soil.

(iii) Azotobacter

- Is a nitrogen-fixing bacterium in the soil;
- that converts nitrogen in the soil;
- into ammonium compounds/amino compounds and protein in the soil;
- converts nitrites to nitrates in the soil.

b. Excretory organs

Insects – malpighian tubules;

Earthworm – nephridium

c. Explanation of diseases

(i) Hepatitis

- Is the inflammation/destruction of liver cells;
- hepatitis can arise from bacteria/virus/an

autoimmune disease;

- or from excess consumption of alcohol.

The symptoms include fatigue;

- loss of appetite/anorexia;
- muscle aches/pains;
- flu;
- fever;
- nausea;
- jaundice;
- and dark coloured urine/stool.

(ii) Kidney stones

- Are small solids/crystals of concentrated minerals/uric acid found in the urine;

- they are formed if there is an excessive loss of water/insufficient water intake;

- so that the solubility of minerals/calcium phosphate/calcium oxalate in the urine is reduced;

- high levels of uric acids/amino acids can lead to kidney stones;

- bacteria may be trapped in the kidney tubules as the kidney stones get bigger;

- symptoms are sharp pain at the flank of back;

- frequent urination;

- nausea;

- sweating;

- kidney stones can be treated by medications;

- change in diet of food rich in calcium;

- surgery;

- drinking plenty of water.

d. (i) Effects of lack of sense receptors in the skin

- Humans will not be able to sense/feel/perceive external stimuli in the environment;

- pain;

- touch;
- a change in temperature;
- pressure;
- danger which could lead to death.

(ii) Layers of the epidermis

- cornified layer/stratum corneum;
- granular layer/stratum granulosum;
- malpighian layer/stratum Malpighi.

e. (i) Metamorphosis

Is the development of some/certain organisms from the egg; to an adult stage; involving three to four stages.

(ii) Type of metamorphosis of listed insects

Insect	Type of metamorphosis
Grasshopper	Incomplete
Cockroach	Incomplete
Butterfly	Complete
Mosquito	Complete
Housefly	Complete

61. . a. Complete the table below.

Organism	Mode of feeding	Feature of mouthpart that adapts organism to mode of feeding

[6 marks]

- b. List five parts of the alimentary canal of an earthworm. [5 marks]
- c. State six ways by which water could be polluted by agricultural practices. [6 marks]
- d. Describe briefly the life cycle of a housefly. [5 marks]
- e. Make a diagram, 6 cm – 8 cm long of the hypogeal germination in a mature maize seedling and label fully. [8 marks]

Solution & Explanation:

a. Table

Organism	Mode of feeding	Feature of mouthpart that adapts organism to mode of feeding
Adult mosquito	Fluid feeding/piercing/sucking	Proboscis/modified long mandible and maxillae
Weevil	Chewing/boring	Possession of pointed short and strong rostrum
Cockroach	Biting/chewing	Well developed

ch		mandibles/strong sharp mandibles
----	--	----------------------------------

b. Parts of alimentary canal of earthworm

- Mouth;
- Pharynx;
- Oesophagus/gullet;
- Crop;
- Gizzard
- Intestine;
- Anus.

c. Ways by which water could be polluted by agricultural practices

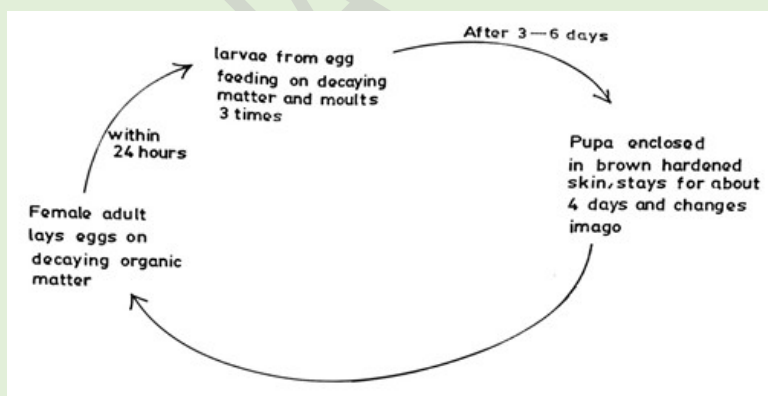
- Farmers add excess fertilizers/organic material to soil which may cause water pollution;
- The inorganic/artificial fertilizers that contain high concentrations of nitrates and phosphates, dissolves in rain water and washed into nearby streams and rivers;
- Nitrates and phosphates accelerate the growth of algae/eutrophication;
- The death/decay of algae in rivers and streams increase the population of bacteria rapidly;
- And cause oxygen level to decrease drastically;
- Killing aquatic organisms;
- Use of chemicals/dynamite for fishing which pollutes water
- Dumping of cleared vegetation on water bodies;
- Slash-burn methods of farming which might wash ash to nearby rivers/water bodies;
- Pesticides/insecticides also sprayed on crops are usually non-biodegradable;
- They are washed into nearby rivers/streams/water bodies;

- Accumulate in the bodies of aquatic organisms;
- And are passed along the food chain;
- The concentration of pesticides/insecticides can reach toxic level in the bodies of the final consumers in the higher trophic levels is causing these organisms to die;

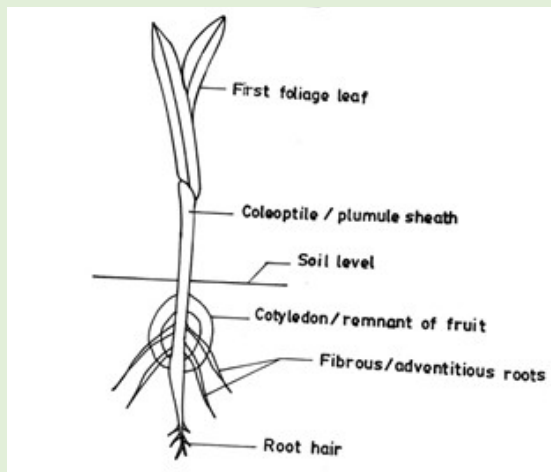
d. (i) Life cycle of a Housefly

- The female housefly lays eggs on decomposing matter/rubbish;
- The eggs hatch into larvae/maggot;
- In a few hours;
- They crawl unto food;
- Which if solid will be digested by enzymes to liquid;
- The larvae have twelve segments on the body;
- The body then contracts;
- The skin hardens and turns brown to enclose the pupa;
- After about four days, the pupa changes into the adult/imago;
- Inside the puparium/brown skin cover;
- The adult fly then emerges;
- And flies away.

OR



e. Diagram of a mature maize seedling



BIOLOGY PAST QUESTIONS 2018 WITH SOLUTIONS

- 1.** The scientist who introduced binomial nomenclature in the classification of organisms was
- **A.** Charles Darwin
 - **B.** Carolus Linnaeus
 - **C.** John Ray
 - **D.** Louis Pasteur

Short Explanation: Carolus Linnaeus, the Swedish naturalist, is the man behind this present way of classifying plants and animals.

Correct Option: Answer is B

2. Which of the following statements is true about arthropods?

- **A.** Prothorax bears only legs
- **B.** Mesothorax bears only legs
- **C.** Metathorax bears only wings
- **D.** Prothorax bears only wings

Short Explanation: The prothorax contains the first pair of legs in the arthropods. It's the first of the three segments of the thorax.

Correct Option: Answer is A

3. What level of organisation is Spirogyra?

- **A.** System
- **B.** Organ
- **C.** Cell
- **D.** Tissue

Short Explanation: Unicellular organisms like the Amoeba, Spirogyra etc fall in the cellular level of organisation.

Correct Option: Answer is C

4. The streaming movement of cytoplasm within the Paramecium is known as

- **A.** Transpiration
- **B.** Digestion
- **C.** Cyclosis
- **D.** Osmosis

Short Explanation: Cyclosis is also called Cytoplasmic streaming. It is defined as the streaming rotary motion of the protoplasm within certain cells and unicellular organisms. Paramecium is an example of a unicellular organism

Correct Option: Answer is C

5. Which of the following organelles is found only in plant cells?

- **A.** Mitochondria
- **B.** Ribosomes
- **C.** Lysosomes
- **D.** Plastids

Short Explanation: Plastids are found only in plants and algae. One common type of plastid is the chloroplast.

Correct Option: Answer is D

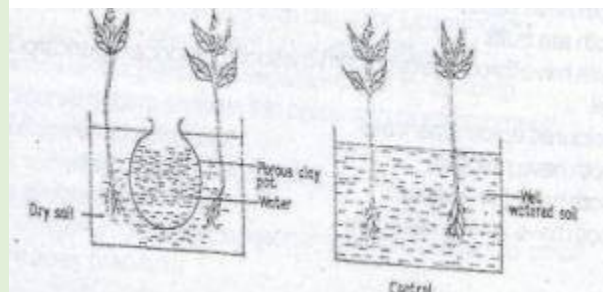
6. Active transport differs from diffusion in that active transport

- **A.** is a very fast process
- **B.** allows the movement of substances against concentration gradient
- **C.** takes place in both light and dark reaction of photosynthesis
- **D.** occurs in both plant and animal tissues

Short Explanation: Diffusion, as well as osmosis depends only on concentration gradient but active transport works against concentration gradient.

Correct Option: Answer is B

7.



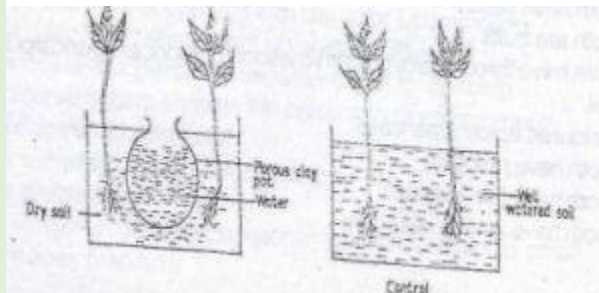
The diagrams above are illustrations of an experimental set-up to demonstrate a type of tropic response in plants. The type of response demonstrated is

- **A.** phototropism
- **B.** geotropism
- **C.** hydrotropism
- **D.** thigmotropism

Short Explanation: From the first diagram, you notice the roots of the plants bending towards the porous water pot, hence this is a form of response in the direction of the stimuli of the water. This is a form of hydrotropism.

Correct Option: Answer is C

8.



The conclusion drawn from the experiment in the diagram above is that

- **A.** shoots of plants are negatively thigmotropic
- **B.** shoots of plants are negatively geotropic
- **C.** shoots of plants are positively phototropic
- **D.** roots of plants are positively hydrotropic

Short Explanation: The roots of the plants in diagram 1 bend towards the porous clay pot while growing and the well-watered soil sees the plants growing straight and erect. The roots of the plants are positively hydrotropic (respond towards the pull of water positively).

Correct Option: Answer is D

9. The odontoid process is found on the

- **A.** axis vertebra
- **B.** atlas vertebra
- **C.** thoracic vertebra
- **D.** sacral vertebra

Short Explanation: The **odontoid process** (also dens or **odontoid** peg) is a protuberance (**process** or projection) of the Axis (second cervical vertebra). It exhibits a slight constriction or neck, where it joins the main body of the vertebra

Correct Option: Answer is A

10. A seedling was made to stand in a solution of red ink for three hours and a transverse section of the stem was examined under the microscope. The process being investigated was

- **A.** the importance of micro elements to plants
- **B.** ascent of water through the xylem
- **C.** diffusion of colored substances
- **D.** distribution of food in plants

Short Explanation:

Correct Option: Answer is C

. Which of the following structures would carry out cellular respiration? A

-
- **A.** germinating seed
- **B.** decolorised leaf
- **C.** dry leaf

- **D.** boiled cowpea

Short Explanation: Germinating seeds respire by using oxygen and producing carbon dioxide gas, they respire in the similar way as animals

Correct Option: Answer is A

12. Which of the following is NOT an excretory product of animals?

- **A.** carbon dioxide
- **B.** urea
- **C.** sweat
- **D.** oxygen

Short Explanation: Animals take in oxygen in respiratory process, taking out carbon dioxide.

Correct Option: Answer is D

13. An example of homeostasis in living organisms is

- **A.** cooling effect resulting from evaporation of water from the body surface
- **B.** root hairs of a plant growing towards the source of light
- **C.** changing of the body colour of chameleon to match the colour of the foliage on which it is resting
- **D.** the release of phosphorus into the phloem of a plant growing in a phosphorus-deficient soil

Short Explanation: *Homeostasis*, any self-regulating process by which *biological* systems tend to maintain stability while adjusting to conditions that are optimal for survival.

Correct Option: Answer is C

14. Ultrafiltration in the kidney takes place in the

- **A.** Renal vein
- **B.** Medulla
- **C.** Loop of Henle
- **D.** Bowman's Capsule

Short Explanation: **Bowman's capsule** (or the **Bowman capsule**, capsula glomeruli, or glomerular **capsule**) is a cup-like sack at the beginning of the tubular component of a nephron in the mammalian kidney that performs the first step in the filtration of blood to form urine. A glomerulus is enclosed in the sac.

Correct Option: Answer is D

- 15.** The thyroid gland is found at the base of the
- **A.** Midbrain
 - **B.** Kidney
 - **C.** Neck
 - **D.** Liver

Short Explanation:

The **thyroid** is a butterfly-shaped gland that sits low on the front of the neck.

- 16.** Which of the following statements about the response of neurones to stimulus is correct?
- **A.** Neurons respond to stimulus of any strength
 - **B.** Neurons respond to all stimuli applied in rapid succession
 - **C.** Neurons respond more rapidly during the absolute refraction period
 - **D.** Intensity of stimulus must reach a threshold value before the neurons can be excited

Short Explanation: Each neuron has a threshold that must be met before an action potential travels the length of the neurons axon. In other words, the neurons don't just begin to act immediately, there is a threshold value

where the intensity of the stimulus must get to in order to yield a response from the neuron. This is why you can consciously pinch yourself and not feel anything serious but when it is unconscious, it gives you serious pain

Correct Option: Answer is D

17. Which of the following practices may lead to infection of the eye? Use of

- **A.** contact lenses
- **B.** convex lenses
- **C.** biconcave lenses
- **D.** concave lenses

Short Explanation: Contact lenses have a way of causing infections to the eyes at some point. It can cause a lack of oxygen since they are worn directly in the eye, especially if worn for long periods at a stretch.

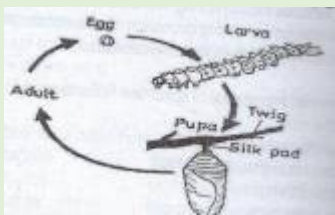
Correct Option: Answer is A

18. The structures for gaseous exchange in breathing roots are

- **A.** stomata
- **B.** lenticels
- **C.** cuticle
- **D.** mitochondria

Short Explanation: Breathing roots, also known as pneumatophores, have portions of the root which grow upward until they project some centimetres above the low-tide level. They have small openings called lenticels in their bark so that air can reach the rest of the plant's root system.

Correct Option: Answer is B



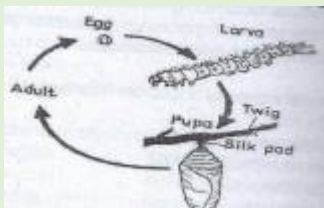
19.

The diagram above is an illustration of the life cycle of an insect. The adult fly in this cycle is called

- **A.** Housefly
- **B.** Butterfly
- **C.** Cockroach
- **D.** Mosquito

Short Explanation: From the diagram, the larva is a caterpillar, hence, the adult is called a Butterfly.

Correct Option: Answer is B



20.

In the diagram above, the larva is also known as

- **A.** Nymph
- **B.** Caterpillar
- **C.** Maggot
- **D.** Chrysalis

Short Explanation: The next stage is the **larva**. This is **also called a caterpillar** if the insect is a **butterfly** or a moth. The job of the **caterpillar** is to eat and eat and eat. As the **caterpillar** grows it splits its skin and sheds it about 4 or 5 times

Correct Option: Answer is B

21. The source of energy required by plants during food production is

- **A.** Photosynthesis
- **B.** Chlorophyll
- **C.** Sunlight
- **D.** Microorganisms

Short Explanation: The food production process (photosynthesis) makes use of the energy of light for the process.

Correct Option: Answer is C

22. One major difference between plant and animals' nutrition is the ability of plants to synthesize

- **A.** Food for plants and animals
- **B.** Water for plants
- **C.** Water for animals
- **D.** Food for plants only

Short Explanation: The food produced by plants serve as food both for the plants and animals (herbivores and in extension, all animals.)

Correct Option: Answer is A

23. By what process is starch converted into maltose?

- **A.** Hydrolysis
- **B.** Condensation
- **C.** Translocation
- **D.** Photosynthesis

Short Explanation: Carbohydrates are broken down into their component sugars by hydrolysis. This type of hydrolysis is called Saccharification

Correct Option: Answer is A

24. The ascent of water in tall trees is mainly due to

- **A.** adhesive forces
- **B.** transpiration pull
- **C.** root pressure
- **D.** cohesive forces

Short Explanation: Transpiration pull or the suction force is the force which aids in drawing the water upward from roots to leaves which is used by the leaves for photosynthesis and excess is released through the stomata

Correct Option: Answer is B

25. The duodenum of a person was surgically removed. Which of the following substances would have their digestion affected?

- **A.** starch and protein only
- **B.** starch and lipids
- **C.** starch, peptones and maltose
- **D.** starch, protein and lipids

Short Explanation: These three (starch, protein and lipids) have a process to undergo in the duodenum (the beginning of the small intestine), therefore, the surgery will affect them

Correct Option: Answer is D

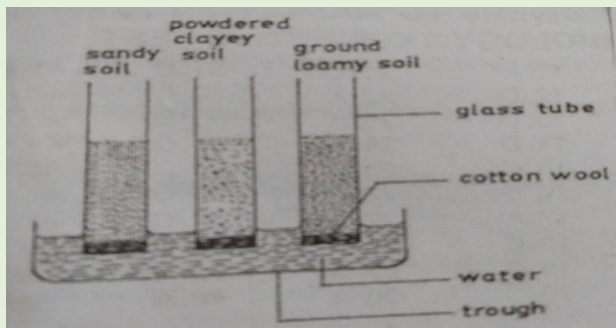
26.

Which of the following dental formulae represents the dentition in rabbits?

- **A. I 2/1 C 0/0 P 3/2 M 3/3**
- **B. I 2/2 C 1/1 P 2/2 M 3/2**
- **C. I 1/2 C 0/0 P 2/3 M 3/3**
- **D. I 2/1 C 0/01 P 3/2 M 3/3**

Short Explanation: The normal dental formula of rabbits is I2/1 C0/0 P3/2 M3/3 for 28 teeth

Correct Option: Answer is A



27.

The diagram above is an illustration of an experimental set-up to demonstrate a property of soil. The property of soil demonstrated is

- **A. particle size**
- **B. capillarity**
- **C. colour**
- **D. texture**

Short Explanation: The setup is to show the capillarity (water rise) of each of the soil type

Correct Option: Answer is B

28. Which of the following statements would be a correct observation at the end of the experiment? Water moves

- **A. highest in sandy soil**
- **B. lowest in loamy soil**
- **C. to the same level in all soil types**
- **D. highest in clayey soil**

Short Explanation: Capillarity in soil refers to the ability of a soil to allow and retain water in it. Clayey soil has the highest tendency to hold water hence, the water would rise highest in it

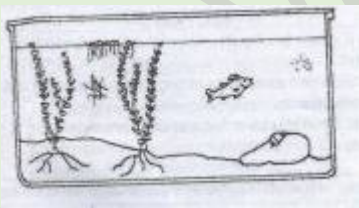
Correct Option: Answer is D

29. The most important factor(s) that influence(s) the ecological niche of an organism is/ are the

- **A.** water, light and predator-prey relationship
- **B.** food and water availability
- **C.** physical environment
- **D.** competition for food and space

Short Explanation: The competition in a territory is the factor that basically determines whether or not a specie will survive or not in that territory. Ecological niche is the total package of the habits of a specie. The competition for food and water is the most important influencer of the ecological niche

Correct Option: Answer is D



30.

The diagram above is an illustration of organisms in an air-tight aquarium. The most important factor needed by the organisms is

- **A.** light energy
- **B.** low humidity
- **C.** oxygen
- **D.** nitrate

Short Explanation:

Correct Option: Answer is A

31. Which of the following instruments is used to determine the turbidity of water?

- **A.** hygrometer
- **B.** hydrometer
- **C.** secchi disc
- **D.** rain gauge

Short Explanation: The *Secchi disk* (or *Secchi disc*), as created in 1865 by Angelo Secchi, is a plain white, circular *disk* 30 cm (12 in) in diameter used to measure water transparency or turbidity in bodies of water.

Correct Option: Answer is C

32. In an ecosystem, the least efficient energy transfer link is from the

- **A.** secondary consumers to decomposers
- **B.** sun to producers
- **C.** primary consumers to secondary consumers
- **D.** producer to primary consumers

Short Explanation: From the food chain, the energy flows from the producers to the primary consumer to the secondary consumers and about 90% of the energy is lost in the process of this transfer. Hence, the energy level by the time it gets to the secondary consumers, there is less energy transferred to the decomposers.

Correct Option: Answer is A

33. Xerophytes have the following characteristics aimed at conserving water except

- **A.** thick cuticle
- **B.** broad leaf surfaces

- **C.** sunken stomata
- **D.** waxy cuticle

Short Explanation: The xerophytes have reduced leaf size in order to reduce the surface area, thereby reducing the water loss in the plant.

Correct Option: Answer is B

34. An association between living organisms in which one lives on and feeds at the expense of the other is known as

- **A.** commensalism
- **B.** predation
- **C.** parasitism
- **D.** mutualism

Short Explanation: A parasite is an organism that lives on or in a host organism and gets its food from or at the expense of its host

Correct Option: Answer is C

35. When large numbers of organisms share limited space and resources, the immediate result is

- **A.** extinction
- **B.** emigration
- **C.** competition
- **D.** immigration

Short Explanation: Competition in a territory is as a result of a hassle for the available resources within the territory. For instance, when the water in a river seems to be drying out, the organisms in the water begin to compete for the water

Correct Option: Answer is C

36. Which of the following disease is caused by a bacterium?

- **A.** ringworm
- **B.** poliomyelitis
- **C.** syphilis
- **D.** malaria

Short Explanation: Syphilis is caused by a bacterium known as *Treponema pallidum*. It is majorly passed on by sex

Correct Option: Answer is C

37. Conservation of wildlife is necessary mainly because

- **A.** wild species cannot resist disease better than their domesticated species
- **B.** many predator wild animals help in the control of pests
- **C.** it prevents extinction of species
- **D.** many people derive pleasure from observing wild animals

Short Explanation: Wildlife conservation is necessary for the prevention of the extinction of wildlife. It helps with the preservation of these wild animals

Correct Option: Answer is C

38. Two unconscious patients X and Y whose blood group genotypes are AO and AB respectively were transfused with the blood from the same donor. Patient X immediately showed signs of difficulty in breathing while patient Y showed no negative reaction. Patients X and Y were likely transfused with blood of genotype

- **A.** OO
- **B.** AO
- **C.** BO
- **D.** AA

Short Explanation: The blood types A and AB can receive blood from blood types A and O but not B. Hence, if there was a problem after transfusion, it was most likely to be from BO

Correct Option: Answer is C

39. Two unconscious patients X and Y whose blood group genotypes are AO and AB respectively were transfused with the blood from the same donor. Patient X immediately showed signs of difficulty in breathing while patient Y showed no negative reaction. What should the hospital have done to prevent X from showing the symptom described above? Patient X should have

- **A.** undergone an agglutination test
- **B.** been asked for the blood group
- **C.** been screened for HIV
- **D.** undergone malaria test

Short Explanation: The basic technique in identification of the antigens and antibodies of blood groups is the agglutination test and can be done to prevent the clumping of the red blood cell. The symptom is due to clumping due to incompatible blood transfusion.

Correct Option: Answer is A

40. The greatest contribution to genetic studies was made by

- **A.** Gregor Mendel
- **B.** Thomas Morgan
- **C.** Charles Darwin
- **D.** Robert Hooke

Short Explanation: Gregor Mendel is considered as the founder of modern science of genetics

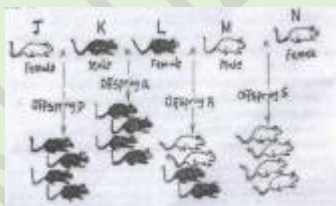
Correct Option: Answer is A

41. The exchange of genes between homologous chromosomes is called

- **A.** crossing over
- **B.** back cross
- **C.** test cross
- **D.** mutation

Short Explanation: Crossing over is the process where homologous chromosomes pair up with each other and exchange different segments of genetic material to form recombinant chromosomes.

Correct Option: Answer is A



42.

The diagrams above are illustrations of the inheritance of coat colour in mice. J, K, L, M and N are parents. The cross between two parent mice gave rise to offspring P, Q, R and S. Which of the offspring are all homozygous?

- **A.** P

- **B.** Q
- **C.** S
- **D.** R

Short Explanation:

Correct Option: Answer is C

43. Genes that remain linked are those

- **A.** close to each other on the chromosomes
- **B.** on the same locus on the chromosomes
- **C.** on the homologous chromosomes
- **D.** which are allelic pair

Short Explanation: Linked genes are genes that are found near each other on the same chromosome.

Correct Option: Answer is A

44. plant cell has 14 chromosomes prior to mitosis. What is the number of chromosomes in the daughter cells?

- **A.** 14
- **B.** 56
- **C.** 28
- **D.** 7

Short Explanation: Mitosis is a process of cell reproduction in which one cell gives rise to two genetically identical daughter cells of the same number. Hence, the daughter cell has the same number as the parent cell prior to the process.

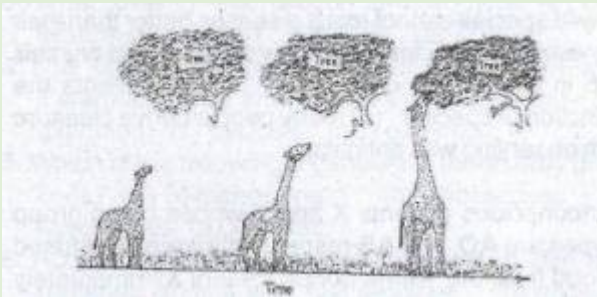
Correct Option: Answer is A

45. The chemical bond that holds the bases of the two strands of DNA together is

- **A.** oxygen bond
- **B.** hydrogen bond
- **C.** electrovalent bond
- **D.** covalent bond

Short Explanation: Hydrogen bonds form between the nitrogenous bases in one strand and the nitrogenous bases in the other strand of DNA

Correct Option: Answer is C



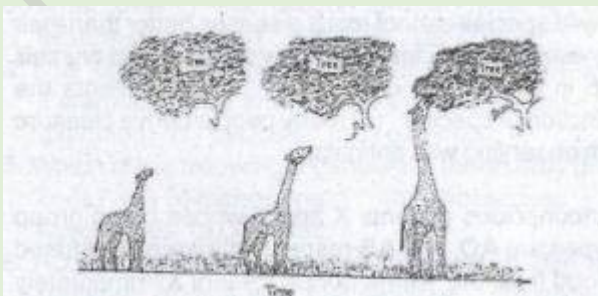
46.

The diagram above is an illustration of one of the theories of evolution. Who proposed the theory in the illustration?

- **A.** Louis Pasteur
- **B.** Gregor Mendel
- **C.** Robert Hooke
- **D.** Jean Lamarck

Short Explanation: This is the law of use and disuse propounded by Jean-Baptiste Lamarck.

Correct Option: Answer is D



47.

The theory supports

- **A.** natural selection
- **B.** struggle for survival
- **C.** independent assortment
- **D.** the use and disuse of body parts

Short Explanation: The theory implies that the more you use a part of your body, with time that part of your body starts adjusting in order to make the work easy

Correct Option: Answer is D

48. In evolution, fossils are naturally preserved in

- **A.** rocks
- **B.** water
- **C.** chemicals
- **D.** trees

Short Explanation: Some sedimentary rocks actually have fossils stored in them. Therefore, fossils are stored in rocks

Correct Option: Answer is C

49. In evolution, fossils are naturally preserved in

- **A.** rocks
- **B.** water
- **C.** chemicals
- **D.** trees

Correct Option: Answer is B

50. Which of the following is **not** a social insect?

- **A.** termite
- **B.** housefly

- **C.** honeybee
- **D.** ant

Short Explanation: Social insects are characterized by Group integration (blending the new members into the already existing social structure), Division of labour and Overlap of generations (having several breeding seasons, hence having multiple generations present at a time)

Correct Option: Answer is B

BIOLOGY PAST QUESTIONS 2018 THEORY WITH SOLUTION

1.(a) *Explain briefly how the structure of each of the following cells relate to their functions:*

(i) Sperm cell

(ii) Palisade cell.

(b) Make a drawing, 8cm to -10cm long of the front view of the female reproductive system in humans and label fully.

(c) State two differences between reproduction in mammals and in amphibians.

(d) State two methods of birth control in humans.

Solution & Explanation 1:

(a) Structure of the cell relating to its function

(i) Sperm cell

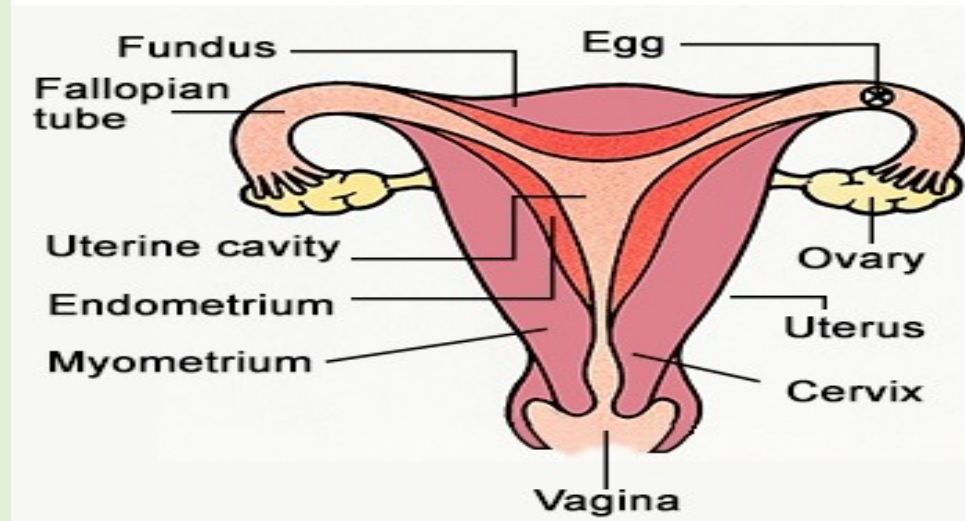
- Has a long/whip-like tail/flagellum, with which it swims to reach the egg/ovum of the female organism.
- Has a head with acrosome; for digesting/penetrating egg membrane
- Has a nucleus in its head; which fuses with the nucleus of the ovum/egg cell during fertilisation to form zygote
- Its middle piece contains mitochondria; for supply of energy/ATP.

(ii) Palisade cell

- Has thin wall; for easy penetration of light
- Has numerous chloroplasts in peripheral cytoplasm for maximum absorption of sunlight

- Has large vacuole containing cell sap; for temporary storage of products of photosynthesis.

(b) Diagram of the front view of the female reproductive system



Mammals	Amphibians
fertilization is internal	fertilization is external
Development of offspring is internal / Viviparous born young ones alive	development of offspring is external / Oviparous/ lay eggs
placenta is present	placenta is absent
shows parental care	does not show parental care
produce fewer eggs / offspring	Produce numerous eggs/offspring

Differences between reproduction in Mammals and Amphibians

(d) Methods of birth control in humans

- Abstinence
- Use of contraceptive pills/injection of hormonal drugs
- Coitus interruptus/withdrawal method
- Safe period/rhythm method
- Sterilization of men/vasectomy
- Sterilisation of women/tubal ligation
- Use of condoms/female condom/male condom
- Use of spermicide/foam
- Use of implant/Norplant
- Use of IUD/intra-uterine devices
- Use of diaphragm.

2. (a). Name the gases involved in the photosynthesis of a plant.

(b) State one role each of the gases named in 2 (a).

(c) (i) What is a variegated leaf?

(ii) Which part of the variegated leaf would test positive when treated with iodine solution?

(iii) State two reason for the answer in 2(c) (ii).

(iv) Name one mineral element required by plants for the formation of the part that would test positive in 2(c)(ii).

(d) The table below indicates different methods by which organism obtain food. Place the following organism under the heading in the table below: Human, Mushroom, Venus flytrap, Waterleaf plant, Tapeworm, Elephant grass, Housefly, Lichen, Spirogyra, Rhizopus.

Holozoic	Parasitic	Symbiotic	Saprophytic	Autotrophic

Mode of Nutrition

Solution & Explanation 2:

(a) Gases involved in the photosynthesis of a plant

- Carbon dioxide/Carbon (IV) oxide
- Oxygen

(b) Role of gases involved in photosynthesis

- (i) Carbon dioxide/Carbon (IV) oxide: raw material for photosynthesis/to produce glucose/starch
- (ii) Oxygen: by-product/given off during photosynthesis.

(c) Variegated leaf

- (i) A leaf with different colour parts/patches/green and non-green parts.
- (ii) Part of variegated leaf that would be positive when treated with iodine solution the green part/patch

(iii) Reasons for the answer in 2(c) (ii)

Presence of chlorophyll/chloroplast; for trapping solar energy; to manufacture starch; which turns blue-black with iodine solution

- (iv) Mineral elements used to manufacture chlorophyll
Magnesium/iron/nitrogen.

(d)

Holozoic	parasiti	symbiot	saprophy	Autotrophic
----------	----------	---------	----------	-------------

	c	ic	tic	
Human	Tapeworm	Lichen	mushroom	water leaf plant
Housefly			Rhizopus	Elephant grass
Venus flytrap				Spyrogyra
				Venus flytrap

3. (a). Explain briefly the following terms:

(i) Renewable natural resources

(ii) Non -renewable natural resources;

(b) Give two examples each of:

(i) Renewable natural resources

(ii) Non -renewable natural resources

(c) State five ways of conserving forests

(d) State four effects of adding animal manure to garden soil.

(e) Give the possible genotypes in humans;

Solution & Explanation 3:

(a) (i) Renewable natural resources

Materials that can be reused/replaced/ regenerated/recycled; after they have been used

(ii) Non-renewable natural resources

Materials that are difficult/impossible to regenerate/ recycled; after being exhausted/depleted.

(b) (i) Renewable natural resources

Air; soil; crops; fisheries; water; vegetation/forest; game/wildlife.

(ii) Non-Renewable natural resources Rocks; coal; minerals/correctly named example; crude oil/petroleum.

(c) Ways of conserving forest

- Prevent forest/bush fires
- Practice selective cutting of trees/avoid indiscriminate felling of trees
- Regulations/laws to control excessive clearing of forest
- Employ forest guards to prevent/arrest intruders
- Use pesticides to kill pests/control plant diseases
- Embark on public education on dangers of deforestation.

(d) Effects of adding animal manure to garden soil

- Animal manure provides nutrients/nitrogenous materials /improves soil fertility
- Animal manure are acted upon by nitrifying bacteria
- To form nitrates for plants
- Manure improves the water-holding properties of the soil by binding the soil particles together/ improves soil structure
- Prevents soil from being blown away by wind/erosion

(e) Possible genotypes in humans

AA, AS, SS

4. a. Explain briefly the following terms:

- gene;
- Hybrid;
- Trait.

(b) Two heterozygous yellow flowers are crossed. Using a genetic diagram, state the phenotypic and genotypic ratios of the first filial generation.

(c) State four transmitted characters in plants.

Solution & Explanation 4:

(a) Explanation of terms

(i) **Gene;** Is a unit of heredity/genetic material; composed of DNA; located on the chromosome.

(ii) **Hybrid** Is an offspring; formed from a cross between parents that are genetically different but of same species.

(iii) **Trait** Is a feature/character; that is possessed by an organism; and can be transferred from parents to offspring.

(b)

(b) Genetic Cross

Parental genotype

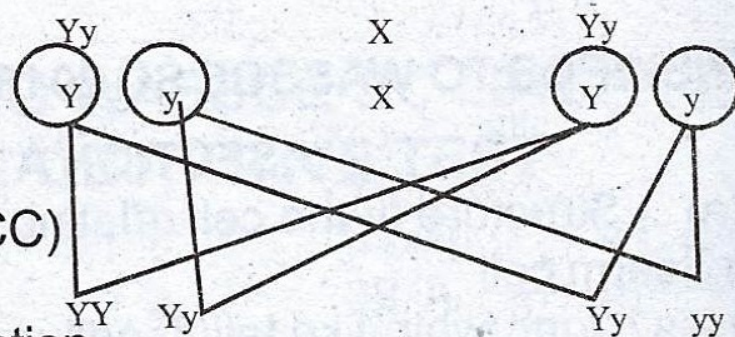
Gametes

Correct crossing (CC)

Offspring/ F_1 generation

Phenotypic ratio

Genotypic ratio



3:1

1:2:1

(c) Transmittable characters in plants

- Type of seed
- Size of fruit
- Colour of flower
- Colour of leaf
- Height of plant

- Texture of seed coat
- Size of plant
- Leaf shape
- Any correctly named transmittable character

5 (a)(1) Explain briefly chemosynthesis as a mode of nutrition.

(ii) Give two examples of organism that carry out chemosynthesis.

(b) List three gases in the atmosphere with their percentage composition.

Gases in the atmosphere	Percentage composition

(c) (i) State four characteristics of a salt marsh habitat.

(ii) Explain briefly how plants are modified for anchorage in a salt marsh habitat.

(d) (i) List four bacterial disease associated with poor food hygiene.

(ii) State three effects of poor food hygiene.

(e) A person was involved in a car accident and became unconscious due to excessive blood loss. Explain briefly how the blood lost could be restored.

(f) State three differences between tillage and bush burning as farming practices.

Tillage	Bush burning

Solution & Explanation 5:

(a) (i) Chemosynthesis;

- The process by which some organisms manufacture their food
- By converting carbon molecules into organic substances
- Using the oxidation of inorganic substances/ hydrogen sulphide/ammonia/methane
- As a source of energy without the use of sun light/solar energy.

(ii) Examples of organisms that carry out chemosynthesis;

- Sulphur bacteria
- Nitrosomonas
- Nitrobacter
- Blue-green algae
- Nitrococcus

(b) (i) Percentage composition of gases in the atmosphere

S/N	Gases in the atmosphere	% composition
1	Oxygen	20.5 - 21.5
2	Nitrogen	78- 79
3	Noble gases / Argon	0.9 - 1.0

4	Carbon dioxide	.03 – 0.05
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(c) (i) Characteristics of salt marsh habitat

- Presence of low oxygen level;
- High salinity
- Soft muddy environment
- Presence of changing water levels
- Flat lowland
- Swampy/water-logged
- Low light'penetration

(i) Modification of plants to salt marsh habitat

Plants grow long/have numerous roots; to hold to the substratum; possess stilt roots for anchorage

(d) (i) Bacterial diseases associated with poor food hygiene

- (Staphylococcal) Food poisoning
- Dysentery; Cholera
- Hepatitis
- Typhoid
- Gastritis
- Botulism
- Tuberculosis

(ii) Effects of bad food hygiene

- May lead to intestinal ulcer
- May lead to serious food poisoning
- Sickness/discomfort
- Frequent stooling/running stomach/diarrhoea
- Loss in economic value/loss in food value
- Death.

(e) Restoring blood loss of an accident victim after a car accident

- The victim must first be rushed to the hospital
- Where the blood sample will be taken

- For agglutination test
- To know the blood group
- Then the compatible blood is transferred to the victim.

(f) Differences between tillage and bush burning.

Tillage	Bush burning
Loosens the soil	compacts the soil
Aerates the soil	leads to poor aeration of the soil
Mixes organic matter with soil/improves soil nutrients	Depletes / destroys soil nutrients / leaching
Enhances the activities of micro organism	Micro fauna / soil organisms are killed
Does not lead to soil / air pollution	leads to air / soil pollution

BIOLOGY PAST QUESTIONS 2017 WITH SOLUTIONS

1. Which of the following groups of animals do not possess a nervous system?

- **A.** Porifera
- **B.** Cnidaria
- **C.** Platyhelminthes
- **D.** Nematoda

Short Explanation: Sponges, the members of the phylum **Porifera**, are a basal Metazoan clade as a sister of the Diploblasts. They are multicellular organisms that have bodies full of pores and channels allowing water to circulate through them, consisting of jelly-like mesohyl sandwiched between two thin layers of cells

Correct Option: Answer is A

2. The structural similarities in Paramecium and Euglena are in the

- **A.** shape of locomotory organs
- **B.** blunt anterior and pointed posterior
- **C.** presence of micro and mega nuclei in both
- **D.** presence of anterior and posterior contractile vacuoles

Short Explanation:

Correct Option: Answer is B

3. Which of the following organisms cannot exist freely on its own?

- **A.** Chlamydomonas
- **B.** Amoeba
- **C.** Paramecium
- **D.** Plasmodium

Short Explanation: Plasmodium is a genus of unicellular eukaryotes that are obligate parasites of vertebrates and insects. The life cycles of **Plasmodium** species involve development in a blood-feeding insect host which then injects parasites into a vertebrate host during a blood meal.

Correct Option: Answer is D

4. Which of the following cell organelles is the site for the production of ATP?

- **A.** Lysosome
- **B.** Nucleus
- **C.** Mitochondrion
- **D.** Ribosome

Short Explanation: The production of ATP happens in the mitochondrion, which likewise exists in both animals and plants.

Correct Option: Answer is C

5. An example of osmosis in plants is the

- **A.** movement of water through the xylem
- **B.** loss of water vapour from the stomata
- **C.** translocation of food through the phloem
- **D.** absorption of water from the soil by the root

Short Explanation: **Plant** roots function as a semipermeable membrane, allowing water to infiltrate the roots. The inside of the root contains minerals, sugars and salts dissolved in water; when water is available in the surrounding soil, **osmosis** causes it to flow through the root wall to its interior.

Correct Option: Answer is D

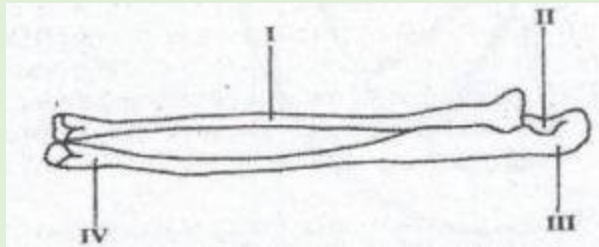
6. The organism that can carry out both autotrophic and heterotrophic modes of nutrition is

- **A.** Chlamydomonas
- **B.** Eudorina
- **C.** Euglena
- **D.** Spirogyra

Short Explanation: **Euglena** use their chloroplasts to produce nutrients through photosynthesis. It is a dual organism as it can act both as an autotroph, making its own food, and heterotroph, feeding on other substance

Correct Option: Answer is C

7.



The diagram above illustrates a part of the human skeleton. It represents the bones of the

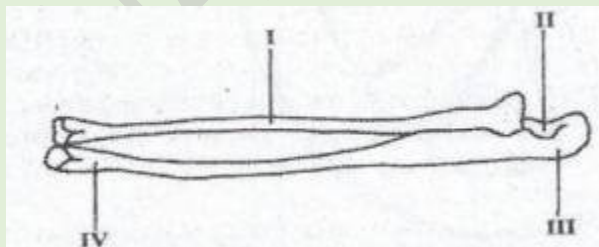
- **A.** upper arm
- **B.** lower arm
- **C.** upper leg
- **D.** lower leg

Short Explanation:

Correct Option: Answer is B

The diagram shows the lower arm which can be seen between the elbow and the wrist.

8.

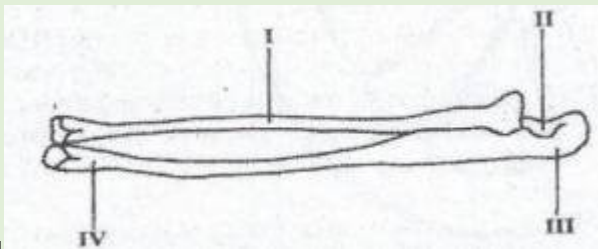


Which of the labelled parts of the diagram of the part of the human skeleton articulates with the head of the trochlea to form a hinge joint?

- **A. I**
- **B. II**
- **C. III**
- **D. IV**

Short Explanation: The part II is attached to the humerus. It forms a hinge joint to make up the hand.

Correct Option: Answer is B



9.

The diagram above illustrates a part of the human skeleton. The labelled part that provides surface for the attachment of the triceps is

- **A. I**
- **B. II**
- **C. III**
- **D. IV**

Short Explanation: The triceps is a large muscle on the back of the upper limb of many vertebrates. The part III is a part of the bone that forms the triceps.

Correct Option: Answer is C



10

The diagram above illustrates the transverse section of a

- **A.** dicotyledonous root
- **B.** monocotyledonous root
- **C.** dicotyledonous stem
- **D.** monocotyledonous stem

Short Explanation:

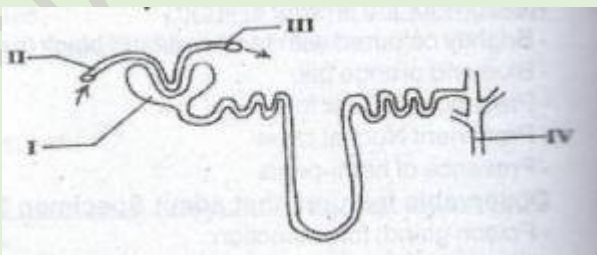
Correct Option: Answer is C

11. Which of the following actions does not occur during exhalation in man?

- **A.** Thoracic cavity decreases in volume
- **B.** Diaphragm flattens
- **C.** Ribs move down and in
- **D.** Air pressure increases in the thoracic cavity

Short Explanation: When you breathe in, or inhale, your diaphragm tightens and moves downward (flattens).

Correct Option: Answer is B



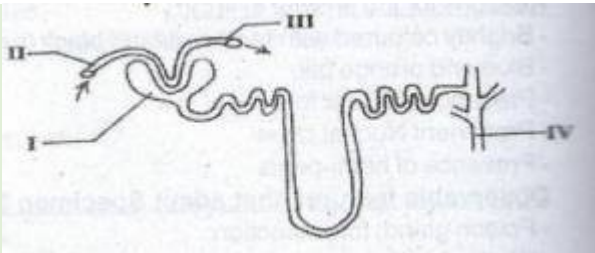
12.

The diagram above is an illustration of the urinary tubule in a mammal. Which of the following substances is found in the part labelled II only?

- **A.** Lipid
- **B.** Glucose
- **C.** Urea
- **D.** Uric acid

Short Explanation:

Correct Option: Answer is C



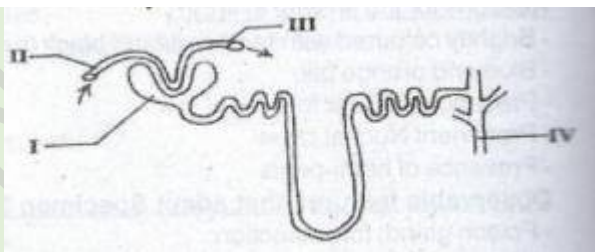
13.

The diagram above is an illustration of the urinary tubule in a mammal. The part which contains lowest concentration of urea is labelled

- **A.** I
- **B.** II
- **C.** III
- **D.** IV

Short Explanation:

Correct Option: Answer is C



14.

The diagram above is an illustration of the urinary tubule in a mammal. Which of the following substances is greater in concentration in the part labelled IV when compared to that part labelled I?

- **A.** Lipid
- **B.** Glucose
- **C.** Urea
- **D.** Uric acid

Short Explanation:

Correct Option: Answer is C

15. Deamination of amino acids in the liver produces

- **A.** blood sugar
- **B.** glycogen
- **C.** bile
- **D.** urea

Short Explanation: deamination is used to break down amino acids for energy. The amino group is removed from the amino acid and converted to ammonia. The rest of the amino acid is made up of mostly carbon and hydrogen, and is recycled or oxidized for energy. Ammonia is toxic to the human system, and enzymes convert it to [urea](#) or uric acid

16. Which of the following glands also serves as an exocrine gland?

Correct Option: Answer is D

- **A.** Ovary
- **B.** Pancreas
- **C.** Adrenal
- **D.** Thyroid

Short Explanation:

This simply means they are the primary reproductive organs. In addition to their role in producing ova, the **ovaries** also have the distinction of being an **endocrine gland** because they secrete hormones—primarily estrogen and progesterone—that are vital to normal reproductive development and fertility

Correct Option: Answer is A

17. Which of the following sequences is the correct route of the transmission of impulses along a reflex arc?

- **A.** Receptor→sensoryneuron→synapse→intermediate neuron→synapse→motorneuron→effector
- **B.** Receptor→motorneuron→synapse→intermediateneuron→synapse→sensoryneuron→effector
- **C.** Effector→intermediateneuron→synapse→motorneuron→synapse→sensoryneuron→receptor
- **D.** Effector→motorneuron→synapse→sensoryneuron→synapse→intermediateneuron→receptor

Short Explanation:

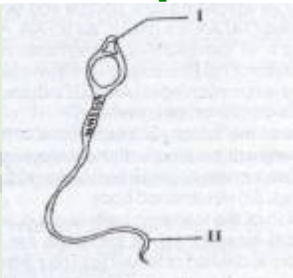
Correct Option: Answer is A

18. When viewing an object that is close to the human eye, the

- **A.** eye lens becomes thin
- **B.** ciliary muscles relax
- **C.** suspensory lens become taut
- **D.** eye lens becomes fat

Short Explanation:

Correct Option: Answer is D



19.

The diagram above is an illustration of the human sperm. The part labelled I is the

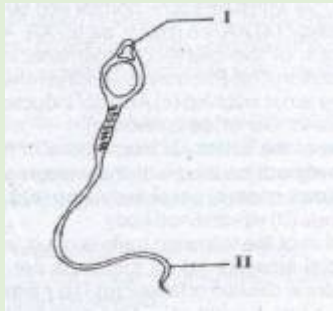
- **A.** acrosome
- **B.** nucleus

- **C.** mitochondrion
- **D.** tail

Short Explanation: The **acrosome** is an organelle that develops over the anterior half of the head in the spermatozoa (sperm cells) of many animals including humans. It is a cap-like structure derived from the Golgi apparatus. In Eutherian mammals the **acrosome** contains digestive enzymes (including hyaluronidase and acrosin).

Correct Option: Answer is A

20.



In the diagram of the human sperm above, the function of the part labelled II is to

- **A.** fuse the ovum during fertilization
- **B.** generate the energy for the sperm to swim to the fallopian tube
- **C.** help the sperm to swim forward
- **D.** dissolve the membrane of the egg during fertilization

Short Explanation: The **tail** of the **sperm**, sometimes called the flagellum, is a slender, hairlike bundle of filaments that connects to the head and middle portion. ... The **tail** gives the **sperm** cell movement. It whips and undulates so that the cell can travel to the egg

Correct Option: Answer is C

21. Which of the following types of placentation is not common amongst syncarpous pistils?

- **A.** Marginal
- **B.** Axile
- **C.** Parietal
- **D.** Free central

Short Explanation: Marginal placentation– The ovary in which the placenta forms a ridge along the ventral suture of the ovary and the ovules develop on two separate rows is known to have **marginal placentation**

Correct Option: Answer is A

22. The complex energy-rich organic matter which living organisms need for life is

- **A.** water
- **B.** air
- **C.** food
- **D.** mineral salts

Short Explanation:

Correct Option: Answer is C

23. Which of the following processes are associated with photosynthesis? I. Energy from sunlight is absorbed II. Carbon dioxide is evolved III. Oxygen is given off IV. Glucose is synthesized

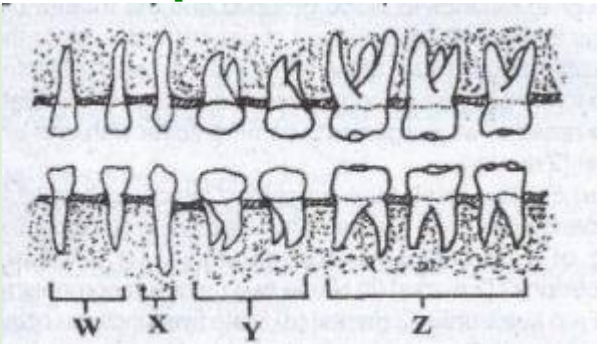
- **A.** I and II only
- **B.** I, II and IV only
- **C.** I, III and IV only
- **D.** I, II, III and IV

Correct Option: Answer is C

24. Which of the following function is associated with calcium in plants? Formation of

- **A.** cell wall
- **B.** ribosomes
- **C.** proteins
- **D.** cell membrane

Correct Option: Answer is A



25

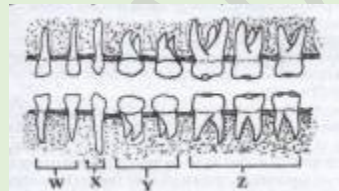
The diagram above shows the dentition of an organism. What organism is this?

- **A.** rabbit
- **B.** cat
- **C.** frog
- **D.** man

Correct Option: Answer is D

26.

The diagrams below show the dentition of an organism. The function of the teeth labelled X is for

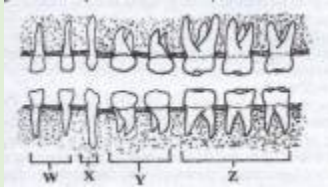


- **A.** biting and grasping
- **B.** chewing and grinding
- **C.** tearing and grasping
- **D.** chewing and tearing

Short Explanation: Canine: They have a sharp, pointed biting surface. Their **function** is to grip and tear food. There are four **canine teeth** in both primary and permanent dentitions. The premolars, unlike the incisors and **canines**, have a flat biting surface.

Correct Option: Answer is C

27.



The teeth which are absent in the young of the organism whose dentition is shown above is labelled

- **A.** W
- **B.** X
- **C.** Y
- **D.** Z

Short Explanation: Premolars and molars

Correct Option: Answer is D

28. The number of individuals that the environment can support over a long period of time is its

- **A.** growth rate
- **B.** total capacity
- **C.** population capacity
- **D.** carrying capacity

Correct Option: Answer is C

29. A good site for secondary succession would be

- **A.** a sand dune
- **B.** a bare rock

- **C.** a land that has just had forest fire
- **D.** a stretch of barren land

Correct Option: Answer is C

30 A population which exceeds its carrying capacity and then collapses is represented graphically by a

- **A.** bell-shaped curve
- **B.** linear curve
- **C.** sigmoid curve
- **D.** skewed curve

Correct Option: Answer is A

31. The natural dwelling place of an organism is called its

- **A.** ecological niche
- **B.** habitat
- **C.** population
- **D.** environment

Short Explanation: a **habitat** is the type of natural environment in which a particular species of organism lives.

Correct Option: Answer is B

32. Which of the following ecological factors will result in food shortage?

- **A.** Drought
- **B.** Low rate of reproduction
- **C.** Emigration
- **D.** Topography

Short Explanation:

Correct Option: Answer is A

33. Soil permeability refers to

- **A.** how easily water passes through the soil

- **B.** the proportion of air in the soil
- **C.** the proportion of water the soil holds
- **D.** how well water rises up the soil

Short Explanation: **Soil permeability** is the property of the *soil* to transmit water and air and is one of the most important qualities to consider for fish culture

Correct Option: Answer is A

34. Which of the following groups of organisms do not have members that can be called autotrophs?

- **A.** Protists
- **B.** Mosses
- **C.** Chlorophyta
- **D.** Fungi

Correct Option: Answer is D

35 Which of the following pyramids gives the most accurate picture of the relationships between at the various trophic levels in a food chain? A pyramid of

- **A.** numbers
- **B.** energy
- **C.** biomass
- **D.** size

Correct Option: Answer is B

36. Which of the following statements is not associated with pyramids of energy?

- **A.** Efficiency of energy transfer between trophic levels can be studied
- **B.** Comparisons of pyramids for different ecosystems can be made
- **C.** More accurate comparisons of trophic levels
- **D.** Representation of numbers of organisms at different trophic levels

Correct Option: Answer is B

37. Which of the following materials is not easily biodegradable?

- **A.** Polythene bag
- **B.** Cotton bag
- **C.** Hides and Skin bag
- **D.** Paper bag

Correct Option: Answer is B

38. Which of the following pairs of organisms is in a mutualistic relationship?

- **A.** Hydra and Sea anemone
- **B.** Hermit crab and Sea anemone
- **C.** Dodder plant and Cocoa plant
- **D.** Mistletoe and Kolanut

Short Explanation: **Hermit crabs and sea anemones** share an unusual and intimate underwater relationship. Young **hermit crabs** will often pick up a young **sea anemone** to attach to their shell and they become partners for life. ... When **hermit crabs** outgrow their shell, they often take the **sea anemone** with them to the new one.

Correct Option: Answer is B

39. The establishment of game reserves and parks is a method of

- **A.** conserving natural resources
- **B.** cultivating game
- **C.** controlling pollution
- **D.** cultural awareness

Correct Option: Answer is A

40 Which of the following practices is not used in conserving water?

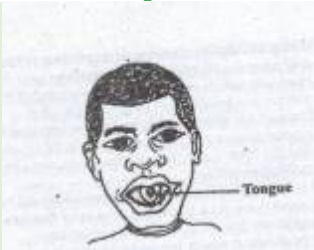
- **A.** Afforestation
- **B.** Damming
- **C.** Deforestation
- **D.** Reforestation

Correct Option: Answer is C

41. Which of the following resources cannot be renewed?

- **A.** Air
- **B.** Petroleum
- **C.** Food crop
- **D.** Soil

Correct Option: Answer is B



42.

The type of variation illustrated above is

- **A.** continuous variation
- **B.** discontinuous variation
- **C.** morphological variation
- **D.** environmental variation

Short Explanation: Tongue rolling is an example of discontinuous **variation**: you either can **roll** your **tongue** or you can't. Other characteristics, for example height and weight, show continuous **variation**.

Correct Option: Answer is B

43. Which of the following characters is not a morphological variation?

- **A.** shape of the nose
- **B.** colour of skin
- **C.** colour of hair
- **D.** tongue rolling

Correct Option: Answer is D

44. An apple plant can produce sexually and asexually and it has 40 chromosomes in its leaf cells. How many chromosomes would be in each gamete and somatic cell?

- **A.** 20 chromosomes in the gametes and 10 in the somatic cells
- **B.** 40 chromosomes in the gametes and 20 in the somatic cells
- **C.** 20 chromosomes in the gametes and 40 in the somatic cells
- **D.** 40 chromosomes in the gametes and 40 in the somatic cells

Correct Option: Answer is C

45. Assuming that A is the gene for normal skin colour and is dominant, while a is the gene for albinism and its recessive, what is the likely genotype of the couple which had 50% normal and 50% albino offspring?

- **A.** AA, aa
- **B.** Aa, aa
- **C.** AA, Aa
- **D.** Aa, Aa

Correct Option: Answer is C

46. Which of the following statements is true about mutation?

- **A.** Phenotype is not involved
- **B.** Genotype is not involved

- **C.** Artificial induction is not possible
- **D.** Genotype is involved

Correct Option: Answer is D

47. One of the structural adaptations of mammals for movement on land is the development of the

- **A.** hollow bones
- **B.** pentadactyl limb
- **C.** fused thoracic bones
- **D.** streamlined body

Short Explanation: pentadactyl limb A **limb** with five digits, characteristic of tetrapod vertebrates (amphibians, reptiles, birds, and mammals). It evolved from the paired fins of primitive fish as an adaptation to locomotion on land and is not found in modern fish

Correct Option: Answer is B

48. Which of the following is NOT characteristic of social insects?

- **A.** All members are identical
- **B.** There is division of labour
- **C.** The members dwell in the same habitat
- **D.** Some members protect the queen

Short Explanation: Social insect, any of numerous species of **insects** that live in colonies and manifest three characteristics: group integration, division of labour, and overlap of generations.

Correct Option: Answer is A

49. The gradual cumulative, adaptive and heritable changes in an organism over a long period of time resulting in new species is called

- **A.** adaptation

- **B.** survival
- **C.** organic evolution
- **D.** interspecific competition

Short Explanation: organic evolution The process by which changes in the genetic composition of populations of organisms occur in response to environmental changes

Correct Option: Answer is C

50 A population which exceeds its carrying capacity and then collapses are represented graphically by a

- **A.** bell-shaped curve
- **B.** linear curve
- **C.** sigmoid curve
- **D.** skewed curve

Correct Option: Answer is C

BIOLOGY PAST QUESTIONS 2017 THEORY WITH SOLUTION

1. (a)(i) What is classification of living things?
(ii) Name the scientist that developed the Binomial System of Classification
(iii) State three reasons why it is necessary to classify living things
(iv) List seven major groups into which taxonomists classify living things in order of hierarchy
- (b) state two features each which viruses have in common with: (i) living things
(ii) non-living things
- (c) Name two viral disease each of:
(i) plants
(ii) humans

Solution & Explanation 1:

Classification of Living things: Is the sorting/ grouping/arranging of living things (in groups); according to their common/similar characteristics/features/shared qualities.

(ii) The scientist who developed the Binomial system of classification: Carl/Carolus Linnaeus/Linnaeus

(iii) Reasons for classification of living things

- To put every living organism into a systematic order/ specific group
- for easy identification of similar living things
- to indicate/Show the relationships between different categories of living organisms
- to show evolutionary trends in different groups of living organisms
- to trace geographical distribution of living organisms
- for easy reference
- for easy research/study.

(iv) Classification of ranks in order of hierarchy

Kingdom

Phylum/Division

Class Order

Family

Genus

Species.

Features of Viruses as (i) living things (ii) nonliving things

(i) Living things

- has protoplasm/protein coat/membrane
- contain DNA/RNA/Nucleic acid

(ii) Non-living things

- appear as crystals
- lack of organelles

(c) Viral diseases of:

(i) Plants:

- Cassava mosaic disease
- Bean mosaic diseases
- cacao swollen shoot disease
- Groundnut rosette
- maize streak
- cucumber mosaic
- any correctly named disease.

(ii) Humans;

- poliomyelitis/polio
- chicken pox
- mumps
- measles
- influenza/flu
- AIDS
- Ebola disease
- infective hepatitis
- Lassa fever
- Avian influenza (Bird flu)
- Swine flu (Bovine flu)
- Rabies
- Zika

2. (a) State four characteristics of enzymes

(b) (i) List two digestive enzymes produced in the duodenum of humans

(ii) Name the substrate that each enzyme listed in 2(b)(i) acts on.

(iii) State the products of each enzyme activity in 2(b)(ii)

(c) (i) State one way in which chlorophyll is important in plants

(ii) Name two microelements that are required for the formation of chlorophyll in plants.

(iii) State one deficiency symptom of each microelement named in 2 (c) (ii) above

(d) Outline the procedure used for testing for starch in a leaf

Solution & Explanation 2:

(a). Characteristic of Enzymes;

- They are biological catalysts alter/speed up the rate of biological reactions
- remain chemically unchanged at the end of a reaction
- their actions are reversible
- they are specific in action/act on specific substrates
- enzymes are required in small amount
- are protein in nature
- they act best at optimum temperature/high temperature/denatures them/ low temperature inactivates them
- specific/sensitive to pH
- actions retarded by poisons/inhibitors
- some required co-enzymes to function.

(b)i. Digestive enzymes produced in the duodenum of humans

- Amylase
- Trypsin/chymotrypsin
- Lipase

ii. Substrates for each enzyme

- Amylase = starch
- Trypsin/chymotrypsin =protein
- Lipase = fats and oil

iii. Product of enzyme activity

- Amylase = maltose
- Trypsin/chymotrypsin =
peptones/polypeptides/peptide
- Lipase =glycerol and fatty acid

(c)i. Importance of chlorophyll in plants

Absorbs/traps sunlight; for photosynthesis

ii. Macro-elements required for the formation of chlorophyll

- Nitrogen
- Magnesium

- Iron

iii. Deficiency symptoms of each macro-element in (c)ii

Nitrogen

- Chlorosis/yellowing of leaves
- Stunted growth
- Poor flower/fruit development

Magnesium

- Poor growth
- Chlorosis/yellowing of leaves

Iron

- Poor/stunted growth
- Chlorosis/yellowing of leaves

(d) Procedure for test for starch

- Detach/pluck off/take leaf to be tested from plant after 4 to 6 hours of exposure to sun to enable it to form starch
- Put leaf in boiling water for about 30 seconds to kill the cells
- Put boiled leaf in 70% warm alcohol/ethylated spirit in water bath to decolorise the leaf Rinse decolorised/brittle leaf in running cold water to soften the leaf tissues/remove alcohol
- Spread out on white tiles/watch glass and add a few drops of iodine solution
- Allow to stand for a few minutes
- It turns blue-black showing/indicating presence of starch.

3. a) Name one instrument used for collecting:

(i) soil organism from a soil sample

(ii) tiny insects from a leaf or a stem.

(b) make a diagram 6 cm – 8 cm long of the instrument named in (a)(ii) above and label fully.

(c) Explain briefly the relationship between three examples of symbiotic organisms

Solution & Explanation 3:

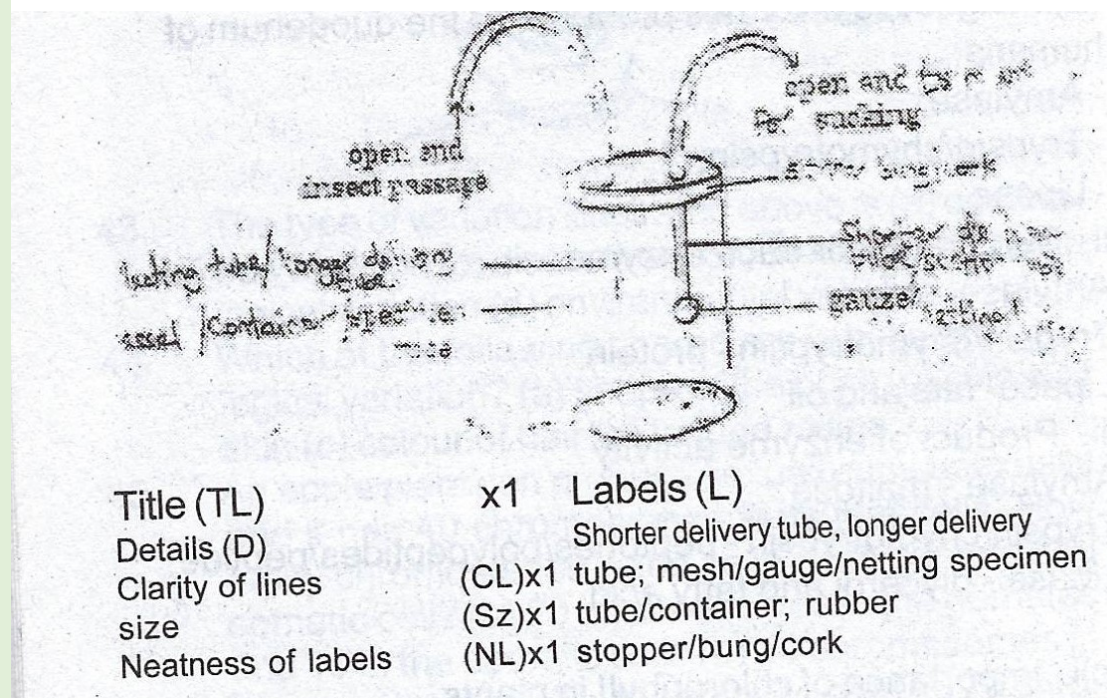
(a)i. Instrument for collecting soil organisms from a soil sample

- Sieve
- Tullgren funnel

ii. Instrument for collecting tiny insects from a leaf or stem

- Pooter

(b) Diagram/drawing of pooter



Quality (0)

Stopper/rubber bung with two openings (RO) x1

Two delivery tubes one longer than the other x1

Inner end of shorter delivery tube with mesh/gauze(IM)
x1

(c) Symbiotic relationships in organisms

Trichonympha Protozoa living in the gut of a termite;

Helps the termite to digest cellulose; while the protozoa trichonympha gets protection and food in return from the termite.

Sea anemone attaching itself to the shell of hermit crab; thereby enjoying pieces of leftover food therein. The hermit crab in turn enjoys protection against predators/ the sea anemones stings prevent the predators from harming the hermit crab.

Algae and fungi in a lichen; Lichen is made up of a fungus and a unicellular alga living together. The green alga produces food/photosynthesis for the association; the fungus protects the alga/absorbs water from the surrounding.

Rhizobium sp Nitrogen- fixing bacteria and root modules of leguminous plants. The Rhizobium fixes atmospheric nitrogen directly into the plant while rhizobium enjoys shelter/food/nutrients from the plant.

4. (a) A child belongs to blood group O and the mother belongs to group B. With the aid of genetic cross, state the possible blood groups of the father.

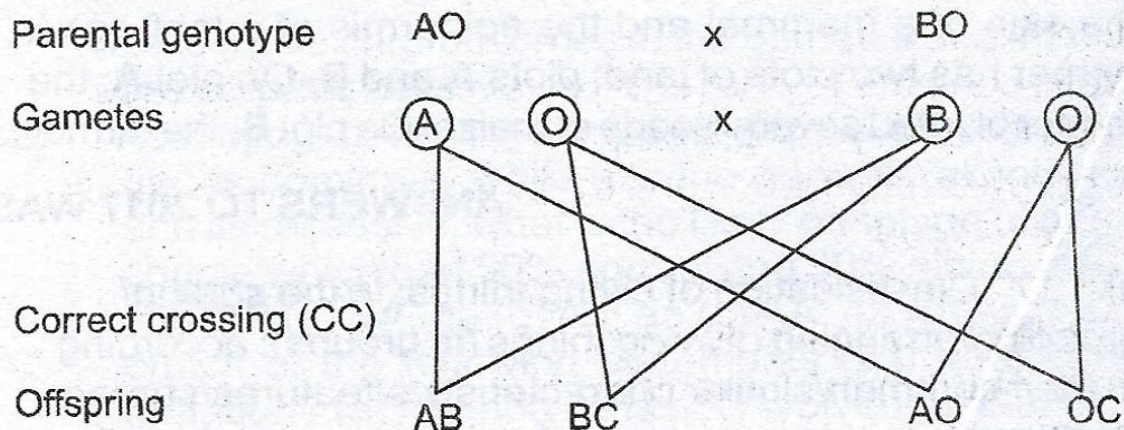
(b) (i) Name two forms of adaptive communication in animals

(ii) State two reasons why organisms communicate with one another.

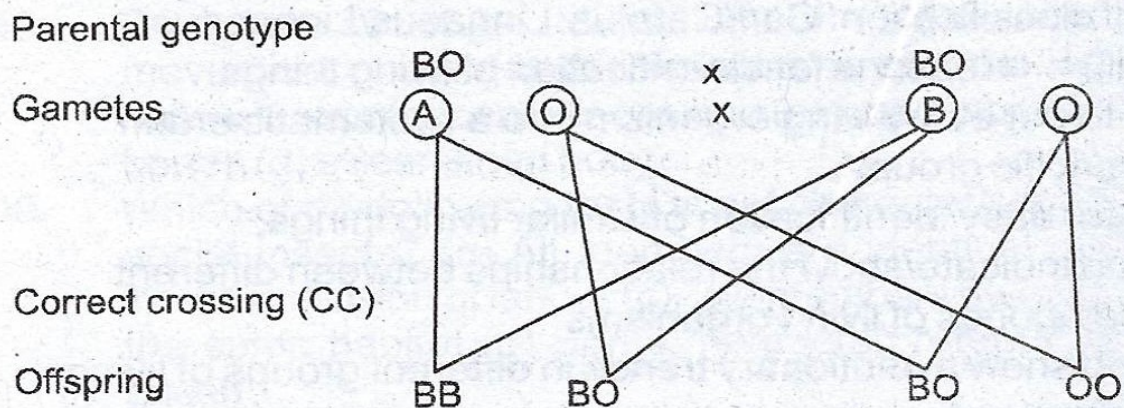
Solution & Explanation 4:

A child belongs to blood group O and the mother belongs to group B. With the aid of a genetic cross. State the possible blood groups of the father

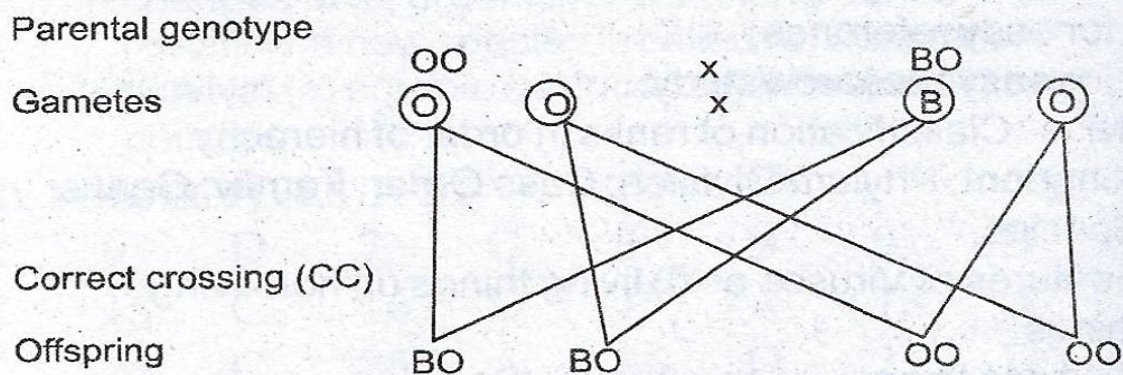
(a) Genetic cross
If the father is group A



If the father is group B



If the father is group O



(b)i. Forms of adaptive communication

- Sound/auditory/acoustic
- visual; smell/chemical/ pheromones
- tactile

- electrical signal
- display.

ii. Reasons why organisms communicate with one another

- for food
- mating /reproduction
- territorial defense
- danger/threats/predator
- tracking routes.

5 (a) Explain briefly the process of perceiving smell in humans.

(b) State the adaptive features of the egg of a domestic fowl

(c) (i) What are autotrophs?

(ii) Name two unicellular organisms that are autotrophs

(d) State five functions of the World Health Organisation

(e) Name two example each of plants and animals that live in a freshwater pond.

(f) In the table below, state five differences between the skin of a mammal and the epidermis of a leaf.

(g) A farmer has two plots of land; plots **A** and **B**. on Plot **A** **the** farmer planted several seeds of maize. On plot B, the farmer planted several seeds of maize and pepper in a small area. What type of competition would be observed on

(i) Plot A

(ii) Plot B

normal and 50% albino offspring?

Solution & Explanation 5:

(a) The process of perceiving smell

- Chemical substances/odour carried in the air pass through the air/nasal cavity
- The molecules dissolve in the moist membrane of the nasal cavity
- Sensory cells are stimulated
- Impulses are formed
- which pass through the nerves ending to the olfactory nerves
- which pass into the cerebrum of the brain/brain
- The brain interprets the impulses
- And the sensation of smell is produced/smell is perceived

(b) Adaptive features of the egg of a domestic fowl;

Porous: shell; for gaseous exchange/provides shelter/protects the embryo against mechanical damage/desiccation

Shell membranes; allow gaseous exchange/for protection

Yolk; contains high level of proteins and fats/provides food for embryonic-growth

Egg white/albumen; consists of water to absorb shock/proteins/food

Air space; provides surface for gaseous exchange
Chalaza; holds the yolk/embryo in position.

(c)i. Autotrophs: Organisms that do not rely on other organisms for food/they manufacture their own food; from using simple inorganic compounds; can be chemosynthetic /photosynthetic

ii. Examples of unicellular autotrophs Chlamydomonas; Euglena; chemosynthetic; bacteria/ sulphur, bacteria; blue-green; algae/Anabaena; Nostoc / Oscillatoria; Chlorella diatoms; Nitrosomonas; Nitrobacteria

(d) Functions of WHO

1. provides grants for medical research
2. Improves public health/promotes environmental health
3. prevents the spread of diseases vaccination
4. provides vaccines/drugs in cases of emergencies
5. Informs/creates awareness/educates the public of any research findings/outbreak of diseases
6. Assists by sending medical personnel/logistical support to countries upon request
7. Helps in the training of medical personnel
8. keeps health statistics
9. promotes nutritional/sanitary products.

(e) Examples of plants and animals that live in fresh water pond

Plants: Water arum; diatoms; Ceratophyllum /hornwort; utricularia/bladderwort. Arrowhead/sedge; Pistia; Salvinia/water fern; water lily; wolffish/duckweed any other correctly.

Animals: Cyclops; Daphnia/water flea; Hippopotamus; crocodile; water snake; tadpoles; pond skaters; zooplanktons; mosquito larvae; water bugs; water scorpions; leeches; water snails; catfish/Tilapia fishes etc.

(f) Differences between skin of a mammal and epidermis of leaf

skin of a mammal	Epidermis of leaf
skin is made up of many layers/cells	made up of a single layer/cells
presence of sweat gland	absence of sweat gland
presence of sebaceous gland	absence of sebaceous gland
Absence of cuticle	presence of cuticle

Absence of guard cell stomata	Guard cells are present / stomata
Has blood cells capillaries	absence of blood vessels / capillaries
synthesizes vitamin D	does not make vitamin D
Fatty cells are present	absence of fatty cells
It is a sensory organ	its not a sensory organ
Hair follicle present	hair follicle absent
erector muscle present	erector muscle absent
presence of sweat pore	absence of sweat pore

(g) Type of competition on

- (i) Plot A: intra specific/intra species competition
- (ii) Plot B: interspecific/interspecies competition

1. In evolution, analogous structures are significant because they show
 - A. physiological diversity
 - B. functional diversity
 - C. genetic diversity
 - D. structural diversity

Short Explanation: In evolutionary biology, the term analogous structures pertain to the various structures in different species having the same function but have evolved separately, thus do not share common ancestor. In comparison, the homologous structures pertain to the structures that show similar morphology and anatomy but have different functions. Moreover, the homologous structures are believed to have evolved from a common ancestor since they show similar development pattern during embryonic development.

Correct Option: Answer is A

2. Which of the following statements reflects the mechanism of natural selection?
 - A. There are variations within every species
 - B. Individuals of every species are genetically identical
 - C. No organism in a species dies before sexual maturity
 - D. Organisms with selective advantages are less likely to survive

Short Explanation: Natural selection is the process whereby organisms better adapted to their environment tend to survive and produce more offspring. The theory of its action was first fully expounded by Charles Darwin,

and it is now regarded as be the main process that brings about evolution.

Correct Option: Answer is A

3. Which of the following animals exhibits territoriality?

- A. Rabbit
- B. Earthworm
- C. Lizard
- D. Toad

Short Explanation: Territoriality is a type of intraspecific or interspecific competition that results from the behavioral exclusion of others from a specific space that is defended as territory. This well-defined behavior is exhibited through songs and calls, intimidation behavior, attack and chase, and marking with scents.

Correct Option: Answer is A

4. The branch of Biology that deals with the principles of classification of organisms is known as

- A. biological index
- B. nomenclature
- C. taxonomy
- D. ecology

Short Explanation: Taxonomy is the science of defining groups of biological organisms on the basis of shared characteristics and giving names to those groups.

Correct Option: Answer is C

5. Which of the following structures is a tissue

- A. Vessel element
- B. Blood
- C. Sieve tube element
- D. Erythrocyte

Short Explanation: In biology, tissue is a cellular organizational level intermediate between cells and a complete organ. A tissue is an ensemble of similar cells from the same origin that together carry out a specific function. Organs are then formed by the functional grouping together of multiple tissues.

Correct Option: Answer is B

6. Which of the following cells are not regarded as specialized?

- A. Sperm cells
- B. Root up cells
- C. Muscle cells
- D. Somatic cells

Short Explanation: Some cells in multicellular organisms are modified to carry out a particular function, such as transporting a certain substance or executing a specific task. These cells are called specialized cells. A somatic or vegetal cell is any biological cell forming the body of an organism; that is, in a multicellular organism,

any cell other than a gamete, germ cell, gametocyte or undifferentiated stem cell.

Correct Option: Answer is D

7. Which of the following pairs of cells carry out the same function?

- A. Cheek cell and red blood cell
- B. Spermatozoon and ovum
- C. Palisade cell and epidermal cell
- D. Root tip cell and guard cell

Short Explanation: A spermatozoon is a motile sperm cell, or moving form of the haploid cell that is the male gamete. A spermatozoon joins an ovum to form a zygote. A zygote is a single cell, with a complete set of chromosomes, that normally develops into an embryo. The egg cell, or ovum, is the female reproductive cell gamete in oogamous organisms. The egg cell is typically not capable of active movement, and it is much larger (visible to the naked eye) than the motile sperm cells. When egg and sperm fuse, a diploid cell (the zygote) is formed, which rapidly grows into a new organism.

Correct Option: Answer is B

8. If Amoeba is placed in a salt solution, the contractile vacuoles would

- A. be bursting more frequently
- B. be more numerous

- C. be formed less frequently
- D. grow bigger before they burst

Correct Option: Answer is C

9. In which of the following habitats is paramecium not found?

- A. Pond
- B. Aquarium
- C. Lake
- D. Puddle

Correct Option: Answer is B

10. The following processes are involved in water movement in the endodermis except

- A. osmosis
- B. vacuolar pathway
- C. diffusion
- D. active transport

Correct Option: Answer is C

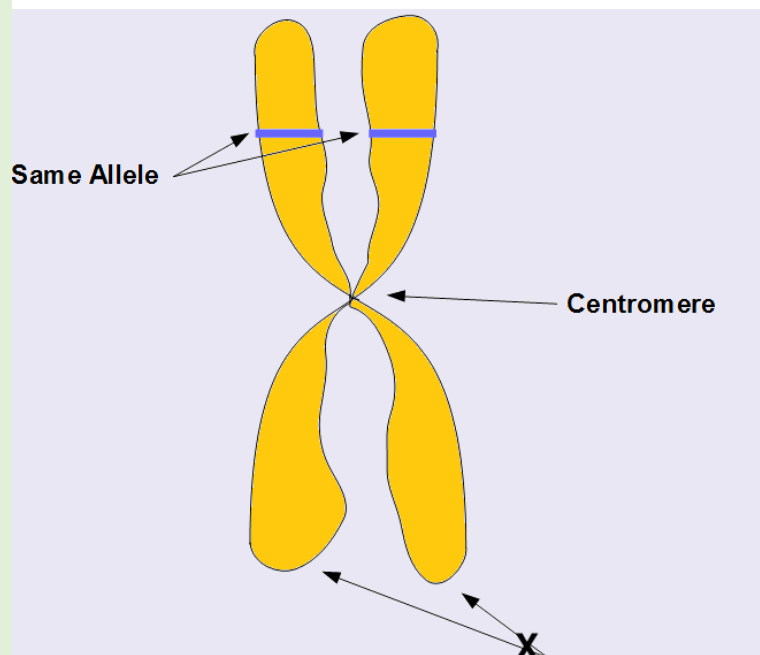
11. Cells that utilize a lot of energy are characterized by the presence of a large number of

- A. vacuoles
- B. mitochondria

- C. endoplasmic reticulum
- D. ribosomes

Short Explanation: Mitochondria are the organelles of the cell that are in charge of producing energy.

Correct Option: Answer is B

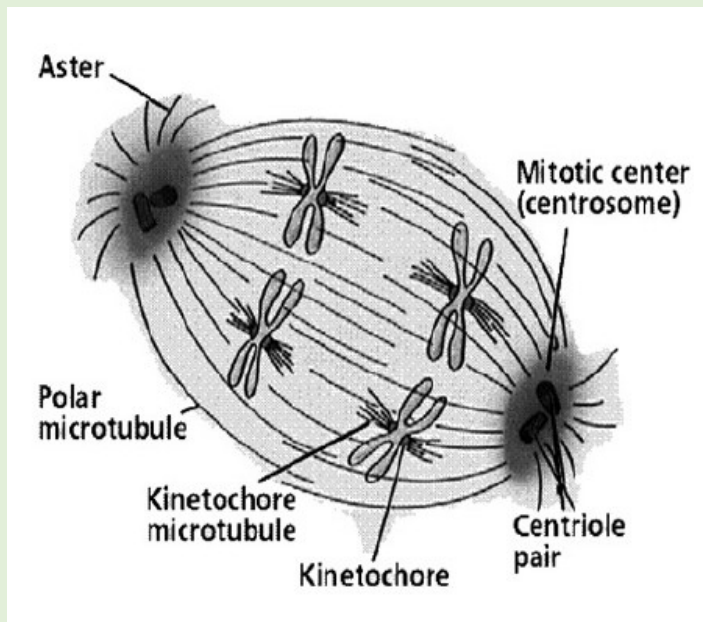


12. The parts labelled X in the diagram are

- A. centriole
- B. chloroplast
- C. chromatid
- D. tonoplast

Short Explanation: Chromatid is each of the two thread-like strands into which a chromosome divides longitudinally during cell division. Each contains a double helix of DNA.

Correct Option: Answer is C



13. The diagram above represents

- A. nucleolus
- B. centromere
- C. centriole
- D. spindle

Short Explanation: Spindle fibers form a protein structure that divides the genetic material in a cell. The spindle is necessary to equally divide the chromosomes in a parental cell into two daughter cells during both

types of nuclear division: mitosis and meiosis. During mitosis, the spindle fibers are called the mitotic spindle.

Correct Option: Answer is D

14. Secondary growth is brought about by the activities of the

- A. phellogen and phelloderm
- B. phellogen and procambium.
- C. vascular cambium and phelloderm
- D. vascular cambium and phellogen

Correct Option: Answer is D

15. A monocot root is different from a dicot root by having

- A. endodermis
- B. cambium
- C. wide pith
- D. root hair.

Correct Option: Answer is C

16. Which of the following statements best describes haemoglobin? It is

- A. yellowish in colour
- B. a red blood cell
- C. an oxygen carrying pigment

D. needed for blood clotting

Correct Option: Answer is C

17. Inhaled air is made warm and moist in the

A. epiglottis

B. nasal cavity

C. trachea

D. mouth

Correct Option: Answer is B

18. Which of the following structures is not involved in respiration?

A. Lung books

B. Mouth

C. Stomach

D. Trachea

Correct Option: Answer is C

19. Filtrate in the Bowman's capsule contains vitamins because

A. only all the amount is required by the body

B. they can be reabsorbed into the blood

C. they have low molecular weight

D. most of them are fat soluble.

Correct Option: Answer is C

20. Pore sweat is produced during muscular exercise because

- A. the contracting muscles produce water
- B. fermentation occurs in muscles
- C. the temperature of the body rises
- D. the muscle fatigues

Correct Option: Answer is C

21. Which of the following neurons may not have myelin sheath?

- A. Effector cells
- B. Intermediate neurons
- C. Motor neurons
- D. Sensory neurons

Correct Option: Answer is A

22. Which of the following is an effector organ?

- A. Tongue
- B. Skin
- C. Nose bridge
- D. Ear glands

Short Explanation: Effector organs are smooth muscle, cardiac muscle and glands that respond to nerve impulses from the central nervous system without conscious thought. They are part of

the automatic, or involuntary, nervous system, along with receptors, afferent nerves and efferent nerves.

Correct Option: Answer is C

23. During sexual reproduction in Paramecium, how many times does the zygote divide to produce eight nuclei?

- A. 1
- B. 2
- C. 3
- D. 4

Correct Option: Answer is C

24. The examples of autotrophic organisms include the following except

- A. Euglena
- B. Chlamydomonas
- C. Hydra
- D. Spirogyra

Correct Option: Answer is C

25. Which of the following organisms feeds both autotrophically and heterotrophically?

- A. Hydra
- B. Paramecium

- C. Mucor
- D. Euglena

Short Explanation: **Euglena**, genus of more than 1000 species of single-celled flagellated microorganisms that feature both plant and animal characteristics.

Correct Option: Answer is D

26. Siphons and gills are structures in aquatic organisms used for

- A. gaseous exchange
- B. buoyancy
- C. excretion
- D. movement

Correct Option: Answer is A

27. The first step in carrying out ecological studies in a terrestrial habitat is

- A. quadrat sampling
- B. mapping the area
- C. collection of organisms
- D. measuring abiotic factors

Correct Option: Answer is B

28. Which of the following components of an ecosystem does not contribute to the biomass of the system?

- A. Producers
- B. Consumers
- C. Micro-organisms
- D. Habitat

Correct Option: Answer is D

29. The group of several populations that interact and share a habitat, form

- A. a species
- B. a community
- C. an ecotype
- D. a tribe

Correct Option: Answer is B

30. The physical and chemical factors which affect the life of organisms in an environment are described as

- A. biotic
- B. edaphic
- C. abiotic
- D. physiographic

Correct Option: Answer is C

31. Which of these factors are common to both aquatic and terrestrial habitats?

- A. Light, rainfall and turbidity
- B. Light, temperature and tide
- C. Light, rainfall and temperature
- D. Rainfall, turbidity and salinity

Correct Option: Answer is C

32. Which of these factors are common to both aquatic and terrestrial habitats?

- A. Light, rainfall and turbidity
- B. Light, temperature and tide
- C. Light, rainfall and temperature
- D. Rainfall, turbidity and salinity

Correct Option: Answer is C

33. Organisms that feed essentially on plants within any ecosystem may be referred to as

- A. carnivores
- B. herbivores
- C. saprophytes
- D. omnivores

Correct Option: Answer is B

34. Which of the following statements about feeding relationships is correct?

- A. In a food web on land, grasshopper feeds on praying mantis and on predator bugs
- B. In a food chain on land, green plant is the producer
- C. In an aquatic food chain mollusc larvae are the producers
- D. In an aquatic food web, various copepods feed on mollusk Larvae and on sand eel

Correct Option: Answer is B

35. Which of these diseases cannot be spread by an insect

- A. Cholera
- B. Malaria
- C. Trypanosomiasis
- D. Measles

Correct Option: Answer is D

36. Air pollution can be reduced by the following methods except

- A. building tall factory chimneys
- B. passing waste gases through filters and absorbers
- C. using lead-free petrol in car
- D. recycling tins, cans and bottles

Correct Option: Answer is D

37. An effective management of natural resources to ensure their continued yield from generation to generation may involve all the following except

- A. protection of wildlife
- B. prevention of habitat destruction
- C. biological control of pests
- D. disposal of raw sewage into the sea

Correct Option: Answer is D

38. Which of the following orders correctly represents the trend in eutrophication?

- A. Lack of oxygen → Death of aquatic organisms → Massive algal growth
- B. Excess nitrates and phosphates → Massive algal growth → Lack of oxygen → Death of aquatic organisms
- C. Heavy mineralization → Lack of oxygen → Massive algal growth
- D. Excess nitrates and phosphates → Lack of oxygen → Massive algal growth → Death of aquatic organisms

Correct Option: Answer is B

39. The following practices can contribute towards the conservation of natural resources except

- A. afforestation
- B. deforestation
- C. contour ploughing

D. establishment of Game Reserves

Correct Option: Answer is B

40. Variation is a factor which

- A. differentiates between individuals of the same species
- B. qualifies an organism to live with others
- C. enables organisms to adapt to the environment
- D. states the external features of an organism

Correct Option: Answer is A

41. Which of the following traits in humans is not a morphological variation?

- A. Height
- B. Shape of head
- C. Colour of skin
- D. Ability to roll the tongue

Correct Option: Answer is D

42. The blood group in humans referred to as a universal recipient

- A. O
- B. A
- C. B
- D. AB

Correct Option: Answer is D

43. If a heterozygous red flowered plant, (Rr) was self-fertilized, the offspring would be expected to be

- A. all red flowered plants
- B. all white flowered plants
- C. red flowered, white flowered plants
- D. white flowered red flowered plants

Correct Option: Answer is D

44. The contrasting pair of genes affecting a trait and located at the same position on homologous chromosomes is referred to as

- A. chromatids
- B. mutants
- C. alleles
- D. chromatin

Correct Option: Answer is C

45. The haploid number of chromosomes in humans is

- A. 48
- B. 46
- C. 24

D. 23

Correct Option: Answer is D

46. A pure breeding white cat (homozygous dominant) mates with a pure breeding black cat (recessive), what would be the fur colour of F1 generation

- A. Homozygous white
- B. Homozygous black
- C. Heterozygous white
- D. Heterozygous black

Correct Option: Answer is C

47. Courtship behaviour in animals may include all of the following except.

- A. howling of a dog
- B. dancing of the worker bee
- C. croaking of a toad
- D. basking in the sun by a lizard

Correct Option: Answer is D

48. Honey-Bee workers carry out the following functions except

- A. laying eggs
- B. building nests
- C. cleaning the nest

D. gathering food

Correct Option: Answer is A

49. The long necks and legs of the present-day giraffes were some of the basis of

- A. Darwins' theory of evolution
- B. Mendel's theory of inheritance
- C. Lamarck's theory of evolution
- D. De Vries theory of evolution

50. A change in the environment of an organism is termed

- A. sensitivity
- B. impulse
- C. response
- D. stimulus

Correct Option: Answer is D

BIOLOGY PAST QUESTIONS 2016 THEORY WITH SOLUTION

- 1.** (a) (i) List three forms in which living cells exist
- (ii) Give one example each of the forms listed in 1 (a)
- (b) Make a diagram, 6 cm — 10 cm long, of a typical plant cell and label fully
- (c) (i) In a tabular form, state three differences between a plant cell and animal cell.
- (ii) State three similarities between a plant cell and animal cell.

Solution & Explanation 1:

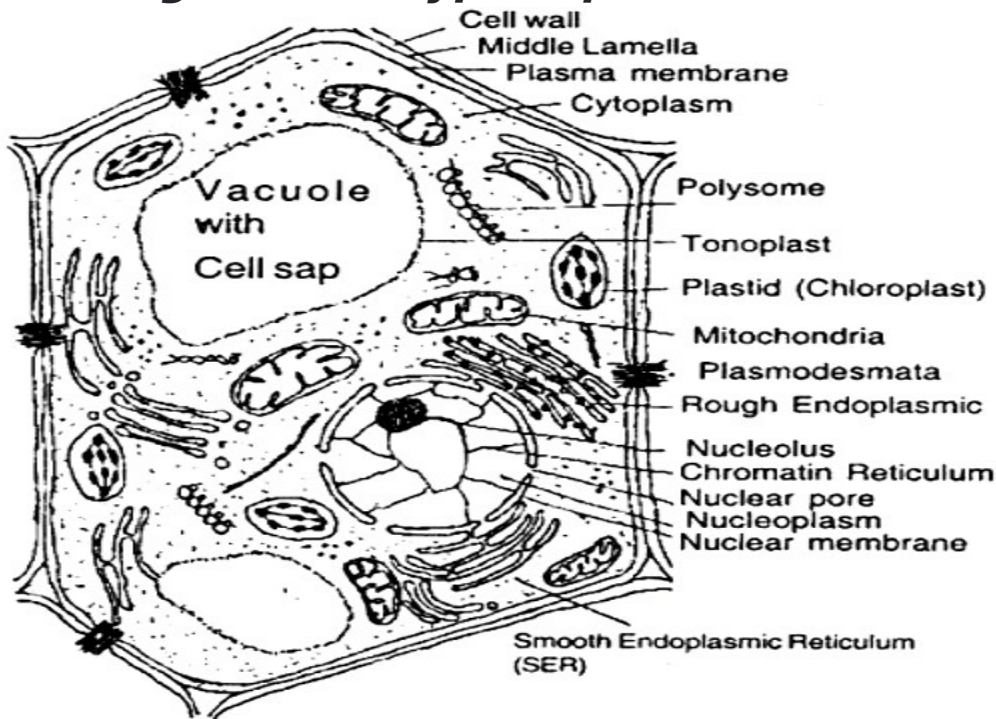
(a) (i) *Forms in which living cells exist*

- Single/free-living
- Colony/colonial
- Filamentous/filament
- Part of a living organism.

(ii) Example

- Single/free-living
-Amoeba/Chlamydomonas/Paramecium/Euglena
- Colony/colonial -Eudorina/Pandorina/Chlorella/Sponges
- Filamentous/filament -
Spirogyra/Chlamydomonas/Ulothrix/Zygnema/Oedogonium
- Part of living organism - cheek cell/onion cell/epidermal cell/parenchyma cell.

(b) Diagram of a typical plant cell



(c) (i) Differences between a plant cell and an animal cell

plant cell	Animal
Has cellulose cell wall / rigid	Lacks cellulose cell wall
Has one large central vacuole	has numerous/vacuole
has chloroplast	lacks chloroplast

Contains starch granules in its cytoplasm	contains glycogen granules in its cytoplasm
Usually larger in size	Usually smaller in size
Does not have lysosome	has lysosome
Does not have centrioles / centrosome	Has centrioles / centrosome
Has a regular / defined cell shape	has an irregular / indefinite cell shape
cytoplasm less dense	cytoplasm more dense
stores lipid as oil	stores lipid as fat

(ii) Similarities in plant cell and animal cell;

They both have cytoplasm; nucleus; vacuole; mitochondrion; Golgi body; (smooth/rough) endoplasmic reticulum; cell membrane/plasma membrane; ribosomes

2. (a) In sequence, name the organs that constitute the alimentary canal of humans

(b) State: (i) three structural differences; (ii) three structural similarities; between the alimentary canal of a bird and human.

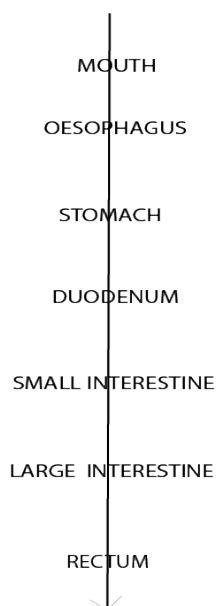
(c) Explain briefly how the dentition of herbivores is adapted for feeding

(d) State two roles of the pancreas in digestion

Solution & Explanation 2:

(a) Organs of alimentary canal of humans in sequence

- Mouth/buccal cavity
- Oesophagus/gullet
- Stomach
- Duodenum
- Ileum/jejunum/small intestine
- Caecum/Colon/Large intestine
- Rectum.



(b)(i) Structural differences between the alimentary canal of a bird and human

Bird	Human
Beak present / mouth modified to beak	Beak absent / present of mouth

Teeth absent	teeth present
Crop present	crop absent
Gizzard present	gizzard absent
Anal pore modified to cloaca	anal pore modified to anus
Shorter alimentary canal	Longer alimentary canal
Appendix absent	appendix present

(ii) Similarities in the alimentary canal of a bird and human;

They both have mouth; oesophagus/gullet; stomach; duodenum; small intestine; large intestine; caecum; rectum.

(c) Adaptation of the dentition of herbivores for feeding

- Elongated/curved incisors; for cutting grass/herbs
- Thick horny pad; against which the lower incisors bit
- Canines are absent/presence of diastema; allows tongue to roll the cropped grass
- Broad surfaces of the cheek teeth/molars; for grinding/chewing/masticating.
- Ridges/furrows on cheek teeth/molars; for grinding/chewing/masticating

(d) Roles of pancreas in digestion

- Produces pancreatic juices which contain enzymes
- Sodium bicarbonate in the pancreatic juice neutralizes the HCL from the stomach/produces alkaline medium
- Pancreatic amylase digests starch to maltose
- Trypsin in the pancreatic juice digests protein to peptides/peptones

- Lipase in the pancreatic juice digests fat droplets to glycerol and fatty acid

3. (a) (i) What is a habitat?

(ii) Explain briefly three roles of a decomposer in an ecosystem.

(b) Explain briefly energy flow in a freshwater habitat.

(c) (i) State three harmful effects of microorganisms to plants

(ii) State six beneficial effects of microorganism to humans.

Solution & Explanation 3:

(a)(i) Habitat

The natural dwelling place of an organism. OR A place/locality where an organism lives successfully

(ii) Role of a decomposer in an ecosystem

Decomposers are organism which bring about the decay of dead parts/whole of plants/animals; to produce manure/humus; they release nutrients in them into the soil; which are absorbed by roots of plants; and are used to build up new plant tissues; they thus help in nutrient

recycling; also release offensive gases/odour which pollute the air.

(b) Energy flow in a freshwater habitat;

Common aquatic plants/Pistia/Spirogyra/any correctly named aquatic plant; using energy from the sun; photosynthesize/produce food; the food is eaten by primary consumers/animals; small aquatic animals would be eaten by bigger aquatic animals/secondary consumers; tertiary consumer/a much bigger fish/any correctly named fish will feed on secondary consumer, in the process of feeding energy flows from one trophic level to another; decreases progressively; dead animals/plants decompose; energy is lost.

(c)(i) Harmful effects of microorganisms to plants

1. They cause diseases
2. They can cause undesirable qualities in plants/plant product
3. They cause spoilage
4. Reduction in yield

(ii) Beneficial effects of micro organisms to humans

- Bacteria/decomposers help to decay/decompose dead organisms to release nutrients
- Bacteria is used in cleaning/mopping up oil spillage
- Used in the fermentation of foods/cocoa beans/garri/locust beans/kenkey production
- Used to make vinegar
- Used in making alcoholic beverages/wine/beer/spirit
- Used in making some antibiotics/penicillin
- Used in the manufacture of insulin

- Used in making cheese/yoghurt/bread/Single Cell Proteins
- Used in making flax/tanning of leather;

4. (a) (i) Explain the term agglutination as used in blood transfusion.

(ii) The table below represents blood transfusion between blood donors and recipients.

		Recipient			
		A	B	AB	O
Donor	A				
	B				
	AB				
	O	+			

(b) Explain how each of the following organisms are adapted for obtaining food.

- (i) Mosquito larva
- (ii) Dodder plant
- (iii) Grasshopper

Solution & Explanation 4:

(a)(i) Agglutination;

The clumping together; of antigens or red blood cells/RBC; with the corresponding antibodies in the plasma; of incompatible blood groups/wrong blood

group.
(ii)

Donor		Recipient			
		A	B	AB	O
	A				
	B				
	AB				
	O				

(b) How each of the following organisms are adapted for obtaining food

(i) **Mosquito larva** - Possess a pair of ciliated mouth brushes; for creating water current to carry water with food particle into the mouth; - Its mouth brushes; filter food out of water and sweep it into the mouth.

(ii) **Dodder plant** - Possess haustoria; for penetrating host plant tissue to obtain food/water

(iii) **Grasshopper**

- Possess mandibles/heavy toothed jaw-like structures; for cutting/crushing food/plant materials/vegetation
- Possess maxillae; for directing food to mandibles/cutting/grinding food
- Labrum; for holding food/ plant materials/vegetation
- Labium; for guiding foods/prevent it from falling off.

- 5.** (a) State two difference between tactic and nastic movements in plants.
- (b) Give two examples each of organisms that show:
 (i) tactic movement
 (ii) nastic movement
- (c) (i) State three ways of caring for the mammalian skin.
 (ii) List three stimuli to which the mammalian skin is sensitive.
- (d) State five effects of high temperature on a terrestrial habitat.
- (e) (i) What is courtship behaviour in animals?
 (ii) List three courtship behaviours in animals.
- (f) Name three organisms that carry out holozoic mode of nutrition.
- (g) Explain briefly how fingerprinting can be used to detect crime.

Solution & Explanation 5:

(a) Differences between tactic and nastic movements in plants

Tactic movement	Nastic movement
movement of whole organism / entire plant	movement shown by part of a plant
movement is directly towards / from stimulus	Non-directional movement
stimuli consist of light / heat / chemical	stimulus is touched /

(b) Examples of organisms that show**(i) Tactic movement**

- Volvox
- Chlamydomonas
- Euglena
- earthworm
- tilapia

(ii) Nastic movement

- Morning glory
- Mimosa
- Pride of Barbados (Caesalpinia)
- Four-o'clock plant
- Morning flower
- sunflower
- Desmodium
- clover

(c)(i) Care of the skin

1. Eating balanced diet rich in vitamins A/B2/E
2. Engaging in regular exercise
3. Washing the skin with clean water and soap daily;
4. Drinking lot of water
5. Moisturizing/rubbing the skin with lotion/cream
6. Avoiding excessive exposure to ultraviolet ray/radiation
7. Avoiding bleaching creams/harsh chemicals
8. Wearing clean clothes next to skin
9. Protect the skin against injury.

(ii) Stimuli to which the mammalian skin is sensitive

- Touch
- Pressure
- Heat
- Cold
- Pain
- Chemicals

(d) Effects of high temperature on terrestrial habitat

1. Causes high transpiration rate in plants
2. Plants wither/wilt
3. Leads to high rate of metabolism in both plants and animals
4. There is a high rate of decay of organic matter
5. Leads to high rate of growth; and shorter length of life cycle in toad/housefly
6. Makes poikilothermic animals to look for shelter from heat/hibernation
7. There is high rate of water loss in animals/sweating/perspiration
8. Leads to high incidence of bush fires
9. Migration/emigration

(e)(i) Courtship behaviour in animals

A type of behaviour/instinctive action in animals; that play an important role in attracting a compatible mate/ opposite sex; of the same species; for copulation/sexual reproduction

(ii) Courtship behaviours in animals

- Pairing

- Territoriality
- Seasonal migration
- Smell/release of attractive sex hormones/odour
- Display of prominent colour patterns
- Mating calls/croaking/vocal displays/sound signal
- Dancing/attractive movements.

(f) Examples of organisms that carry out holozoic mode of nutrition

- Goat
- rat
- lizard
- frog
- bird
- human
- any correctly named example; pitcher plants; sundew; bladderwort.

(g) Use of fingerprinting in crime detection

No two individuals have the same/identical pattern/contour of fingerprints/hence it is used for crime detection; fingerprints left behind by a criminal at the scene of a crime/door/table/weapon/on victim; provide a valuable clue in the detection and subsequent arrest of a criminal; the fingerprints left on the scene of crime are photographed; and the crime suspect's fingerprints are taken and compared with the fingerprint at the scene of crime; if it matches then the suspect is the criminal.

BIOLOGY PAST QUESTIONS 2015 WITH SOLUTIONS

- 1.** A tissue is composed of a group of
- **A.** similar cells
 - **B.** tissues
 - **C.** systems

- **D.** related organs

Short Explanation: In biology, **tissue** is a cellular organizational level between cells and a complete organ. A **tissue** is an ensemble of similar cells and their extracellular matrix from the same origin that together carry out a specific function.

Correct Option: Answer is A

2. The population of different organisms that exist together in a habitat is called

- **A.** biomes
- **B.** biosphere
- **C.** community
- **D.** ecology

Short Explanation: community consists of different organisms living together in a habitat.

Correct Option: Answer is C

3. Which of the following serves as the brain box in man?

- **A.** skeleton
- **B.** skull
- **C.** head
- **D.** spinal column

Correct Option: Answer is B

4. Which organ removes the largest amount of excess water from the body?

- **A.** Liver
- **B.** Lung

- **C.** Kidney
- **D.** skin

Short Explanation: Kidneys: Remove urea, salts, and excess water from the blood

Correct Option: Answer is C

5. The following parts are involved in breathing process, except

- **A.** mouth
- **B.** nostrils
- **C.** ribs
- **D.** diaphragm

Short Explanation: The act of breathing has two stages – inhalation and exhalation:

- Inhalation – the intake of air into the lungs through expansion of chest volume.

- Exhalation – the expulsion of air from the lungs through contraction of chest volume.

Inhalation and exhalation involves muscles:

- Rib muscles = the muscles between the ribs in the chest.

- Diaphragm muscle

Correct Option: Answer is A

6. Which of these is not a sampling method for determining population size?

- **A.** capture and re-capture method
- **B.** complete census
- **C.** Transect method
- **D.** systematic sampling

Correct Option: Answer is D

7. A group of organisms of the same kind inhabiting the same environment is called

- **A.** ecosystem
- **B.** habitat
- **C.** population
- **D.** species
- **Short Explanation:** a group of living organisms consisting of similar individuals capable of exchanging genes or interbreeding
- **Correct Option: Answer is D**

8. The light sensitive cells in the human eye is called the

- **A.** Choroid
- **B.** Sclerotic layer
- **C.** retina
- **D.** cones

Short Explanation: **Cone** cells, or **cones**, are photoreceptor cells in the retinas of vertebrate **eyes** including the human **eye**. They respond differently to light of different wavelengths, and are thus responsible for color vision, and function best in relatively bright light, as opposed to rod cells, which work better in dim light.

Correct Option: Answer is D

9. The earliest form of life in animal kingdom is the

- **A.** class
- **B.** amphibian
- **C.** aves pisces

- **D. reptilia**
- **Correct Option: Answer is C**

10. Which of the following disease is not sexually transmitted?

- **A. Gonorrhea**
- **B. Herpes**
- **C. Influenza**
- **D. Syphilis**
- **Short Explanation:** The flu is caused by influenza **viruses** that infect the nose, throat, and lungs. These **viruses** spread when people with flu cough, sneeze or talk, sending droplets with the virus into the air and potentially into the mouths or noses of people who are nearby.
- **Correct Option: Answer is C**

11. Which of the following is not a monocotyledonous seed?

- **A. cowpea**
- **B. maize**
- **C. millet**
- **D. wheat**
- **Short Explanation:** Cowpea is a dicot plant
- **Correct Option: Answer is A**

12. The lowest unit of classification is the

- **A. class**
- **B. genus**
- **C. phylum**
- **D. species**



- **Short Explanation:**
- **Correct Option: Answer is D**

13. One of the following diseases is caused by fungi

- **A.** cassava mosaic
- **B.** coffee leaf rust
- **C.** leaf blight of cassava rosette
- **D.** disease of groundnut
- **Short Explanation:** Hemileia vastatrix is a fungus of the order Pucciniales (previously also known as Uredinales) that **causes coffee leaf rust** (CLR), a disease that is devastating to susceptible **coffee** plantations.
- **Correct Option: Answer is B**

14. The by-product of photosynthesis is

- **A.** CO₂
- **B.** O₂
- **C.** C₆H₁₂O₆
- **D.** H₂O

- **Short Explanation:** In photosynthesis, solar energy is harvested as chemical energy in a process that converts water and carbon dioxide to **glucose**. Oxygen is released as a byproduct
- **Correct Option: Answer is B**

15. Gaseous exchange in the lungs takes place in the

- **A.** alveoli
- **B.** bronchi
- **C.** bronchioles
- **D.** trachea
- **Short Explanation:** **Alveoli** are tiny air sacs in your lungs that take up the oxygen you breathe in and keep your body going. Although they're microscopic, **alveoli** are the workhorses of your respiratory system. You have about 480 million **alveoli**, located at the end of bronchial tubes.
- **Correct Option: Answer is A**

16. In eukaryotic cells, cellular respiration takes place in the

- **A.** nucleus
- **B.** cytoplasm
- **C.** centrioles
- **D.** mitochondrion

Correct Option: Answer is D

17. One of the following causes Ebola fever

- **A.** Bacteria
- **B.** Fungi
- **C.** Protozoa
- **D.** Viruses

Correct Option: Answer is D

18. The theory of survival of the fittest was propounded by

- **A.** Darwin
- **B.** Hooke
- **C.** Lamarck
- **D.** Linnaeus
- **Correct Option: Answer is A**

19. The deficiency of vitamin D leads to

- **A.** beriberi
- **B.** pellagra
- **C.** ricket
- **D.** scurvy
- **Short Explanation:** Vitamin D deficiency can lead to a loss of bone density, which can contribute to osteoporosis and fractures (broken bones). Severe vitamin D deficiency can also lead to other diseases. In children, it can cause **rickets**. **Rickets** is a rare disease that causes the bones to become soft and bend.
- **Correct Option: Answer is C**

20. One of these is used for excretion in earthworm

- **A.** Contractile vacuole
- **B.** Flame cell
- **C.** Malpighian tubule
- **D.** Nephridium
- **Short Explanation:** The nephridium is an invertebrate organ, found in pairs and performing a function similar to the vertebrate kidney. Nephridia remove metabolic wastes from

an animal's body. In addition to two basic categories, other types may exist.

- **Correct Option: Answer is D**

21. The following group of plants is the most advanced except

- **A.** bryophyte
- **B.** pteridophyta
- **C.** spermatophyte
- **D.** thallophytic

Correct Option: Answer is D

22. Which of the following hormonal glands is located on top of the kidney?

- **A.** Adrenal
- **B.** gonads
- **C.** pancreas
- **D.** thyroid
- **Correct Option: Answer is A**

23. One of these is present in both plant and animal cell?

- **A.** Cell membrane
- **B.** Cell wall
- **C.** Chloroplast
- **D.** Large cell vacuole

Correct Option: Answer is A

24. When the solute concentration of the cell and its surrounding medium are the same, the solution is said to be

- **A.** hypertonic
- **B.** hypotonic
- **C.** isotonic
- **D.** acidic

Correct Option: Answer is C

25. The primary consumer in the food chain below is

Green plants → Grasshopper → Lizard → Snake → Hawk

- **A.** Grass hopper
- **B.** green plants
- **C.** hawk
- **D.** lizard
- **Correct Option: Answer is A**

26. Angiosperms belong to the class

- **A.** bryophyte
- **B.** pteridophyta
- **C.** spermatophyte
- **D.** thallophytic

Correct Option: Answer is C

27. In the kidney, both useful substances and wastes are removed from the blood by

- **A.** Filtration
- **B.** selective absorption
- **C.** dialysis machine
- **D.** excretion
- **Short Explanation:** **kidneys** act like a **filter** to remove wastes and extra fluid from your body. Your **kidneys filter** about 200 quarts of blood each day to make about 1 to 2 quarts of urine. The urine contains wastes and extra fluid. This prevents buildup of wastes and fluid to keep your body healthy.
- **Correct Option: Answer is A**

28. Which of the following is not part of the mammalian male reproductive organ?

- **A.** Epididymis
- **B.** Vas deferens
- **C.** testis
- **D.** Vulva
- **Short Explanation:** The **vulva** is the external part of the female genitalia. It protects a woman's sexual organs, urinary opening, vestibule and vagina and is the center of much of a woman's sexual response.
- **Correct Option: Answer is D**

29. Which of the following reagent is used for testing the presence of protein in food?

- **A.** Benedict solution
- **B.** Fehling's Solution
- **C.** Millon's reagent
- **D.** Sudan III

Short Explanation: Millon's reagent is an analytical **reagent** used to detect the presence of soluble proteins.

- **Correct Option: Answer is C**

30. The breaking down of food in the alimentary canal is called

- **A.** digestion
- **B.** egestion
- **C.** excretion
- **D.** ingestion

Correct Option: Answer is A

31. The followings are the functions of the kidneys, except

- **A.** Regulation of water content in the blood
- **B.** Maintenance of blood PH and homeostasis
- **C.** Excretion of CO₂
- **D.** Removal of waste products from the blood

Correct Option: Answer is C

32. The following disrupts the balance in an ecosystem except

- **A.** afforestation
- **B.** deforestation
- **C.** migration
- **D.** pollution

Correct Option: Answer is A

33. The following animals are viviparous expect

- **A.** Cow
- **B.** Dog
- **C.** Goat
- **D.** Pigeon

Short Explanation: Viviparous (of an animal) bringing forth live young which have developed inside the body of the parent.

Oviparous (of an animal) producing young by means of eggs which are hatched after they have been laid by the parent, as in birds. Therefore, Pigeon is the correct option.

Correct Option: Answer is D

34. The following animal is an invertebrate except

- **A.** flat worms
- **B.** round worms
- **C.** chordate
- **D.** protozoan
- **Correct Option: Answer is C**

35. The following are kidney diseases except

- **A.** diuresis
- **B.** hepatitis
- **C.** nephritis
- **D.** oedema

Correct Option: Answer is B

36. The gland that releases hormones into the blood are part of the

- **A.** digestive system
- **B.** endocrine system
- **C.** circulatory system
- **D.** respiratory system

• **Correct Option: Answer is B**

37. One of the following is called emergency hormone in man.

- **A.** Adrenalin
- **B.** Prolactin
- **C.** Testosterone
- **D.** Thyroxine

Correct Option: Answer is A

38. Which of these responses is not voluntarily controlled by the brain?

- **A.** Dancing
- **B.** Eating
- **C.** Sneezing
- **D.** Writing

Correct Option: Answer is C

39. The scientific study of life is

- **A.** Botany
- **B.** Biology
- **C.** Zoology

- **D.** Ecology

Correct Option: Answer is B

40. The artery supplying the liver with blood is called

- **A.** hepatic
- **B.** mesenteric
- **C.** renal
- **D.** subclavian

Correct Option: Answer is A

41. One of the following exist as colony

- **A.** euglena
- **B.** paramecium
- **C.** spirogyra
- **D.** volvox

Correct Option: Answer is D

42. The translucency spot in food test shows the presence of

- **A.** fats
- **B.** starch
- **C.** water
- **D.** protein

Correct Option: Answer is A

43. The following are examples of oviparous animals except

- **A.** crocodile
- **B.** dove
- **C.** rat
- **D.** tilapia

Short Explanation: Oviparous (of an animal) producing young by means of eggs which are hatched after they have been laid by the parent, as in birds.

Viviparous (of an animal) bringing forth live young which have developed inside the body of the parent. Rat is a mammal and Viviparous.

Correct Option: Answer is B

44. The part of cell that is responsible for the production of energy is

- **A.** cell membrane
- **B.** cytoplasm
- **C.** mitochondrion
- **D.** nucleus

Correct Option: Answer is C

45. Hormones that help regulate blood calcium levels are produced by the

- **A.** Adrenal
- **B.** thyroid
- **C.** pancreas
- **D.** parathyroid

Correct Option: Answer is D

46. Which of the following statements is not true of a fish?

- **A.** respire with the use of gills
- **B.** covered with scales
- **C.** they are homeothermic
- **D.** they are poikilothermic

Correct Option: Answer is C

47. Which of the following is NOT a waste product of plants?

- **A.** Auxins
- **B.** Gum
- **C.** Resins
- **D.** tanins

Correct Option: Answer is A

48. An association between bacteria and root nodules of legume is called

- **A.** Commercialism
- **B.** mutualism
- **C.** parasitism
- **D.** saprophytism

• **Short Explanation:** The term **mutualism** refers to a relationship in biology or sociology that is mutually beneficial to two living things

• **Correct Option: Answer is B**

49. Which of these is not detected by the nerve endings of the skin?

- **A.** cold
- **B.** pain
- **C.** pressure

- **D.** weakness

• **Short Explanation:**

• **Correct Option: Answer is D**

50. Behaviour conditioning was described by

- **A.** Felix Dujartin
- **B.** Gregor Mendel
- **C.** Ivan Pavlov
- **D.** Mathias Schnelden

Correct Option: Answer is C

BIOLOGY PAST QUESTIONS 2015 THEORY WITH SOLUTION

- 1.** (a) (i) Explain sexual reproduction in Rhizopus
- (b) A student suddenly stepped on a big snake, he cried for help and ran away
- (i) Name the hormone produced in the body for the reaction
- (ii) In which part of the human body is the hormone named in 1(a)(i) Produce?
- (iii) State five changes that must have taken place as a result of the hormone named in 1(a)
- (c) In a tabular form, state three differences between the excretion in flowering plants and excretion in humans
- (i) In a tabular form, state three differences between the excretion in flowering plants and excretion in humans

(ii) List two organelles found in a plant cell that are responsible for the excretion of metabolic wastes.

Solution & Explanation 1:

(a)(i) Sexual Reproduction in Rhizopus are:

- It is usually between two strains of hyphae
- Known as + and -/plus and minus/opposite mating types;
- The hyphae come in contact;
- And produce small projections on their walls;
- Which develop into progametangia with several nuclei;
- The two progametangia enlarge;
- Each progametangium develops a cross wall to divide into two parts; the gametangium and suspensor;
- Subsequently the wall dissolves/breaks; and their contents/nuclei fuse to form a zygote;

(b) (i) Name of hormone is Adrenalin/Adrenaline

(ii) Part where hormone is produced is Adrenal gland/Medulla

(iii) Changes that must have taken place as a result of the hormone any of the following:

- Increase in heart beat;
- Increase in rate of respiration;
- Arterioles of skin and alimentary canal constrict;
- Dilation of blood vessels that cause more blood to flow to the muscles/heart/lungs;
- Increase in muscular tone/voluntary muscles become tensed;

- Glycogen is converted to glucose/increase in blood sugar level;
- Momentary freight/fear/anxiety/heavy breathing;
- Spleen releases more Red Blood Cells from its reserve;
- Increase in blood pressure.

(c)i Differences between excretion in flowering plant and excretion in human

Flowering Plant	Human
• No definite excretory organ, • Waste products include: • Tannins, resin, gum, alkaloid, etc	Definite excretory organs Presence of kidney, Liver, Skin. Waste products include: Urea, Urine, Salts etc.
Excretory products produced in small quantities	Excretory products produced in large quantities
Excretory products released at lower rate	Excretory products released at higher rate
• Most Excretory products produced, which are Tannins, resin, gum, alkaloid, etc are stored	Most Excretory products produced, which are urea, urine, salt etc are completely removed.
Some excretory product like oxygen, carbon dioxide can be reused	Some excretory product like urea, urine cannot be reused

(c)ii. two organelles found in a plant cell that are responsible for the excretion of metabolic wastes are :

- a) Mitochondria
- b) Chloroplast

- 2.** (a) Explain how the leaf of a flowering plant is adapted for photosynthesis?
(b) Describe briefly the mode of feeding in tapeworm
(c) Name one mineral element each needed for the proper functioning:
i. Bones
ii. Red blood cells
iii. Thyroid gland
- (d) (i) State three benefits of including roughage in the diet of human
(ii) State three reasons why proteins are important to human

Solution & Explanation 2:

(a) Adaptation of leaf for photosynthesis

- Broad/flat leaf lamella; exposes large surface area for maximum absorption of light;
- Palisade mesophyll cells contain a lot of chloroplast/chlorophyll; for maximum absorption of light;
- Position of leaf to stem; for optimum reception of light rays/sunlight;
- Thin lamella; allows light penetration into leaf tissue of mesophyll;

- Palisade cells are at right angle/perpendicular to the upper epidermis/chloroplasts are arranged in the periphery of the cells; all for easy activation of the chloroplasts for photosynthesis;
- Large vacuoles of the palisade cells; store photosynthetic products;

(b) Mode of feeding of tapeworm

Parasitic mode of feeding; lives in the gut of man/pigs; has hooks/suckers; which enable it attach itself to the lining of the intestine of the host; has a flattened tape-like/ribbon-like body; provide large surface area for easy absorption of digested food; no alimentary/digestive system.

(c) Mineral elements necessary for the proper functioning of

(i) Bones: Calcium; phosphorus; fluorine

(ii) Red blood cells: Iron; copper

(iii) Thyroid gland

Iodine

(d) (i) Benefits of including roughages in the diet of humans

- Stimulates bowel movement/peristalsis;
- Reduces blood cholesterol/bowel cancer/gall stones/appendicitis/obesity/diabetes mellitus;
- They do not contribute to weight gain/high blood sugar;

(ii) Importance of proteins to humans

- For the repair of damaged tissues;
- replacement of worn-out tissues;
- For body building;
- For growth;

3. (a)(i) What is sewage?

(ii) State five effects of releasing untreated sewage into a stagnant water body.

(b) Explain surface terracing as a soil conservation method.

(c) Explain the term adaptation.

(d) Explain two ways each by which the following organisms adapt to their habitats: (i) hydrophytes; xerophytes.

Solution & Explanation 3:

(a)(i) waste water and excrement conveyed in sewers.

(ii) effects of releasing untreated sewage into a stagnant water body are as follows:

- Spread of water borne diseases such as microbes and pathogen
- Makes water become unsuitable for consumption
- Increased decomposition
- Death of aquatic animal

(b) Surface terracing as a soil conservation method

- A method of controlling erosion on a gentle/steep slope of hill;
- flat horizontal areas are constructed on which crops are cultivated;
- following the contour;
- steps/walls are built along the contours;
- ploughing is also done along the contours;
- reduces the speed of water running down the slopes.

(c) **Adaptation** is the evolutionary process whereby an organism becomes better able to live in its habitat or habitats. 2. Adaptedness is the state of being adapted: the degree to which an organism is able to live and reproduce in a given set of habitats.

(d) (i) Adaptation of Hydrophytes

- Some hydrophytes lack/have reduced roots/root hairs/root caps;
 - o do not need firm anchorage in water/wetland;
 - o presence of large air space/aerenchyma in roots/leaves; provide support for buoyancy;
 - o presence of hairs on the leaves; to prevent blockage of the stomata;
 - o stomatal pores; occur only at the upper epidermis of the leaves to aid transpiration/loss of water;
 - o numerous adventitious roots and root hairs; aid the absorption of water/mineral salts;
 - o small size of the plant; for buoyancy/floating in water;
 - o waxy upper surface of the leaf; to prevent clogging of the leaf by water;

(ii) Adaptation of Xerophytes

- possession of thick succulent leaves/stem; for storing water;
- reduction of leaves to spines; to reduce water loss/rate of transpiration;
- thick waxy cuticle on epidermis; to reduce water loss/transpiration;
- possession of long tap root system; to obtain water from great depth of soil;

4. (a) Explain the following terms:

- (i) Test Cross
- (ii) Monohybrid Cross

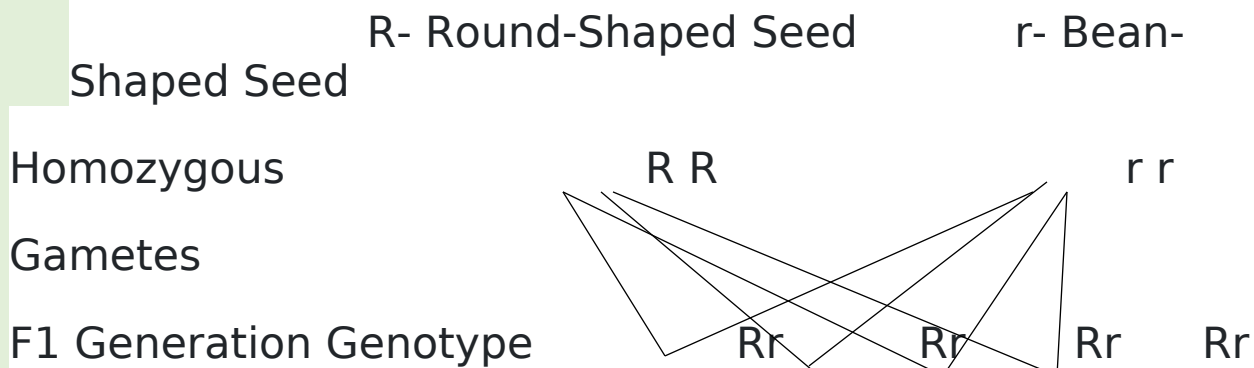
(b) In a mango plant, the allele for bean-shaped seed is r and is recessive to round-shaped seed R . with the aid of a genetic diagram, determine the genotypes of the offspring if a homozygous bean shaped parent is crossed with a homozygous round -shaped parent.

Solution & Explanation 4:

(a) (i) **Test cross:** This is the crossing/mating of an organism of unknown genotype; with a homozygous recessive/double recessive organism; to determine the genotype of an organism.

(ii) **Monohybrid cross:** Is a genetic cross between parents; that differ in the alleles they possess; for one particular gene

(b) crosses



BIOLOGY PAST QUESTIONS 2014 WITH SOLUTION

1. The lowest level of organization in living organisms is

- **A.** organ
- **B.** cell
- **C.** system
- **D.** tissue

Short Explanation: A **cell** is the **smallest unit** of a **living** thing. A **living** thing, whether made of one **cell** (like bacteria) or many **cells** (like a human), is called an organism. Thus, **cells** are the basic building blocks of all organisms

Correct Option: Answer is B

2. Which of the following is the most complex according to their cellular level of organization?

- **A.** Heart
- **B.** Hair
- **C.** Euglena
- **D.** Hydra

Correct Option: Answer is A

3. Which of the following organisms is multicellular?

- **A.** Chlamydomonas
- **B.** Spirogyra
- **C.** Amoeba
- **D.** Euglena

Short Explanation: A spirogyra is a multi-cellular organism that is also an informal grouping of diverse eukaryotic organisms. All other options are unicellular organisms.

Correct Option: Answer is B

4. In bryophytes, sex organs are produced in the

- A. protonema
- B. sporophyte
- C. gametophyte
- D. rhizoid

Short Explanation: The **sex organs** in **bryophytes** are multicellular. The male **sex organ** is called antheridium. They produce biflagellate antherozoids. The leafy shoots belong to the gametophytic phase and produce **sex organs** when they mature.

Correct Option: Answer is C

5. Seed plants are the most dominant vegetation on land because of

- A. their motile gametes
- B. their ability to photosynthesize
- C. efficient seed dispersal
- D. availability of water

Correct Option: Answer is C

6. Which of the following is an arboreal organism?

- A. Elephant
- B. Fish
- C. Antelope
- D. Bird

Correct Option: Answer is D

7. $i^{2/1}$, $c^{0/0}$, $pm^{3/2}$, $m^{3/3}$. The general formula above represents that of

- **A.** an omnivore
- **B.** a detritus feeder
- **C.** a carnivore
- **D.** an herbivore

Correct Option: Answer is D

8. A circulatory system is very essential in mammals but not in smaller organisms like Amoeba because

- **A.** Amoeba lives in freshwater
- **B.** Diffusion is sufficient to transport materials in Amoeba
- **C.** Amoeba lacks blood containing haemoglobin
- **D.** Amoeba exhibits anaerobic respiration

Correct Option: Answer is B

9. In vascular plants, the sieve tubes and companion cells are present in the

- **A.** cambium
- **B.** cortex
- **C.** xylem
- **D.** phloem

Correct Option: Answer is D

10. The stomata of leaves are similar in function to the

- A. pharynx of humans
- B. scales of fish
- C. spiracle of insects
- D. trachea of toads

Correct Option: Answer is C

11. The use of moist skin for respiration in amphibians is known as

- A. cellular respiration
- B. cutaneous respiration
- C. buccal respiration
- D. pulmonary respiration

Short Explanation: *Cutaneous respiration*, or **cutaneous** gas exchange, is a form of **respiration** in which gas exchange occurs across the **skin** or outer integument of an organism rather than gills or lungs.

Correct Option: Answer is B

12. Water in plants is removed as water vapour through the process of

- A. diffusion
- B. osmosis
- C. evaporation
- D. transpiration

Short Explanation: **Transpiration** is the process of water movement through a plant and its evaporation from aerial parts, such as leaves, stems and flowers.

Correct Option: Answer is D

13. An example of an organ of perennation in plants is

- **A.** rhizome
- **B.** seed
- **C.** petal of a flower
- **D.** calyx of flower

Correct Option: Answer is A

14. Alternation of generation is a feature shown in

- **A.** mosses
- **B.** fungi
- **C.** grasses
- **D.** conifers
- **Short Explanation:** All land plants have alternation of generations. In **mosses** and their relatives (Bryophytes), the haploid gametophyte is the dominant generation, and the diploid sporophytes are sporangium-bearing stalks growing from the gametophytes.
- **Correct Option: Answer is A**

15. Coordination and regulation of body activities in mammals are achieved by the

- **A.** nerves and muscle
- **B.** nerves and hormones
- **C.** nerves only
- **D.** hormones only

Correct Option: Answer is B

16. The Cerebellum of the Brain controls

- **A.** Reflex Action
- **B.** Muscular Activity
- **C.** Emotional Expressions
- **D.** the Endocrine System
- **Short Explanation:** Coordination of voluntary movements. Most movements are composed of a number of different muscle groups acting together in a temporally coordinated fashion. One major function of the **cerebellum** is to coordinate the timing and force of these different muscle groups to produce fluid limb or body movements.
- **Correct Option: Answer is B**

17. The part of the brain responsible for peristalsis is the

- **A.** Olfactory Lobe
- **B.** Medulla Oblongata
- **C.** Hypothalamus
- **D.** Thalamus

Short Explanation: The medulla oblongata is responsible for regulating several basic functions of the autonomic nervous system, including **respiration**, cardiac function, vasodilation, and reflexes like vomiting, coughing, sneezing, and swallowing (peristalsis).

Correct Option: Answer is B

18. Which of the following instruments is used for measuring atmospheric pressure?

- **A.** Hydrometer
- **B.** Hygrometer
- **C.** Thermometer
- **D.** Barometer

Short Explanation: A barometer is a scientific instrument that is used to measure air pressure in a certain environment.

Correct Option: Answer is D

19. The influence of soil on organisms in a habitat is referred to as

- **A.** Edaphic
- **B.** Physiographic
- **C.** Biotic
- **D.** Topographic

Short Explanation: Edaphology is concerned with the influence of soils on living things, particularly plants. Edaphology includes the study of how soil influences humankind's use of land for plant growth as well as people's overall use of the land.

Correct Option: Answer is A

20. The genetic make-up of an organism is described as

- **A.** Allele
- **B.** Chromosome
- **C.** Phenotype
- **D.** Genotype

Short Explanation: A genotype is an organism's complete set of heritable genes, or genes that can be passed down from parents to offspring.

Correct Option: Answer is D

21. The major limiting factor of productivity in the aquatic habitat is

- **A.** Food
- **B.** Temperature
- **C.** Water
- **D.** Sunlight

Correct Option: Answer is A

22. Which of the following group of organism's feeds directly on green plants?

- **A.** Primary Consumers
- **B.** Secondary Consumers
- **C.** Producers
- **D.** Decomposers

Correct Option: Answer is A

23. A characteristic feature of tropical rainforest is that it

- **A.** Contains trees with narrow leaves
- **B.** Contains large number of plant species
- **C.** Contains fewer number of plant species
- **D.** Has total annual rainfall of less than 50cm

Correct Option: Answer is B

24. The study of how and why population size change over time is

- **A.** Population estimation
- **B.** Population dynamics
- **C.** Population ecology
- **D.** Population Cycle
- **Short Explanation:** **Population dynamics** is the branch of life sciences that studies the size and age composition of **populations** as dynamical systems, and the biological and environmental processes driving them (such as birth and death rates, and by immigration and emigration).
- **Correct Option: Answer is B**

25. A severe and long dry season is a characteristic feature of

- **A.** Sahel Savanna
- **B.** Mangrove Swamps
- **C.** Sudan Savanna
- **D.** Guinea Savanna

Correct Option: Answer is A

26. Which of the following is a nitrogen-fixing blue-green algae of soil?

- **A.** Rhizobium
- **B.** Nitrosomonas
- **C.** Clostridium
- **D.** Anabaena
- **Short Explanation:** **Anabaena** is a genus of filamentous cyanobacteria that exist as plankton.

They are known for nitrogen-fixing abilities, and they form symbiotic relationships with certain plants, such as the mosquito fern.

- **Correct Option: Answer is D**

27. The soil with highest water-retaining capacity is

- **A.** Clayey Soil
- **B.** Stoney soil
- **C.** Sandy soil
- **D.** Loamy Soil

Correct Option: Answer is A

28. The causative agent of Poliomyelitis is

- **A.** Virus
- **B.** Fungus
- **C.** Protozoan
- **D.** Bacterium

Short Explanation: Poliovirus, the **causative agent** of **polio** (also known as **poliomyelitis**), is a serotype of the species Enterovirus C, in the family of Picornaviridae. **Poliovirus** is composed of an RNA genome and a protein capsid.

Correct Option: Answer is A

29. One of the ways of controlling noise pollution in urban areas is

- **A.** by siting industries away from residential areas
- **B.** that fuel should be completely combusted by engines

- **C.** by planting trees on both sides of the road
- **D.** by wearing ear devices

Correct Option: Answer is A

30. A constituent of the exhaust fumes from electricity generating sets which causes serious pollution is

- **A.** Carbon (II) Oxide
- **B.** Water Vapour
- **C.** Ozone
- **D.** Carbon (IV) Oxide

Correct Option: Answer is A

31. Which of the following is true of small pox?

- **A.** It is transmitted by bacteria
- **B.** It can effectively be controlled with antibiotics
- **C.** It can effectively be controlled by vaccination
- **D.** It is a water-borne infection

• **Correct Option: Answer is A**

32. A pollutant that is mostly associated with acid rain is

- **A.** Nitrogen (IV) Oxide
- **B.** Ozone
- **C.** Fluorine
- **D.** Carbon (IV) Oxide
- **Short Explanation: Acid rain** is caused by a chemical reaction that begins when compounds like sulfur **dioxide** and **nitrogen oxides** are released

into the air. These substances can rise very high into the atmosphere, where they mix and react with water, oxygen, and other chemicals to form more **acidic pollutants**, known as **acid rain**.

Correct Option: Answer is A

33. When the adults have reach a certain degree of weakness, the process of binary fission is replaced by conjugation in

- **A.** Paramecium
- **B.** Euglena
- **C.** Amoeba
- **D.** Plasmodium

Correct Option: Answer is A

34. Whorls, arches, loops and compounds are types of variation in

- **A.** Colour
- **B.** Finger prints
- **C.** Hair Colour
- **D.** Blood group

Correct Option: Answer is B

35. A couple has 10 children, all female. Which of the following best explains the situation

- **A.** The sex determination was by the man's X chromosome
- **B.** The man's sperm count is low
- **C.** The woman is not capable of producing male children

- **D.** The sex determination was by the man's Y chromosome

Correct Option: Answer is A

36. A biological agent with antiviral property is

- **A.** Interferon
- **B.** enzyme
- **C.** antibiotic
- **D.** disinfectant

Short Explanation: All **interferons** share several common effects: they are **antiviral** agents and they modulate functions of the immune system

- **Correct Option: Answer is A**

37. One of the advantages of outbreeding is

- **A.** pests tolerance
- **B.** disease resistance
- **C.** fast growth
- **D.** tall height

Correct Option: Answer is B

38. An individual with blood group AB can receive blood from those in blood group(s)

- **A.** A, B, AB, O
- **B.** A, AB and O only
- **C.** AB only
- **D.** A and B only

Short Explanation:

If your blood type is:

You can receive from:

A Positive	A+, A-, O+, O-
B Positive	B+, B-, O+, O-
AB Positive	All blood types
O Negative	O- only

Correct Option: Answer is A

39. The stream-lined shape of fishes is an adaptation for

- **A.** Securing mates
- **B.** easy movement
- **C.** obtaining food
- **D.** defense and attack

Correct Option: Answer is B

40. An example of a poikilothermic organism is a

- **A.** Lizard
- **B.** Cockroach
- **C.** rabbit
- **D.** bird

Short Explanation: A poikilotherm is an animal whose internal temperature varies considerably. It is the opposite of a homeotherm, an animal which maintains thermal homeostasis. **Poikilothermic** animals include types of vertebrate animals, specifically some fish, amphibians, and reptiles, as well as many invertebrate animals.

Correct Option: Answer is A

41. All living organisms are constantly involved in a struggle for existence. This was proposed by

- **A.** Morgan
- **B.** Darwin
- **C.** Lamarck
- **D.** Wallace

Short Explanation: The **struggle for existence** is a natural history [metaphor]. It refers to the competition between living things to survive. This, and the similar phrase **struggle** for life, were used over 40 times by Charles **Darwin** in the Origin of Species.

- **Correct Option: Answer is B**

42

42. Adaptive radiation is illustrated in

- **A.** modified insect mouthparts
- **B.** dentition in mammals
- **C.** wings in birds and bats
- **D.** appendages in insects

Short Explanation: In evolutionary biology, **adaptive radiation** is a process in which organisms diversify rapidly from an ancestral species into a multitude of new forms, particularly when a change in the environment makes new resources available, creates new challenges, or opens new environmental niches.

- **Correct Option: Answer is C**

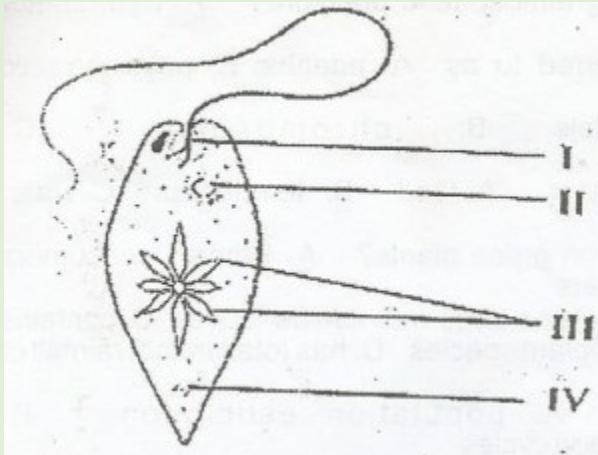
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43. i. Growth is mainly apical ii. Growth is specific with definite shape iii. Growth is throughout life. Which of the above correctly describes the growth pattern in plants?

- **A.** i, ii and iii only
- **B.** ii and iii only
- **C.** i and ii only
- **D.** i and iii only

- **Correct Option: Answer is D**

44

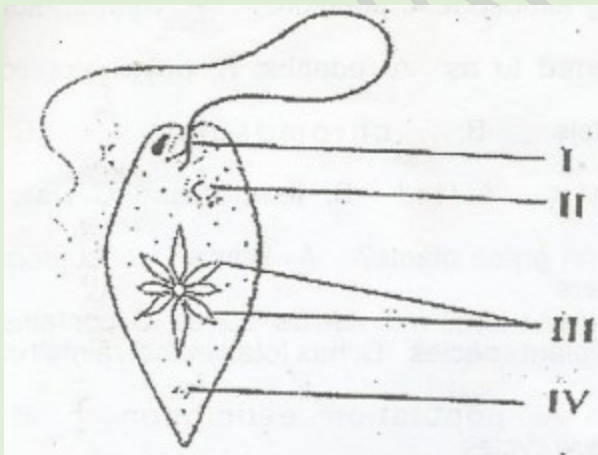


44. Use the diagram to answer this question. The part labelled II is the

- **A.** nucleus
- **B.** eyespot
- **C.** basal granule
- **D.** contractile vacuole

Correct Option: Answer is D

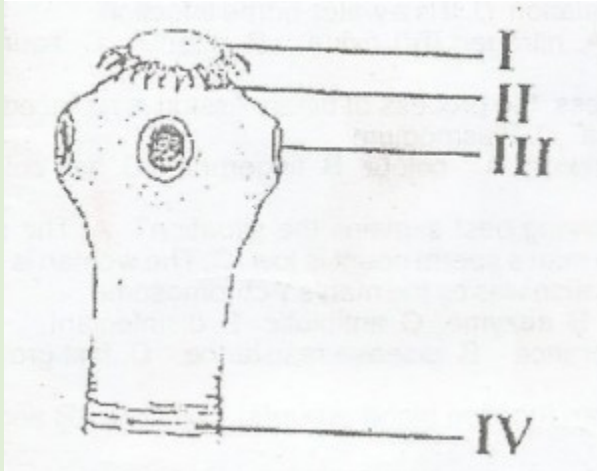
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45. Use the diagram above to answer this question. The part responsible for photosynthesis is labelled

- **A.** III
- **B.** IV

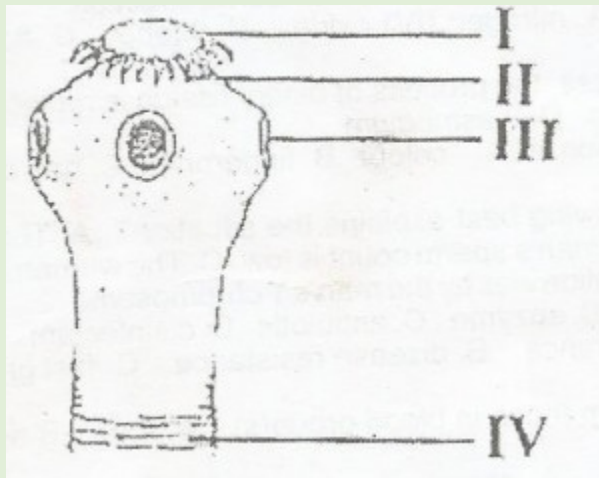
- C. I
- D. II
- **Correct Option: Answer is A**



46. Use the diagram above to answer this question.
The young proglottid is represented by

- **A. III**
- B. IV
- C. I
- D. II

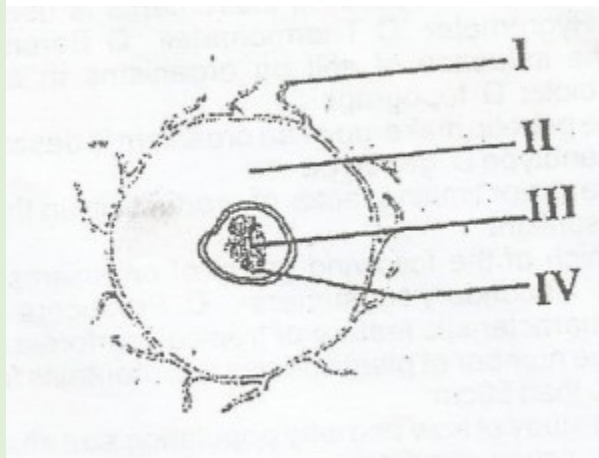
Correct Option: Answer is B



47. In the diagram, the organs for attachments to the lining of the host's intestine are labelled

- **A.** ii and iii
- **B.** iii and iv
- **C.** i and ii
- **D.** i and iii

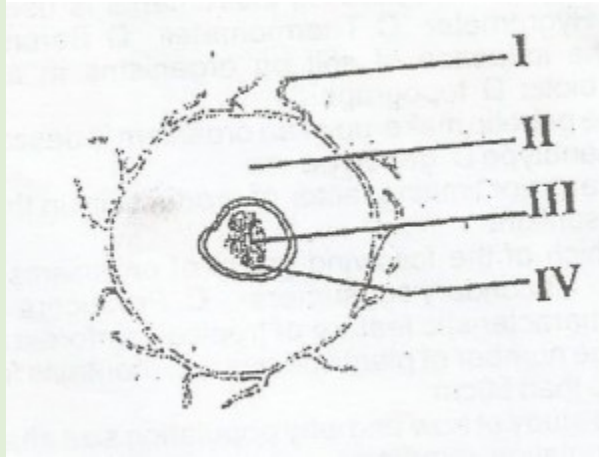
Correct Option: Answer is A



48. In the diagram, the part labelled I is the

- **A.** xylem
- **B.** phloem
- **C.** root hairs
- **D.** cortex

Correct Option: Answer is C



49. The diagram is the transverse section of a

- **A.** monocotyledonous stem
- **B.** dicotyledonous stem
- **C.** monocotyledonous root
- **D.** dicotyledonous root

Correct Option: Answer is D

50. Active transport can be defined as the movement of

- **A.** substances from a region of high concentration to a region of low concentration
- **B.** substances from a region of low concentration to a region of high concentration across a living cell membrane
- **C.** substances across a living cell membrane using energy from the cell
- **D.** movement of molecules in a medium

Short Explanation: Active transport. In cellular biology, **active transport** is the **movement** of molecules across a cell membrane from a region of lower

concentration to a region of higher concentration—
against the concentration gradient.

Correct Option: Answer is B

**BIOLOGY PAST QUESTIONS 2014 THEORY WITH
SOLUTION**

1. (a) (i) List three characteristics of living organisms.

(ii) Explain briefly the characteristics listed in (a)(i) above.

(b) In a tabular form, state four differences between plants and animals.

(c)(i) Name one organism that exhibits both plant and animal features.

(ii) State four animal features and two plant features possessed by the organism named in (c)(i) above.

Solution & Explanation 1:

(a)(i) Characteristic of Living organisms:

- Movement/locomotion
- Excretion
- Respiration
- Reproduction
- Irritability/sensitivity/response
- Growth/development
- Nutrition/feeding
- Competition
- Adaptation
- Life span/death

(ii) **Movement:** Living things change the position of the whole body/parts of the body; from one place to another; in search of food/shelter/mates, etc.

Excretion: This involves the removal/getting rid of waste products of cell metabolism; membrane wastes are poisonous if allowed to accumulate.

Respiration This is the breakdown/burning of food substances; by aerobic/anaerobic respiration; to release energy needed for all processes of life.

Reproduction: This is the ability of living organisms to produce new individuals of their own kind; this ensures the continuity of the species of an organism.

Irritability: Response/ reaction of organisms; towards changes in the environment; to ensure survival. Nutrition: the process involved in obtaining/manufacturing food; can be autotrophic/heterotrophic; and utilizing it for growth and maintenance/life processes.

Growth: This is the irreversible increase; in size and weight of an organism; which leads to complexity and maturation.

Competition: Ability of organisms to struggle; for all necessity of life; to survive in their environment.

Adaptation: Ability of organisms to adjust; to changes in environment; for survival.

Life span/death: All living organisms have definite period of existence; all organisms die.

b Differences between plants and animals;

Animals	plants
move from place to place, locomotion, move with the whole body	Are fixed/do not move from place to place, and move only parts of the body, root/stem/leaf
Heterotrophic mode of nutrition that depend on already-made food	Autotrophic mode of nutrition/ manufacture their own food using the

	energy from the sun
No chlorophyll/chloroplast	chlorophyll/chloroplast present
No cell wall	Has a cell wall made of cellulose
sense organs present	sense organs absent
Growth is limited/definite, growth in all parts of the body	growth is unlimited/indefinite .growth is apical or terminal
stores carbohydrates as glycogen	stores carbohydrates as starch
Has special excretory body	No special excretory body
Response is fast, short termed	response very slow/short termed
Respiratory organs present	No respiratory organs

(c)(i) Name of organism:

Euglena/Chlamydomonas.

(ii) Animal features: For Euglena

- Gullet for feeding.
- Has flagellum for locomotion in water.
- Myonemes for movement.
- Has pellicle that gives it a definite shape.
- Has eye spot to detect light intensity.
- Has contractile vacuole for osmoregulation and excretion.

Plant features:

- Definite shape.
- Has chloroplasts containing chlorophyll used to absorb light.
- Paramylum granules.
- Pyrenoid.

2. (a) Define the following in

(i) elimination

(ii) excretion

(iii) secretion.

(b) Describe the digestion of boiled fish in the gut of humans

(c) Make a diagram 8cm - 10cm long of a longitudinal section of a villus and label fully.

Solution & Explanation 2:

(a) Definition of terms:

(i) **Elimination:** Removal of undigested food; from the body of living organisms.

(ii) **Excretion:** Removal of waste products of metabolism; from the cells/tissues of an organism; may be toxic if retained in the body.

(iii) **Secretion:** Production and release of substances; useful to the body; by specialized glands of living organism.

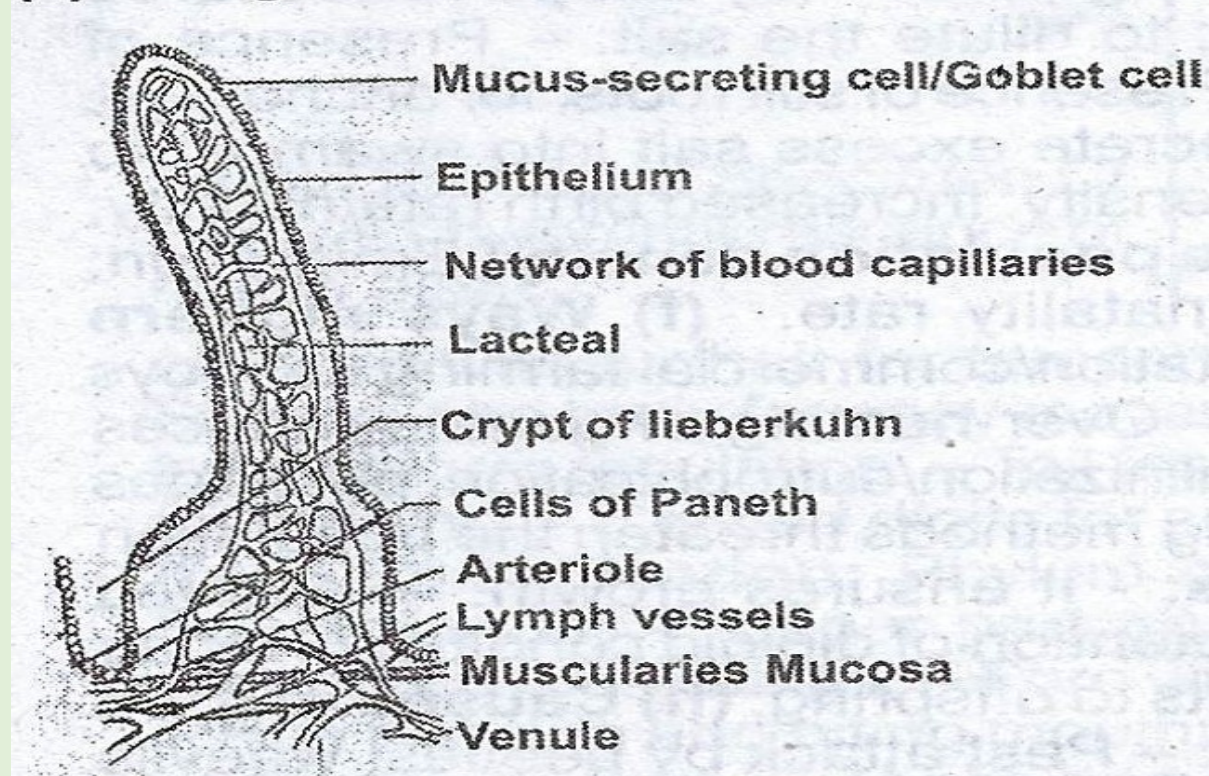
(b) Digestion of boiled fish in the gut of

humans: Boiled fish is a source of protein; chewed and swallowed into the stomach; digestion of protein starts from the stomach; stomach secretes gastric juice; containing pepsin, rennin and hydrochloric acid; pepsin in acidic medium acts on the boiled fish and convert it to peptides; peptides move into the duodenum/small intestine; into which pancreatic juice is secreted; undigested protein is acted upon by trypsin in alkaline medium; into peptides/peptones, erepsin finally acts on

peptides/peptones in alkaline medium; and converts it into amino acids.

(c) Diagram of villus

(c) Diagram of villus



Title TL: - Longitudinal section of villus. Size: (8 — 10cm long).

Quality (Q): Clarity of lines. Neatness of labels.

Details: Epithelium layer shown (EL) Conical shape of villus (one cell thick) (CV).

Labels (L): Epithelium layer blood capillary, lacteal, lymphatic vessels, goblet cells, venule, arteriols, crypts of Lieberkuhn, cells of Paneth, muscularis mucosa.

3. (a) Explain briefly the following ecological terms:

(i) biosphere

(ii) habitat

(b) Describe the mode of nutrition in a named carnivorous plant.

(c) State three reasons why the dispersal of fruits and seeds are important.

(d) State six methods of conserving soil.

Solution & Explanation 3:

(a)(i) Biosphere: Total/whole area, part of the earth/lithosphere/ hydrosphere/water, and atmosphere/air, inhabitable by living things/where life exists.

(ii) Habitat: A place/home; within an environment, where an organism naturally lives/dwells.

(b) Mode of nutrition of named carnivorous plants:

- - Carnivorous plants have green leaves and are able to manufacture their own food carry out photosynthesis.
- They have roots to absorb water and mineral salts from the soil.
- Usually grow in places with little nitrogenous salts.
- Use insects/other small animals as source of Nitrogen.
- They have devices/modified leaves with bright colour/scent/sugary liquids/sticky wet surfaces for attracting and trapping insects.
- Once the insects is trapped, the plant secretes enzymes to kill and digest it.
- The soluble nitrogenous compounds are absorbed/assimilated by the plant.

Name of carnivorous plants: -

- Butterwort/*Pinguicula*
- Bladderwort/ *Utricularia*

- Sundew/Drosera
- Venus fly-trap/Dionaea
- Pitcher plants/Nepenthes/Sarracenia.

(c) Importance of fruits and seed dispersal: -

- Prevents overcrowding of plants.
- Prevents undesirable competition for space/light/water and nutrients.
- Prevents diseases from spreading among plant species.
- Results in colonization of new areas with new plants.
- Results in adaptation to adverse/harsh/unsuitable environment.

(d) Soil conservation method: -

- Contour ploughing/ridges are made along the contour of a sloping land/to prevent water running down the hill/slope/and washing away soil.
- Strip cultivation/the land is divided into strips along the contours of the slopes/crops are planted in rows that alternate with the strips planted with cover crops/grasses.
- Terracing/construction of terraces which create barriers to hinder rapid water flow down the slope.
- Mulching/materials left after weeding or harvest are left to lie on the soil/ protecting the soil against agents of erosion.
- Afforestation/cover cropping/growing grasses, trees and shrubs to cover and protect the soil.
- Crop rotation/growing crops on a piece of land in sequence/season after season to maintain its fertility.
- Storm drains/construction of barriers/to stop free-flow of surface water through farms.
- Application of manure/fertilizer/to stabilize the soil.
- Avoid bush burning.

- Avoid over grazing.

4. (a)(i) Explain the term co-ordinance.

(ii) Calculate the number of individuals with co-dominant blood group.

(iii) What is the total number of individuals in the table that are able to donate blood to an accident victim with blood group B?

(b) A man whose blood group is heterozygous A is married to a woman whose blood group is AB. With the aid of a genetic diagram, suggest the possible blood groups of their children.

A survey to determine blood groups was carried out on 250 people living in a community. The results are represented in the table below.

Blood group	Percentage
A	8.0
B	14.0
AB	32.8
O	45.2

Solution & Explanation 4:

(a)(i) Co-dominance: Co-dominance occurs when both the alleles in a heterozygous individual are fully expressed; that is both alleles exert their effects so that the hybrid has a phenotype; that is different from what the parents exhibit.

(ii) Calculation of individuals with co-dominant

blood group: Mention of co-dominant blood group is AB. Evidence of calculation 32.8% of 250 OR $32.8/100 \times 250 = 82$ individuals/ people.

(iii) Total number of individuals that can donate blood to blood group B:

The total number of individuals will be from groups B and O.

—14% of the population is Blood group B 14% of 250 OR $14/100 \times 250 = 35$ individuals/people.

— 45.2% of the population is Blood group O 45.2% of 250 OR $45.2/100 \times 250 = 113$ individuals/people.

— Total number of people = 113 of group O+ 35 of group B = 148 Individuals/people can donate blood to the accident victim in B. OR The total number of individuals from (B and O) blood group B and O

— $14\% + 45.2 = 59.2\%$.

— 59.2% of 250. $59.2/100 \times 250 = 148$ people/individuals.

(b)

(b) Crossing:

Present
Parent genotype
Gametes

Correct crossing

Offspring
Phenotype

Man
AO
A
O

Woman
AB
A
B

AA
AO
BO
AB

A - 2
AB - 1
B - 1 } All listed to score

Likely blood group is A, B and AB. OR Punnet square

Man
Woman
Parent: Genotype AO AB

	MALE	
	A	O
FEMALE	AA	AO
	AA	BO

AA, AO, AB, BO
A - 2, AB - 1, B - 1. Likely blood group A, AB, B

5. (a) State five reasons why animals move from place to place.

(b) State one function each of the following structures found in plants:

(i) epidermis

- (ii) phloem
- (iii) sclerenchyma.

(c) Give two examples each of the following types of organisms:

- (i) parasitic plants
- (ii) saprophytes.

(d) State two adaptive features of plants which inhabit salt water swamp.

(e) State two factors each which result in

- (i) increase
- (ii) decrease, in population density.

(f) Mention four ways in which modern agricultural activities may threaten the survival of species.

(g) State three reasons why mitosis is important to living organisms

(h) State three causes of food spoilage.

Solution & Explanation 5:

(a) Reasons why animals move from place to place:

- (i) To search for food.
- (ii) Escape predators.
- (iii) Find mates for reproduction.
- (iv) To disperse/reduce competition/overcrowding by moving to new locations.
- (v) Avoid dangers. (vi) Find new and favourable habitats.

(b) Function of structures found in plant:

(i) Epidermis: For protection of inner cells from injury, infection, prevent excessive loss of water.

(ii) Phloem: Responsible for the transportation of enzymes/hormones; manufactured food.

(iii) Sclerenchyma: Strengthens the stem/provide mechanical support; makes plants flexible.

(c) Examples of organism:

(i) Parasitic plants: Viscum/mistletoe; dodder plant/Cassytha; Cuscuta; phytophisor; peronospora.

(ii) Saprophyta: Bacteria; fungi/Rhizopus; Mucor, Aspergillus; Mushroom/Agaricus.

(d) adaptive features of plants in salt water swamps: -

- Succulent stem/root/store water; to dilute the salt.
- Presence of pneumatophores/breathing roots; for breathing atmosphere air/oxygen.
- Presence of stilt roots; for anchorage.
- Viviparous seedling; to ensure their survival/continuity.
- Some a secrete excess salt into swamps; to maintain balance of body salt.

(e) Factors that: (i) Increase population density:

- Increased birth rate/natality
- immigration, increase a food availability
- low mortality rate.

(ii) Decrease population density:

- Emigration
- diseases
- drought/famine/natural disaster
- high mortality rate
- low natality rate.

(f) Ways modern agricultural activities may threaten the survival of species: -

- Deforestation/commercial farming destroys habitats of species.
- Destruction/loss of natural vegetation/habitat.
- Over hunting of games reduces population of species.
- Improper/Indiscriminate application of chemical/salinization eutrophication, pesticides destroys species.
- Lack of afforestation practices.
- Inappropriate fishing methods threaten the population of fish.

(g) Reasons why mitosis is important to living organisms: -

- It ensures growth.
- Involved asexual reproduction e.g. fission, budding, spore formation.
- Ensures retention of diploid condition. Repair of tissues within organisms.
- Transfer of exact genetic code from parents to offspring.

(h) Causes of food spoilage: -

- Microbial attacks by bacteria or fungi/or saprophytic action.

- Pest attack by beetles, weevils that destroy stored grains.
- Improper storage which results in deterioration and spoilage of stored foods.
- Destruction of stored fruits and meat by some worms, usually larva of flies.
- Delay in transportation due to bad roads.
- Lack of market facilities for the commodity.
- Poor harvesting method/damage to fruits/vegetables/tubers during harvesting.
- Poor unhygienic handling/processing/exposure to dust.
- Storage in contaminated containers.
- Use of contaminated water to wash food.

BIOLOGY PAST QUESTIONS 2013 WITH SOLUTIONS

- 1.** The process in which complex substances are broken down into simpler ones is referred to as
- **A.** anabolism
 - **B.** catabolism
 - **C.** metabolism
 - **D.** tropism

Short Explanation: Catabolism is the set of metabolic pathways that breaks down molecules into smaller units that are either oxidized to release energy or used in other anabolic reactions. Catabolism breaks down large molecules into smaller units.

Correct Option: Answer is B

- 2.** The organ which is sensitive to light in Euglena is the
- **A.** gullet
 - **B.** flagellum

- **C.** chloroplast
- **D.** eyespot

Correct Option: Answer is D

3. The organelles present in cells that are actively respiring and photosynthesizing are

- **A.** lysosomes and ribosomes
- **B.** Golgi apparatus and endoplasmic reticulum
- **C.** nucleus and centrioles
- **D.** mitochondria and chloroplast

Short Explanation: Mitochondria are the "powerhouses" of the cell, breaking down fuel molecules and capturing energy in cellular respiration. **Chloroplasts** are found in plants and algae. They're responsible for capturing light energy to make sugars in photosynthesis.

Correct Option: Answer is D

4. *Taenia solium* can be found in

- **A.** cow
- **B.** goat
- **C.** dog
- **D.** pig

Short Explanation: Tapeworm infection is caused by eating the raw or undercooked meat of infected animals. Cattle usually carry *Taenia saginata* (T saginata). **Pigs** carry *Taenia solium* (T solium). In the human intestine, the young form of the **tapeworm** from the infected meat (larva) develops into the adult **tapeworm**

Correct Option: Answer is D

5. Which of the following describes a characteristic of arthropods?

- **A.** The organism finds it easy to grow freely
- **B.** the organism has a pair of jointed appendages
- **C.** the body is not divided into a number of segments seg
- **D.** the body is covered by chitin

Short Explanation: Characteristics of arthropods include:

- A segmented **body** (Figure below) with a head, a thorax, and abdomen segments.
- Appendages on at least one segment. ...
- A nervous system.
- A hard **exoskeleton** made of **chitin**, which gives them physical protection and resistance to drying out.

Correct Option: Answer is D

6. Which of the following distinguishes a butterfly from a moth?

- **A.** the wings of butterfly rest horizontally but those of moth rest vertically
- **B.** Both are active during the day
- **C.** they have similar antennae
- **D.** the abdomen of moth is fatter than that of butterfly

Correct Option: Answer is A

7. Which of the following types of feathers is used for flight in birds?

- **A.** Quill
- **B.** Filoplume
- **C.** covert
- **D.** Down

Correct Option: Answer is A

8. The plants that grow in deserts or very dry areas are referred to as

- **A.** mesophytes
- **B.** hydrophytes
- **C.** epiphytes
- **D.** xerophytes

Short Explanation: A xerophyte is a species of plant that has adaptations to survive in an environment with little liquid water, such as a desert or an ice- or snow-covered region in the Alps or the Arctic. Popular examples of xerophytes are cacti, pineapple and some Gymnosperm plants.

Correct Option: Answer is D

9. Which of the following is the simplest living organism?

- **A.** Paramecium
- **B.** Virus
- **C.** Amoeba
- **D.** Chlamydomonas

Correct Option: Answer is B

10. The structural adaptation of desert plants for water conservation is

- **A.** broad leaves with numerous stomata
- **B.** spongy mesophyll
- **C.** spiny leaves
- **D.** prominent stomata in leaves

Correct Option: Answer is C

11. The long and sharp clawed feet of birds is an adaptation for

- **A.** crushing seeds
- **B.** scooping mud
- **C.** tearing flesh
- **D.** grasping prey

Correct Option: Answer is D

12. During the manufacture of food by plants, which of the following organism use energy from the sun?

- **A.** anabaena
- **B.** sulfur bacteria
- **C.** nitrosomonas sp
- **D.** nitrobacter sp

Short Explanation: They perform oxygenic **photosynthesis**, very similar to that of eukaryotic plants and algae, by utilizing water as a reductant source, consequently producing molecular oxygen.

Correct Option: Answer is A

13. Movement of minerals and chemical compounds with a plant occurs during

- **A.** osmosis
- **B.** translocation
- **C.** transpiration

- **D.** diffusion

Short Explanation: Translocation is the movement of materials in **plants** from the leaves to other parts of the **plant**. Nutrients, mainly sugars, are created in the leaves during photosynthesis. These are then transported throughout the **plant** through phloem, which are a long series of connected cells.

Correct Option: Answer is B

14. The enzyme that is present in the saliva is

- **A.** rennin
- **B.** lipase
- **C.** pepsin
- **D.** ptyalin

Short Explanation: Ptyalin is a form of amylase found in the saliva of humans and some other animals.

Correct Option: Answer is D

15. Plants that have special devices for trapping and digesting insects are

- **A.** carnivorous
- **B.** symbiotic
- **C.** parasitic
- **D.** saprophytic

Correct Option: Answer is A

16. The process of transforming the chemical energy of cellular fuels into the high energy bonds of ATP in plants is

- **A.** autotrophism

- **B.** photosynthesis
- **C.** photolysis
- **D.** respiration

Correct Option: Answer is D

17. Fungi are referred to as heterotrophs because they

- **A.** are filamentous
- **B.** lack chlorophyll
- **C.** have mycelium
- **D.** lack roots

Short Explanation: Fungi are Heterotrophic

Because fungi cannot produce their own food, **they** must acquire carbohydrates and other nutrients from **the** animals, plants, or decaying matter on which **they** live. **The fungi** are generally considered **heterotrophs** that rely solely on nutrients from other organisms for metabolism.

Correct Option: Answer is B

18. An example of a parasitic protozoan is

- **A.** Paramecium
- **B.** Plasmodium
- **C.** Euglena
- **D.** Chlamydomonas

Correct Option: Answer is B

19. Which blood cells are involved in the immune response of vertebrates?

- **A.** Phagocytes

- **B.** lymphocytes
- **C.** erythrocytes
- **D.** monocytes

Correct Option: Answer is B

21

20. The blood circulatory system of vertebrates consists of

- **A.** heart, arteries, capillaries and veins
- **B.** heart, aorta, capillaries and veins
- **C.** heart, aorta, arteries and veins
- **D.** heart, vena cava, arteries, and veins

- **Short Explanation:** Blood Circulatory systems consists of:
- The **heart**. The **heart** pumps **blood** around the body. ...
- The right side of the **heart**. ...
- The left side of the **heart**. ...
- **Blood vessels**. ...
- **Arteries**. ...
- **Capillaries**. ...
- **Veins**

Correct Option: Answer is A 22

21. A plant tissue that carries water and mineral salts is the

- **A.** cambium
- **B.** xylem
- **C.** cortex
- **D.** phloem

Short Explanation: **Xylem** is a type of tissue in vascular plants that transports water and some nutrients from the roots to the leaves.

Correct Option: Answer is B

23

22. Which of the following helps in the clotting of blood?

- **A.** Red blood cells
- **B.** White blood cells
- **C.** Plasma
- **D.** Platelets

Short Explanation: **Platelets** are tiny blood cells that help your body form clots to stop bleeding. If one of your blood vessels gets damaged, it sends out signals to the **platelets**. The **platelets** then rush to the site of damage. they form a plug (clot) to fix the damage.

Correct Option: Answer is D

24

23. The part of the mammalian skin involved in excretion is the

- **A.** sweat glands
- **B.** malpighian layer
- **C.** sebaceous gland
- **D.** horny layer

Correct Option: Answer is A25

24. Which of the following forms about 55% of the volume of the blood in man?

- **A.** leucocytes
- **B.** platelets
- **C.** plasma
- **D.** erythrocytes

Short Explanation: **Plasma** is the clear, straw-colored liquid portion of blood that remains after red blood cells, white blood cells, platelets and other cellular components

are removed. It is the single largest component of human blood, comprising about 55 percent, and contains water, salts, enzymes, antibodies and other proteins.

Correct Option: Answer is C

25. Which of the following is a waste product of an insect?

- **A.** Alkaloids
- **B.** Uric acid
- **C.** Sweat
- **D.** Mucilage

Short Explanation: *Uric acid* is a waste substance excreted by *insects*. *Uric acid* is high in nitrogen but also contains carbon, oxygen and hydrogen

Correct Option: Answer is B

26. The main structure in vertebrates that supports and protects the body is the

- **A.** skeleton
- **B.** ligament
- **C.** muscle
- **D.** joint

Correct Option: Answer is A

27. The chitin in the exoskeleton of many arthropods is strengthened by

- **A.** lids
- **B.** proteins
- **C.** calcium compounds

- **D.** organic salt

Correct Option: Answer is C

28. The transfer of pollen grains from the anther to a stigma is

- **A.** propagation
- **B.** placentation
- **C.** pollination
- **D.** fertilization

Short Explanation: Pollination is the act of transferring pollen grains from the male anther of a flower to the female stigma. The goal of every living organism, including plants, is to create offspring for the next generation.

Correct Option: Answer is C

29. The male reproductive organ of a flower is the

- **A.** carpel
- **B.** stamen
- **C.** petal
- **D.** sepal

Correct Option: Answer is B

30. The gland that is found just below the hypothalamus is the

- **A.** parathyroid
- **B.** adrenal
- **C.** pituitary
- **D.** thyroid

Correct Option: Answer is C

31. The most important plant hormone is

- **A.** cytokinin
- **B.** abscisic acid
- **C.** auxin
- **D.** gibberellin

Short Explanation: Auxins have a cardinal role in coordination of many growth and behavioral processes in the plant's life cycle and are essential for plant body development.

Correct Option: Answer is C

32. The sensory cell that responds to dim light is referred to as the

- **A.** cone
- **B.** lens
- **C.** rod
- **D.** iris

Short Explanation: A **rod** cell is sensitive enough to **respond** to a single photon of **light** and is about 100 times more sensitive to a single photon than cones. Since **rods** require less **light** to function than cones, they are the primary source of visual information at night (scotopic vision).

Correct Option: Answer is B

33. The absence of anti-diuretic hormone in humans results in

- **A.** decreasing dehydration
- **B.** drastic dehydration

- **C.** eliminating dehydration
- **D.** increasing dehydration

Correct Option: Answer is D

34. Oestrogen is a hormone that is synthesized in the

- **A.** ovaries
- **B.** testes
- **C.** anterior pituitary
- **D.** adrenal cortex

Short Explanation: Estrogen, or **oestrogen**, is the primary female sex hormone. It is responsible for the development and regulation of the female reproductive system and secondary sex characteristics.

Correct Option: Answer is A

35. The eye defect caused by the development of cloudy areas in the lenses is

- **A.** presbyopia
- **B.** glaucoma
- **C.** cataract
- **D.** astigmatism

Short Explanation: A **cataract** is a dense, cloudy area that forms in the lens of the eye. A **cataract** begins when proteins in the eye form clumps that prevent the lens from sending clear images to the retina.

Correct Option: Answer is C

36. A pollutant that is biodegradable is

- **A.** crude oil

- **B.** heavy metals
- **C.** cellophane
- **D.** sewage

Correct Option: Answer is D

37. A tropical disease caused by Trypanosoma is

- **A.** sleeping sickness
- **B.** river blindness
- **C.** yellow fever
- **D.** malaria

Short Explanation: Sleeping sickness is caused by two types of parasites Trypanosoma brucei rhodesiense and Trypanosoma brucei gambiense. T b rhodesiense **causes** the more severe form of the illness. Tsetse flies carry the infection.

Correct Option: Answer is A

38. The solid part of the ecosystem is referred to as the

- **A.** atmosphere
- **B.** hydrosphere
- **C.** biosphere
- **D.** lithosphere

• **Short Explanation:** A lithosphere is the rigid, outermost shell of a terrestrial-type planet, or natural satellite, that is defined by its rigid mechanical properties

• **Correct Option: Answer is D**

39. Which of the following is caused by *Treponema palladium*?

- **A.** Gonorrhea
- **B.** Leprosy
- **C.** Tuberculosis
- **D.** Syphilis

Short Explanation: The cause of **syphilis** is a bacterium called *Treponema pallidum*. The most common route of transmission is through contact with an infected person's sore during sexual activity.

Correct Option: Answer is D

40. To which blood group do universal recipients belong?

- **A.** B
- **B.** A
- **C.** O
- **D.** AB

Correct Option: Answer is D

41. The clumping together of red blood cells is

- **A.** agglutination
- **B.** fussion
- **C.** transfusion
- **D.** compatibility

Short Explanation: **Agglutination** is the clumping of particles. The word **agglutination** comes from the Latin *agglutinare* (glueing to). **Agglutination** is the process that occurs if an antigen is mixed with its corresponding antibody called isoagglutinin.

Correct Option: Answer is A

42. Physiological adaptation to very dry conditions in animals demonstrates

- **A.** rejuvenation
- **B.** xeromorphism
- **C.** hibernation
- **D.** aestivation

Correct Option: Answer is D

43. One of adaptation of Cactus opuntia to conserve water is the reduction of

- **A.** internodes
- **B.** stem to leaves
- **C.** leaves to spine
- **D.** flower size

Correct Option: Answer is C

44. Which of the following structure is adapted for feeding in a bird of prey?

- **A.** Hooked beak and sharp claws
- **B.** Smooth beak and strong claws
- **C.** Big beaks and strong feet
- **D.** Pointed beak and strong claws

Correct Option: Answer is A

45. The special pigment for colour change in chameleon is

- **A.** melanin

- **B.** carotenoid
- **C.** chromatin
- **D.** chromatophore

Short Explanation: Chromatophores are pigment-containing and light-reflecting cells, or groups of cells, found in a wide range of animals including amphibians, fish, reptiles, crustaceans and cephalopods. Mammals and birds, in contrast, have a class of cells called melanocytes for coloration.

Correct Option: Answer is D

46. The behavioural adaptation in social insects could best be described as

- **A.** symbiosis
- **B.** saprophytism
- **C.** parasitism
- **D.** commensalisms

Correct Option: Answer is A

47. In which of the following levels of classification are the members most similar?

- **A.** Order
- **B.** Genus
- **C.** Species
- **D.** Phylum

Correct Option: Answer is C

48. Spirogyra is regarded as a multicellular plant because

- **A.** the cylindrical cells are linked end to end

- **B.** its cells are linked together by cytoplasmic strands
- **C.** its cells are large
- **D.** it is an algae containing a large vacuole

Correct Option: Answer is B

49. Which of the following pairs of structures in living organisms do not perform similar functions?

- **A.** Lungs and spiracles
- **B.** Root hairs and mammalian hairs
- **C.** Feathers and scales
- **D.** Contractile vacuole and Kidneys

Correct Option: Answer is B

BIOLOGY PAST QUESTIONS 2013 THEORY WITH SOLUTION

1. (a) (i) What is respiration?

(ii) In a tabular form, state four differences between gaseous exchange and aerobic respiration

(b)(i) Explain the term residual air.

(ii) What is the importance of residual air to mammals?

(c) State four characteristic features associated with respiratory structures.

- (d)(i) What is oxygen debt?
(ii) Outline three activities that can result in oxygen debt.

Short Explanation:

(a) (i) Respiration: Respiration is the process by which living things breakdown/oxidise food/nutrients/glucose/ organic substances; to release energy/ATP for metabolic activities of the body; with or without the use of oxygen.

Gaseous exchange	Aerobic respiration
oxygen is used to oxidise glucose	organism and CO_2 moves out
process takes place outside the cell	process occurs inside the cell
physical process	chemical process
No energy is released	energy is released
No enzymes involved	enzymes involved

(b) (I) Residual Air: During breathing out/expiration/exhalation; not all the air is removed from the lungs; the air left inside the lungs during breathing out is residual air.

(ii) Importance of Residual Air: Air is left in the lungs so that the lungs do not collapse; air is always left inside the lungs so that more oxygen can be removed (diffuse) into the blood stream.

(c) Characteristic features of respiratory structures: -

- They have wide/large surface area

- they are moist
- thin walled membranes
- they are highly vascularised/ richly supplied with blood
- highly permeable to gases.

(d)(i) Oxygen debt: The build up of lactic acid; during strenuous exercise is poisonous and makes the muscles ache; oxygen is needed to break down/ get rid of the lactic acid, that is why athletes continue breathing heavily even after the exercise is over; the oxygen needed to get rid of the lactic acid is known as oxygen debt.

(ii) Activities that can result in oxygen

debt: Running; boxing; swimming /any correct example of strenuous activities.

2. (a) What is photosynthesis?

(b) List: (i) four external factors

(ii) two internal factors; that affect the rate of photosynthesis.

(c)(i) List the major products of the light dependent stage of photosynthesis.

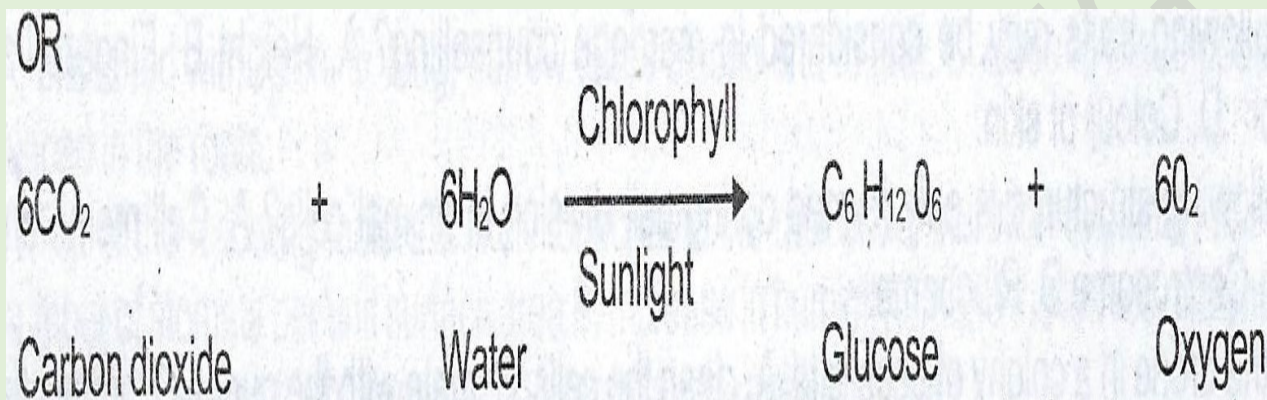
(ii) State the importance of each of the products listed in (c)(i) above.

(d)(i) Explain why there are no green plants at the lower depths of some lakes.

(ii) State why decomposers are important to flowering plants.

Short Explanation:

(a) **Photosynthesis:** The process by which green plants; manufacture/ synthesize their food/organic compounds; from carbon dioxide and water; in the presence of sunlight/solar energy; giving off oxygen as a byproduct.
OR



(b) (i) External Factors: Light intensity/sunlight; carbon dioxide concentration; temperature; pollution; water

(ii) Internal factors: Hormones; chlorophyll/presence of chlorophyll/ebloroplast; enzymes; number/state of stomata

(c)(i) Major products of light-dependent stage of photosynthesis: - ATP (Adenosine triphosphate)/energy, - H^+ (Hydrogen ion), - OH^- (Hydroxyl ion), - NADPH₂ (reduced nicotinamide adenine dinucleotide phosphate), - O_2 (oxygen gas).

(ii) Importance of the products: -ATP to generate energy for the dark stage reaction, - $2\text{H}^+ + \text{NADP} \rightarrow \text{NADPH}_2$, - OH^- (Hydroxyl ion) $\rightarrow \text{H}_2\text{O}$, O_2 gas, NADPH₂ - to generate hydrogen ions/to reduce carbon dioxide / CO_2 to form carbohydrate in the dark stage; - O_2 - given off into the atmosphere for animals/other organisms to use for respiration.

(d) (i) Why green plants are absent at the lower depth of some lakes: Light intensity decreases with depth in water bodies; so at certain depths in water, there is no light for photosynthesis to occur.

(ii) Importance of decomposers to flowering plants: Decomposers (fungi/bacteria) break down dead organic matter; and release the nutrients in them; 002 in the soil/water/environment; which is absorbed by plants for growth.

3. (a)(i) What is ecological succession?

(ii) Describe the process of ecological succession on a bare plot of land.

(b) Explain the following terms:

(i) biosphere

(ii) population

(iii) community

(iv) ecosystem.

(c) Name one instrument each used for measuring:

(i) relative humidity

(ii) wind speed

Short Explanation:

(a)(i) Ecological succession: A gradual and orderly change/replacement; of different species of plant and animal communities, in a given area; over a period of time, until the conditions Of the area became gradually modified by their activities, so that a climax/equilibrium/balanced community is reached/established.

(ii) Process of ecological succession on a bare piece of land: -

- The first living organisms on a bare plot of land are the pioneers/algae/mosses
- as the pioneers live and die they add nutrients to the soil
- this creates conditions that are more suitable/favourable to other plants/ferns/herbs/lower animals/insects/ worms which could not live there before
- as they also live and die, their remains improve the soil conditions farther
- and so perennial herbs/grasses and other animals/lizards, snakes, toads, rats are able to live there
- subsequently a point is reached at which the soil can support the growth of taller / species/shrubs/ trees and higher animals
- and higher animals/birds/various mammals may be found at this stage
- a climax community is established which is in equilibrium with the environment.

(b)(i) **Biosphere:** - Parts of the earth/air, water, soil; in which life exists.

(ii) **Population:** - Total number of organisms; of the same species; which live in a given area; at a given time.

(iii) **Community:** -A group of different living organisms/ population; living together in a habitat; and interacting with each other.

(iv) **Ecosystem:** - A natural unit; made up of living organisms and their non-living components; interacting together; to produce a stable system.

(c)(i) Instrument or measuring relative

humidity: Hygrometer/wet-dry bulb/paper hygrometer.

(ii) Instrument for measuring wind

speed: Anemometer.

4. (a)(i) What is conservation?

(ii) State six factors responsible for the decline of the abundance and variety of wildlife.

(b) Outline six ways in which the government can improve the situation in (a)(ii) above.

(c)(i) What is Eutrophication?

(ii) State two causes of eutrophication.

Short Explanation:

(a)(i) **Conservation:** Planned/wise/judicious use/management of the earth's natural resources; in order to ensure sustenance/availability/continuity; and preservation of the quality of the natural/original nature of the environment.

(ii) Factors responsible for decline of abundance and variety of wildlife:

- Commercial hunting which kills off animals indiscriminately
- industrialization which involves clearing of forests
- for various purposes/construction/farming/urbanization
- destruction of habitats of wildlife by human activities
- bush burning/noise pollution leads to wildlife migration

- environmental pollution which releases dangerous materials that degrades environment making it unbearable for the wildlife
- massive/excessive tree felling to make charcoal/domestic uses
- lack of awareness/inadequate education of populace on the need for conservation
- lack of adequate funding to enforce conservation policy.

(b) Ways by which the Government can improve the situation:

- establishment of forestry departments
- establishment of Environment Protection Agency to improve the situation
- creation of Parks and Games Reserves/any correctly named example; to preserve/maintain plants and animals in their natural environment
- conservation education carried out through tree planting! campaigns against bush burning/ tree felling /poaching
- conservation laws to regulate tree felling; poaching/hunting of wild animals and bush burning
- establishment of zoological gardens preserving endangered species.

(c)(i) **Eutrophication:** Excessive growth of plant life/algae/algal bloom; in water courses/bodies, caused by an increase in organic nutrients and minerals; leading to the death of aquatic organisms.

(ii) **Causes of eutrophication:** Overuse of fertilisers/plant nutrients which drain off into water bodies; sewage discharge; was from fish farming.

5. (a) Describe the carbon cycle.

(b) State the functions of:

(i) nitrogen fixing bacteria

(ii) nitrifying bacteria

(iii) denitrifying bacteria; in nature.

Short Explanation:

(a) Description of the carbon cycle: Carbon occurs in the air/atmosphere/as carbon dioxide/dissolved in the oceans/seas/water bodies; and is circulated continuously; through the activity of microorganisms/ decay bacteria/saprophytic organisms/fungi; which decompose the dead organisms; plants/animals release carbon dioxide and water; the carbon dioxide released increases the concentration of carbon in the atmosphere/ocean/sea/water bodies; this is further increased by respiration in plants and animals; combustion of wood/coal/peat/oil/fuel; some of the excess carbon dioxide is absorbed by the sea/water bodies and also used by plants in photosynthesis; organisms in seas/oceans/water bodies release carbon dioxide into the atmosphere/water.

(b) Functions of;

(i) **Nitrogen-Fixing Bacteria;** Nitrogen fixing bacteria in the nodules of legumes; those living freely in soil; convert atmospheric nitrogen; into ammonia/ammonium compounds.

(ii) **Nitrifying Bacteria:** They live freely in the soil; oxidize/breakdown the ammonium compounds; to nitrites; which are then oxidized by others/Nitrobacter/named bacteria; to nitrates.

(iii) **Denitrifying Bacteria:** They live freely in the soil; these break down/oxidise ammonium compounds; into

nitrogen; and oxygen; which is released into the atmosphere.

6. (a)(i) What is excretion?

(ii) List the excretory organs in humans and name one waste product excreted by each organ

(B) Name the excretory organs in:

(i) insects

(ii) earthworms.

Short Explanation:

(a)(i) Excretion: The elimination of waste products; of metabolism; from the body of an organism.

(ii) Excretory organs in Humans with their waste products.

Excretory Organs in humans

Waste

Kidney -

Nitrogenous

wastes/urea/water/salts/urine

Skin -

Urea/SaltsNVater/Sweat

Liver -

Bile/excess cholesterol

Lungs -

CO₂/water vapour.

(b) Functioning of kidney as excretory: Blood from the renal artery enters the bowman's capsule; through the afferent arteriole; ultra filtration occurs in the Bowman's capsule; high pressure in the glomerulus causes the filtration of small molecules/amino acids/ glucose/ minerals salts/ water/ urea; out of the blood plasma into the cavity of the capsule; the filtrate/glomerular filtrate passes through the nephron; where selective reabsorption of useful metabolites/glucose/amino acids take place; in the

descending/first proximal convoluted/coiled tubule; the reabsorption of salts and water is regulated in the loop of Henle; and the ascending/second/distal convoluted tubule/large waste molecules like creatine secreted at loop of Henle; depending on blood concentration; the filtrate continues into the main collecting duct/tube; where further reabsorption of water and salt takes place; the remaining fluid in the duct becomes urine.

(c) Excretory organs in;

- (i) Insects → Malpighian tubules
- (ii) Earthworm → Nephridia/Nephridium

BIOLOGY PAST QUESTIONS 2012 WITH SOLUTIONS

1. The science of life is referred to as

- **A.** anatomy
- **B.** biology
- **C.** ecology
- **D.** phylogeny

Short Explanation: **Biology** is the science of life. Its name is derived from the Greek words "bios" (life) and "logos" (study). **Biologists** study the structure, function, growth, origin, evolution and distribution of living organisms.

Correct Option: Answer is B

2. The epidermis of a leaf is considered an example of a tissue because the cells

- **A.** allow sunlight to pass through.
- **B.** are covered by a waxy cuticle
- **C.** all possess a similar structure and function
- **D.** allow water to enter them by osmosis

Correct Option: Answer is C

3. The organism at the organ level of organisation of life is

- **A.** Euglena
- **B.** Spirogyra
- **C.** Ginger
- **D.** Tapeworm

Correct Option: Answer is C

4. The disadvantage of complexity of organisation in the higher organisms is that it leads to

- **A.** mutual interdependence between component cells
- **B.** internal structural specialization
- **C.** increased adaptation to environment
- **D.** slower rate of diffusion of oxygen to individual cells

Correct Option: Answer is D

5. In which of the following multicellular organisms is interdependence of cells maintained?

- **A.** Spirogyra
- **B.** Volvox
- **C.** Eudorina
- **D.** Pandorina

Correct Option: Answer is B

6. Which of the following processes occurs in the second phase of meiosis?

- **A.** Homologous chromosomes separate
- **B.** Two daughter cells are formed
- **C.** Fertilization occurs
- **D.** Segregation of genes occurs

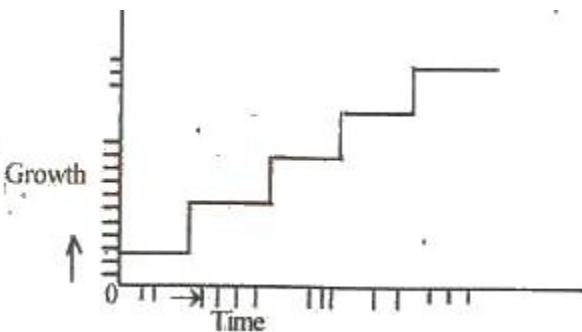
Correct Option: Answer is A

7. The muscle cell fluid of an athlete was tested immediately after a 10m race and was found to contain a high

concentration of lactic acid. Explain what could have caused this. The

- **A.** athlete must have eaten food containing lactic acid just before the race
- **B.** Athlete must have injected lactic acid into his blood just before the race
- **C.** athletes muscle must have carried out anaerobic respiration during the race
- **D.** athlete must have inhaled lactic acid from the environment during the race

Correct Option: Answer is C



8. The graph above represents the growth pattern of an animal. Which of the following groups of animals shows this pattern of growth?

- **A.** Cestoda
- **B.** Reptilia
- **C.** Amphibia
- **D.** Insecta

Correct Option: Answer is D

9. The pulmonary circulation involves movement of the blood to and from the

- **A.** brain
- **B.** Kidney
- **C.** Liver
- **D.** lungs

Short Explanation: The pulmonary circulation is the portion of the circulatory system which carries deoxygenated blood away from the right ventricle, to the lungs, and returns oxygenated blood to the left atrium and ventricle of the heart.

Correct Option: Answer is D

The table below shows the blood composition of four individuals. Use the information contained in the table to answer questions below.

Individual	Protein	Red blood cells	White blood cells	Platelets
I	Normal	Normal	High	Normal
II	Normal	Normal	Normal	Normal
III	Normal	Normal	Low	Normal
IV	Normal	Normal	Normal	Low

10. Which of the individual is likely to be suffering from haemophilia?

- **A.** I
- **B.** II
- **C.** III
- **D.** IV

Correct Option: Answer is D

The table below shows the blood composition of four individuals. Use the information contained in the table to answer questions below.

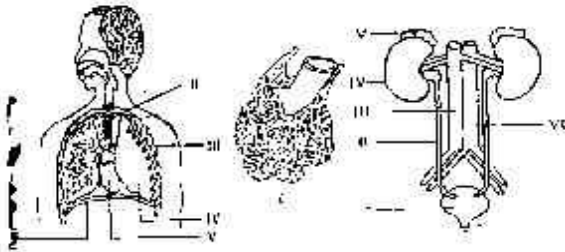
Individual	Protein	Red blood cells	White blood cells	Platelets
I	Normal	Normal	High	Normal
II	Normal	Normal	Normal	Normal
III	Normal	Normal	Low	Normal
IV	Normal	Normal	Normal	Low

11. Which of the individual is likely to have HIV?

- **A.** I
- **B.** II
- **C.** III

- **D. IV**

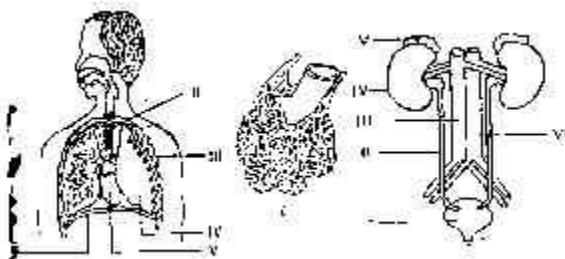
Correct Option: Answer is C



12. The diagrams above are part of the human lungs. Which of the parts is diagram Z taken from

- **A. II**
- **B. III**
- **C. IV**
- **D. V**

Correct Option: Answer is C



13. The diagram above is an illustration of the human excretory system. Which of the labelled parts contains the highest amount of nitrogenous waste?

- **A. I**
- **B. II**
- **C. IV**
- **D. VI**

Correct Option: Answer is C

14. Increasing adrenalin content of the blood would be expected to decrease the flow of blood to the

- **A.** liver
- **B.** heart
- **C.** lungs
- **D.** brain

• **Short Explanation:**

• **Correct Option: Answer is B**

15. The following are all examples of a simple reflex action, except

- **A.** swallowing
- **B.** sneezing
- **C.** walking
- **D.** blinking

Correct Option: Answer is C

16. Impulses are conducted towards the nerve cell through the

- **A.** myelin sheath
- **B.** dendrons
- **C.** axons
- **D.** synaptic nodes

Correct Option: Answer is B

17. A student dissecting an animal specimen discovered that it had two auricles and one ventricle. The animal is likely to be a/an

- **A.** fish
- **B.** reptile
- **C.** amphibian
- **D.** bird

Correct Option: Answer is C

18. The function of the convex lens in the correction of eye defects is to

- **A.** diverge light rays from far objects to focus image on the retina
- **B.** converge light rays to focus image behind the retina
- **C.** converge light rays to focus time image before the retina
- **D.** converge light rays from far objects to focus their image on the retina

Correct Option: Answer is D

19. Which of the following parts of the eye contains light sensitive cells?

- **A.** Choroid
- **B.** iris
- **C.** Retina
- **D.** Lens

- **Short Explanation:** The **retina** is a thin layer of tissue that lines the back of the eye on the inside. It is located near the optic nerve. The purpose of the **retina** is to receive light that the lens has focused, convert the light into neural signals, and send these signals on to the brain for visual recognition

- **Correct Option: Answer is C**

20. The hair on the human skin will stand erect when the person is

- **A.** in an air-conditioned room
- **B.** strolling in the sun
- **C.** taking a hot drink
- **D.** doing strenuous exercise

Correct Option: Answer is A

21. When fertilization occurs that leads to formation of a zygote, which of the following processes takes place?

- **A.** The tail of the sperm fuses with the egg cell
- **B.** Both tail and head of the sperm fuse with the egg cell
- **C.** The nucleus of the sperm fuses with the egg cell
- **D.** The egg cell develops on its own without the sperm

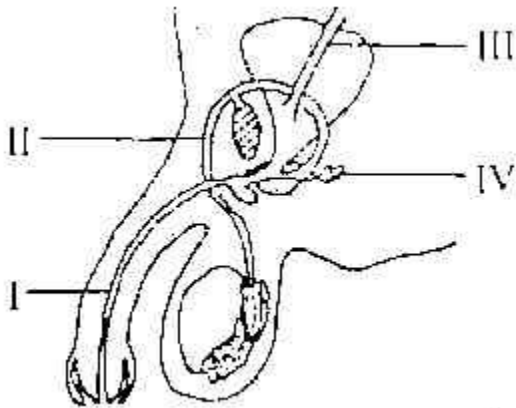
Correct Option: Answer is C

21. Which of the following structures in a flower develops into the seed?

- **A.** Testa
- **B.** Integument
- **C.** Ovule
- **D.** Ovary
- **Short Explanation:** Fertilization occurs when one of the sperm cells fuses with the egg inside of an **ovule**. After fertilization occurs, each **ovule develops** into a **seed**. Each **seed** contains a tiny, undeveloped plant called an embryo. The ovary

surrounding the **ovules** **develops** into a fruit that contains one or more **seeds**.

- **Correct Option: Answer is C**



22. An illustration of a human male reproductive system is shown above. Which of the labelled parts can be cut to-ensure permanent sterility?

- **A. I**
- **B. II**
- **C. III**
- **D. IV**

Correct Option: Answer is B

23. The brightly coloured petals in some plants is an adaptive colouration for

- **A. feeding**
- **B. pollination**
- **C. dispersal**
- **D. shedding**

Correct Option: Answer is B

24. Squirrels easily kill some trees because

- **A.** they inject poisonous substances into the trees.
- **B.** most trees are susceptible to squirrel bites
- **C.** they feed voraciously on the roots
- **D.** they gnaw the bark and destroy

Correct Option: Answer is D

25. The major role played by accessory pigments during photosynthesis is the phloem.
to

- **A.** add colour to plants but not to trap light energy
- **B.** trap light energy that chlorophyll cannot absorb
- **C.** transfer energy from chlorophyll to other pigments
- **D.** prevent chlorophyll from absorbing unnecessary light

Correct Option: Answer is A

26. The main raw materials required for photosynthesis are

- **A.** oxygen and water
- **B.** oxygen and carbon dioxide.
- **C.** oxygen and chlorophyll
- **D.** carbon dioxide and water

Short Explanation: The raw materials of photosynthesis, **water** and **carbon dioxide**, enter

the cells of the leaf, and the products of photosynthesis, **sugar** and **oxygen**, leave the leaf
Correct Option: Answer is D

27. What is the benefit of thunder and lightning to farmers?

- **A.** Addition of nitrates to the soil
- **B.** Aeration of the soil
- **C.** Removal of harmful bacteria from the soil
- **D.** Addition of rain water to the soil

Short Explanation: These **benefits** include assisting **farmers** by helping plants grow. ... **Lightning** strikes help dissolve this unusable nitrogen in water, which then creates a natural fertilizer that plants can absorb through their roots

Correct Option: Answer is A

28. Which of the following substances must be present in the soil for growth of a healthy green plant?

- **A.** Amino acids
- **B.** Carbon
- **C.** Glucose salts
- **D.** Magnesium ions
- **Short Explanation:** **Plants** use **magnesium ions** to make chlorophyll in their leaves. Like in nitrate deficiency, the **plant** is limited in terms of its photosynthetic ability and the **plant** growth is compromised. **Magnesium** is a limiting factor in healthy **plant** growth.
- **Correct Option: Answer is D**

29. The purpose of proteins in the diet of a mammal is to

- **A.** promote growth and repair cells
- **B.** break down molecules
- **C.** regulate the flow of chyme
- **D.** serve as co-factors or enzymes

Correct Option: Answer is D

30. Excess carbohydrates can be stored in the muscles in the form

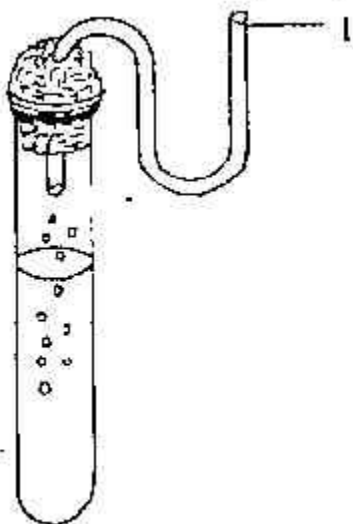
- **A.** cellulose
- **B.** glycogen
- **C.** chitin
- **D.** lipids
- **Short Explanation: Glycogen** is a multibranched polysaccharide of glucose that serves as a form of energy storage in animals, fungi, and bacteria. The polysaccharide structure represents the main storage form of glucose in the body.
- **Correct Option: Answer is B**

31. The bones of the legs of a six years old boy are observed to be weak and crooked. Which of the following elements were deficient in his mother's diet at pregnancy?

- **A.** Sodium and iron
- **B.** Iron and manganese
- **C.** Calcium and phosphorus
- **D.** Magnesium and sulphur

Correct Option: Answer is C

33.



32. The diagram above is an illustration of an experimental setup to test the action of yeast. What test would be used to confirm the nature of the gas given off at?

- **A.** Insert a glowing splint
- **B.** Pass the gas through lime water
- **C.** Insert a piece of litmus paper
- **D.** Smell the gas

Correct Option: Answer is B

33. Which of the following digestive enzymes would be greatly affected if the liver fails to produce bile?

- **A.** Amylase
- **B.** Cellulase
- **C.** Lipase
- **D.** rotease

Correct Option: Answer is B

34. The appropriate food to be given to the child in the picture above is

- **A.** beans
- **B.** garri
- **C.** bread
- **D.** avocado pear

Correct Option: Answer is A

35. Which of the following food substances will have little or no calorific value?

- **A.** Proteins
- **B.** Carbohydrates
- **C.** Lipids
- **D.** Water

Short Explanation: Water has no calorific value and it also absorbs **heat** being vaporized during combustion.

Correct Option: Answer is D

37

36. A space containing a group of interactive organisms is called

- **A.** community
- **B.** habitat
- **C.** ecosystem
- **D.** population

Short Explanation: Community, also called **biological community**, in **biology**, an interacting group of various species in a common location.

Correct Option: Answer is A

38

37. The area around the earth where life activities go on is referred to as

- **A.** biosphere
- **B.** lithosphere
- **C.** habitat
- **D.** ecosystem

Short Explanation: Biosphere, relatively thin life-supporting stratum of Earth's surface, extending from a few kilometres into the atmosphere to the deep-sea vents of the ocean.

Correct Option: Answer is A

39

38. The following are abiotic components of the ecosystem **except**

- **A.** bacteria
- **B.** topography
- **C.** water
- **D.** wind

Correct Option: Answer is A

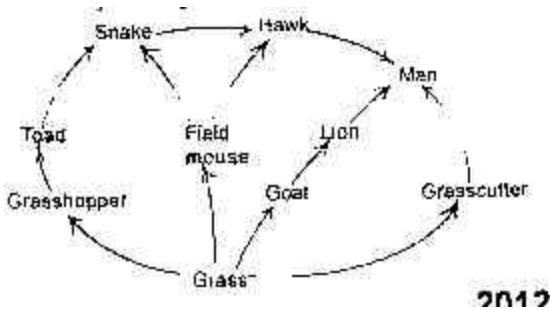
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39. Which of the following instruments would be most appropriate in comparing the air pressure at different places in the habitat?

- **A.** Hygrometer
- **B.** Barometer
- **C.** Photometer
- **D.** Air pressure disc

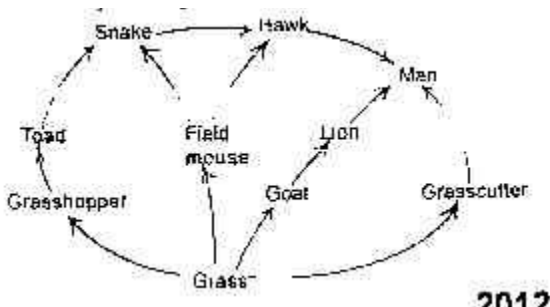
- **Short Explanation:** A **barometer** is a scientific instrument that is used to measure air pressure in a certain environment. Pressure tendency can forecast short term changes in the weather.

Correct Option: Answer is B



40. The best title for the above diagram is

- **A.** terrestrial food web
- **B.** terrestrial food chain
- **C.** aquatic food web
- **D.** aquatic food chain
- **Short Explanation:** The **terrestrial food web** links creatures on land, from the tiniest microbes in the soil to the large mammals of the forests.
- **Correct Option: Answer is A**



41. How many food chains are in the diagram?

- **A.** Three
- **B.** Four
- **C.** Five

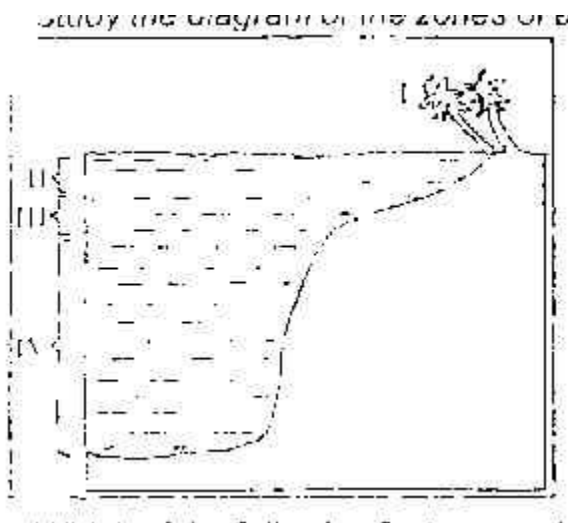
- **D.** Six

Correct Option: Answer is C

42. Plants with breathing roots are characteristic features of vegetation in

- **A.** tropical grassland
- **B.** temperate grassland
- **C.** saltwater swamp
- **D.** Tropical rain forest

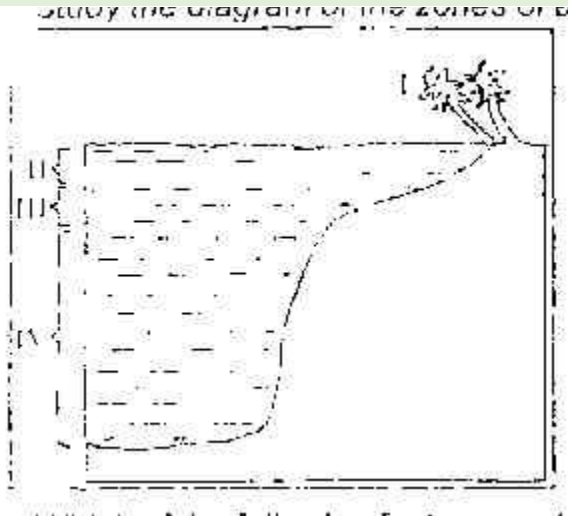
Correct Option: Answer is C



43. In which of the zones can very few organisms be found

- **A.** I
- **B.** II
- **C.** III
- **D.** IV

Correct Option: Answer is D



44. Which of the following factors exerts a major effect on the distribution of living organisms in the habitat?

- **A.** Salt
- **B.** pH
- **C.** Light
- **D.** Rainfall

Correct Option: Answer is C

45. The use of water closet and septic tanks is preferred to the use of bucket and pit latrines because

- **A.** it is cost effective
- **B.** contaminated faeces are completely kept away from flies
- **C.** the water for flushing contains chlorine which kills the germs
- **D.** It prevents flies from breeding

Correct Option: Answer is B

46. The detection of *vibro cholera* in public drinking water is a sign of

- **A.** clean water
- **B.** outbreak of disease
- **C.** water pollution.
- **D.** decayed organisms in the water

Correct Option: Answer is C

47. Which of the following factors would be least likely to act as a density dependent limiting factor in a population of mice?

- **A.** Build-up of waste products
- **B.** Predation
- **C.** Parasitism
- **D.** Unfavourable climate

Correct Option: Answer is C

48. The primary cause of species endangerment is

- **A.** disappearing habitats
- **B.** increase in temperatures
- **C.** acid rain
- **D.** exposure to hazardous wastes

Correct Option: Answer is A

49. The following pairs of traits are examples of continuous and discontinuous variations respectively except

- **A.** eye Colour and body weight,
- **B.** height and blood group
- **C.** skin colour and fingerprints
- **D.** bodyweight and tongue rolling.

BIOLOGY PAST QUESTIONS 2012 THEORY WITH SOLUTION

1. (a) Explain briefly four factors that affect the diffusion of substances.

(b) Explain the following terms:

(i) active transport

(ii) transpiration.

(c) State four ways by which plants can reduce high rate of transpiration.

(d) State the features of red blood cells and how these features adapt the cell to perform its functions.

Short Explanation:

(a) **Factors affecting the rate of diffusion:** - Presence of concentration gradient; when concentration of substance is high the rate of movement of molecules will increase/molecules move from the region of higher concentration to the region of lower concentration; leading to an increase in the rate of diffusion. Surface area: Large surface area will allow more molecules of substance to diffuse. State/nature of the diffusing substance: Molecules in gaseous state diffuse faster than

those in liquids or solid states/molecules in solid state diffuse slower than those of liquid and gas. Size of the molecules/particles: The smaller molecules will move/diffuse faster than larger molecules. Temperature: Increase in temperature increases the movement of molecules/increases in diffusion/decrease in temperature slows down movement of molecules/decreases diffusion.

(b)(i) **Active transport:** This is the movement of molecules from a region of low concentration to a region of high concentration/against a concentration gradient. Across a living membrane. Using energy from the cell.

(ii) **Transpiration:** Loss of water in the form of vapour; from the aerial parts of plants.

(c) **Reduction of Transpiration:** Closure of stomata during the day. Reduction of leaf size/leaf modified into spines. Dropping of leaves in the dry season. Development of thick cuticle. Presence of hairs over leave surfaces. Presence of sunken stomata.

(d) **Red Blood Cells:** Biconcave shape/disc-like, increases its surface area for oxygen uptake. Elastic membrane, enables it to flow easily through the blood vessels. Presence of hemoglobin; for oxygen uptake. Thin membrane; for easy diffusion of oxygen/carbon-dioxide.

2. (a) Describe briefly the role of the stomach in digestion.

(b) (i) Name three parts of plants in which food can be stored.

(ii) Give one example in each case.

(c) Explain briefly how the level of sugar in the mammalian blood can be regulated.

Short Explanation:

(a) **Role of the stomach in digestion:** The stomach receives food when the sphincter at the entrance of the stomach relaxes. The muscular walls of the stomach contract and relax, to churn/mix the food, and further break down the particles. The stomach produces gastric juices, which contain dilute hydrochloric acid/pepsin/rennin. The dilute hydrochloric acid neutralizes the action of the salivary amylase, and kills bacteria present in the food. It creates a suitable condition/pH for the action of pepsin, and activates inactive pepsinogen to form pepsin. The pepsin breaks down the protein to polypeptide/peptides. While the rennin coagulates/curdles milk to enhance digestion by pepsin. The mucus produced by the stomach wall lines the wall of the stomach to prevent the action of the acid on the stomach wall. The mucus produced by the stomach lubricates the movement of the food in the alimentary canal.

(b)(i) **Parts of plants where food is stored:** Leaves. Stems. Roots. Seed. Fruits.

(ii) **Examples:**

Leaves: Onions, shallot, garlic.

Stems: Yam, sugarcane, ginger, Irish potatoes.

Roots: Cassava, carrots, sweet potatoes.

Seed: Groundnut, beans, coconut, cashew.

Fruits: Maize, guava, orange, pawpaw, cashew, palm nut, mango.

(c) **Sugar regulation in mammals:** The liver controls the blood glucose levels. It converts excess glucose in the blood to glycogen. The process is stimulated by the hormone insulin, which is secreted by the pancreas. When the blood glucose level falls below normal. The pancreas secretes/glucagons. This stimulates the conversion of glycogen into glucose, which is released into the blood.

3. (a) Explain briefly the following terms:

(i) conservation

(ii) endangered species.

(b) State: (i) five reasons why conservation of forests is important;

(ii) four ways by which forest reserves can be conserved.

(c)(i) What is the importance of decomposers in the ecosystem?

(ii) Name one plant and one animal decomposer.

Short Explanation:

(a) **Conservation:** Conservation is the protection/preservation/restoration, and wise management/use of natural resources. Planning and management of resources from loss/waste/harm in order to achieve continuous use/ Sustainable quality of living for mankind OR Conservation is the wise/judicious/intelligent use of our natural resources to ensure their availability, and preserve the quality of the environment.

(ii) **Endangered species:** Are organisms that have become depleted/rare. Due to human activity/natural disaster/epidemic. Those habitats have become so disturbed. That they may be lost forever/become extinct.

(b)(i) Necessity to conserve forests: Economic importance:

Forests produce timber for the construction industry/buildings/furniture/boats/paper.

Domestic: Firewood for charcoal, medicines, gum/resins/rubber/tannins/chemicals/turpentine.

Biological importance: Provide food/ shelter for many animals, reduce wind erosion/desertification, serves as wind breaks/checking strong winds, cause precipitation of water from damp air/can increase amount of rainfall in an area, protect source of streams/rivers, help in maintaining CO₂ / O₂ balance in the atmosphere/the plants remove CO₂ for photosynthesis/supply O₂ /purifies the atmosphere.

Educational: For ecological studies, to prevent certain species from getting extinct. Recreational/ Tourism: Serve as tourist attraction.

(ii) Methods for the conservation of

forests: Regulations made to control excessive deforestation, only mature trees should be felled. Planting new forest trees/afforestation where they have been completely destroyed. Discouraging extensive use of natural forests as a source of timber since animals and some other plants are also destroyed/displaced. Replacing harvested/felled mature trees by seedlings in plantations! reforestation. Prevention of forest fires. Controlling plant diseases by using pesticides/chemicals to kill pests. Using trained forest rangers/guards to guard the forests against intruders /fight fires. Educating the local population on the dangers of deforestation/carrying out awareness campaigns against deforestation. Establishment of forest reserve.

(c)(i) **Importance of decomposers in the ecosystem:** They break down organic materials that are difficult for organisms to digest, they return organic matter to the environment in an inorganic form, so that they can be re-assimilated by plants/recycling nutrients.

(ii) **Plant decomposers:** Fungi, Rhizopus, Bacteria.
Animal decomposers: Earthworms, Termites.

4. (a) (i) What is a gene?

(ii) Differentiate between the terms genotype and phenotype

(b) Explain the following terms:

(i) hybrid

(ii) pure breeding

(iii) nucleotide.

(c) In garden pea seeds, smooth seed coat is dominant over rough seed coat. With the aid of a genetic diagram, determine the result expected if a homozygous rough pea is crossed with a smooth seed coat plant whose parents were rough coated.

Short Explanation:

(a)(i) What is a gene? Genes are units of heredity located in the chromosomes which are responsible for the coding of a particular protein to transfer trait or character from parents to offspring.

(ii) Differentiate between the terms genotype and phenotype: Genotype is the genetic make-up of genes in an organism that is independent of the environment while a phenotype is all observable features, appearance

of the organism that is determined by the environment and the genotype.

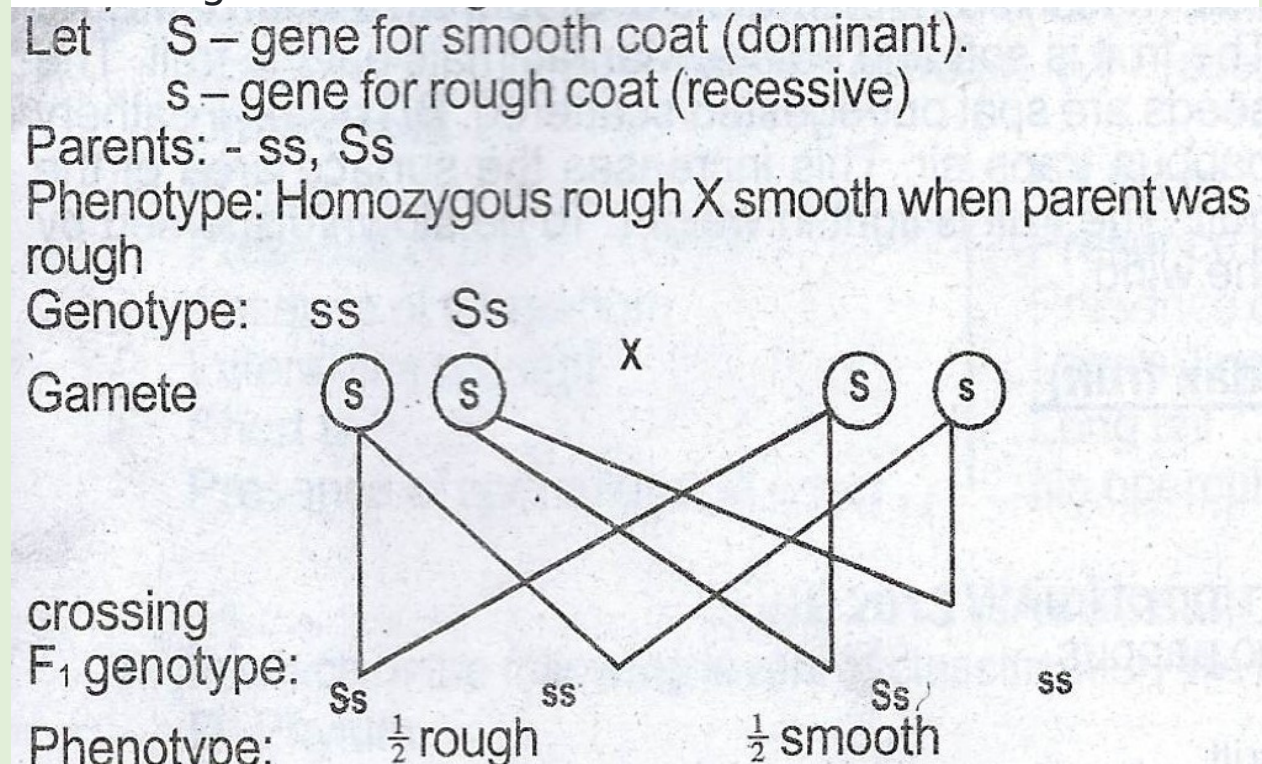
(b) Explain the following terms:

(i) **Hybrid:** Offspring produced from a cross between two parents with contrasting characters.

(ii) **Pure-breeding:** A process of producing a particular genotype over many generations.

(iii) **Nucleotides:** The basic building block of a nucleic acid consisting of a ribose, deoxyribose, protein, a phosphate group and a nitrogen base.

(c) In a garden, pea seeds, smooth seed coat is dominant over rough seed coat. With the aid of a genetic diagram, determine the result expected if a homozygous rough pea is crossed with a smooth seed coat plant whose parents were rough coated.



5. (a) Explain the following terms:

(i) disease

(ii) symptoms of diseases.

(b)(i) List two physical and two chemical barriers that prevent pathogens from penetrating the body of an organism.

(ii) Explain how vaccination protects the body from contracting infectious diseases.

(c) Distinguish between an antibody and an antigen.

(d) Name the causative agents of:

(i) Malaria

(ii) Cholera

(iii) AIDS.

Short Explanation:

(a)(i) **Disease:** A condition in which the normal functions of an organism/part of the body/cell/tissue/organ is impaired/disturbed.

(ii) **Symptom:** Any observable effect on an organism: Caused by the presence and irritation of a disease-causing agent/pathogen.

(b)(i) Physical barriers:

(i) Skin

(ii) Hairs

(iii) Cuticle

(iv) Epidermis

(v) Bark

(vi) Cell wall lining.

Chemical barriers:

(i) Mucus

(ii) Saliva

- (iii) Tears
- (iv) Gastric/ HCL and intestinal secretions
- (v) Phenolic compound
- (vi) Gums
- (vii) Tannins
- (viii) Wax
- (ix) Oily secretions! sebum

(ii) **How vaccination protects the body:** The vaccine/attenuated pathogen is administered into the body. The body responds by producing copious antibodies. The next time the body is exposed to the same antigen/infection/pathogen antigen, the memory cells allow a more rapid and heightened/vigorous response to antibody production. Protection of the body against future infections.

(c) Differences between antibodies and antigens

- (I) Antibodies is a protein molecule. Antigens May not be a protein molecule
- (ii) Antibodies is Produced in response to a particular foreign substance/antigen while antigens Stimulates formation of antibodies
- (iii) Antibodies Usually found in the plasma fluid of the blood, antigens are Usually produced in/introduced into the body.
- (iv) Antibodies are Protective in functions, antigens Leads to an immune response/immunity.

(d) Causative Agent:

- (i) **Malaria:** Plasmodium vivax/Plasmodium/malaria/species
- (ii) **Cholera:** Vibrio cholera.
- (ii) **AIDS:** Human Immuno-deficiency virus/HIV

- 6.** (a) (i) Describe epigeal germination of a seed.
(ii) In a tabular form, state three differences between epigeal germination and hypogeal germination.
- (b)(i) What is seed dormancy?
(ii) State three ways by which dormancy in seeds can be broken.
- (c) State six advantages of using contraceptives in human populations.

Short Explanation:

(a)(i) Description of Epigeal Germination:

- (i) Seed absorbs/imbibes water and swells.
- (ii) Testa splits and the radicle emerges.
- (iii) The hypocotyl of the germinating seed elongates.
- (iv) Pushing the seed out of the soil.
- (v) The cotyledon(s) become exposed to sunlight.
- (vi) Turns green to photosynthesize.
- (vii) The plumule develops into a green shoot.
- (viii) The cotyledon(s) dry up/shrink and fall off.

(ii) Differences between epigeal and hypogeal germination:

Epigeal germination	Hypogeal germination
Hypocotyl elongates	Epicotyl elongates
cotyledon	cotyledon remains in the soil
cotyledon(s) turns green / photosynthesize	cotyledon does not photosynthesize

plumule does not emerge simultaneously with the radicle	plumule emerges simultaneously with radicle
Energy with the radicle from cotyledon	energy derived from endosperm

(b)(i) **Seed dormancy:** An inactive period of a seed. During which growth slows/completely ceases. Due to certain internal or external factors.

(ii) **Ways of breaking dormancy:** Removal/corrosion of testa. Treatment of seed with acid/digestive enzymes/water. Exposure to alternating high and low temperature. Addition of dormancy-breaking hormones/named hormones. Exposure to fire. Exposure to light.

(c) Advantages of using contraceptives in human population:

- (i) It controls birth rate.
- (ii) Reduces over-population.
- (iii) Promotes maternal health/reduces maternal death.
- (iv) Promotes infant survival rate.
- (v) Enables parents to determine family size.
- (vi) Prevents unwanted pregnancy.
- (vii) Enhances good standard of living for the family.
- (viii) Some prevent STI or STDs.

BIOLOGY PAST QUESTIONS 2011 WITH SOLUTION

1. The branch of Biology that deals with the principles of classification of organisms is known as

- **A.** biological index
- **B.** nomenclature
- **C.** taxonomy
- **D.** ecology

Short Explanation: In biology, taxonomy is the science of naming, defining and classifying groups of biological organisms on the basis of shared characteristics

Correct Option: Answer is C

2. Which of the following structures is a tissue

- **A.** Vessel element
- **B.** Blood
- **C.** Sieve tube element
- **D.** Erythrocyte

Short Explanation: Blood is considered a connective **tissue** because it has a matrix. The living cell types are red **blood** cells, also called erythrocytes, and white **blood** cells, also called leukocytes. The fluid portion of whole **blood**, its matrix, is commonly called plasma

Correct Option: Answer is B

3. Which of the following cells are not regarded as specialized?

- **A.** Sperm cells

- **B.** Root up cells
- **C.** Muscle cells
- **D.** somatic cells

Correct Option: Answer is D

4. Which of the following pairs of cells carry out the same function?

- **A.** Cheek cell and red blood cell
- **B.** Spermatozoa and ovum
- **C.** Palisade cell and epidermal cell
- **D.** Root tip cell and guard cell

Correct Option: Answer is B

5. If Amoeba is placed in a salt solution. the contractile vacuoles would

- **A.** be bursting more frequently
- **B.** be more numerous
- **C.** be formed less frequently
- **D.** grow bigger before they burst

Short Explanation: When an **amoeba** is **put** into **salt** water with enough concentration, some of its organelles, like the contractile vacuole, are damaged. As the **amoeba** prevents the **salt** from entering, instead the **solution** will pull water out of the **amoeba**, concentrating the **salts** inside. When this happens the **amoeba** will appear to shrink.

Correct Option: Answer is C

6. In which of the following habitats is Paramecium not found

- **A.** Pond
- **B.** Aquarium
- **C.** Lake
- **D.** Puddle²

Correct Option: Answer is B

7

7. The following processes are involved in water movement in the endodermis except

- **A.** osmosis
- **B.** vacuolar pathway
- **C.** diffusion
- **D.** active transport²

Correct Option: Answer is C

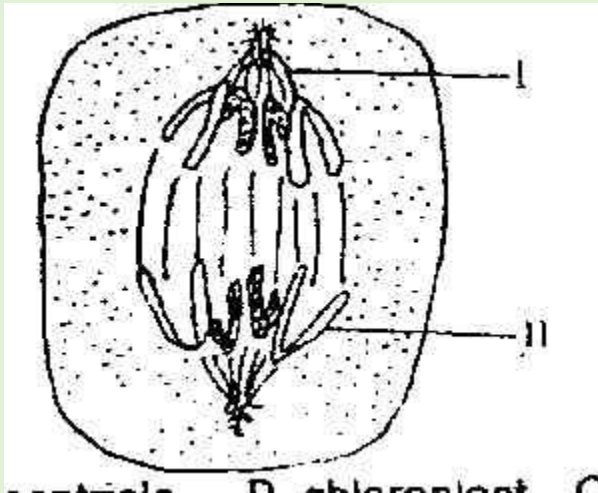
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8. Cells that utilise a lot of energy are characterized by the presence of a large number of

- **A.** vacuoles
- **B.** mitochondria
- **C.** endoplasmic reticulum
- **D.** ribosomes

Correct Option: Answer is B

9

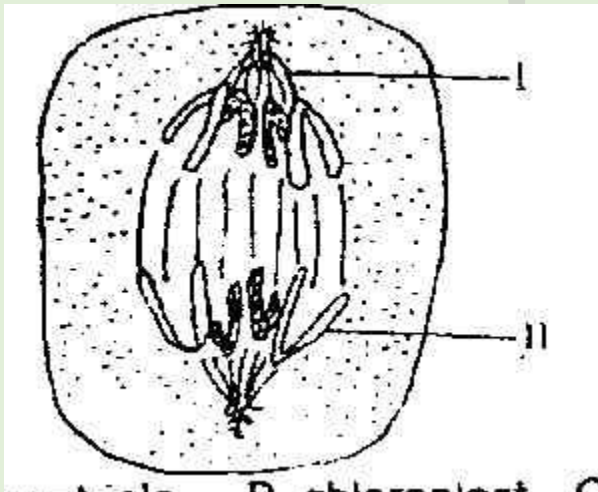


9. The part labelled II in the diagram is

- **A.** centriole
- **B.** chloroplast
- **C.** chromatid
- **D.** tonoplast

Correct Option: Answer is C

10



10. The part labelled I is called the

- **A.** nucleolus
- **B.** centromere
- **C.** centriole
- **D.** spindle

• **Correct Option: Answer is D**

11. Secondary growth is brought about by the activities of the

- **A.** phellogen and phelloderm
- **B.** phellogen and procambium.
- **C.** vascular cambium and phelloderm
- **D.** vascular cambium and phellogen

Short Explanation:

• **Correct Option: Answer is B**

12. A monocot root IS different from a dicot root by having

- **A.** endodermis
- **B.** cambium
- **C.** wide pith
- **D.** root hair.

Correct Option: Answer is D

13. Which of the following statements best describes haemoglobin? It is

- **A.** yellowish in colour
- **B.** a red blood cell
- **C.** an oxygen carrying pigment
- **D.** needed for blood clotting

Short Explanation: Hemoglobin or haemoglobin + - o- + globulin, abbreviated Hb or Hgb, is the iron-containing oxygen-transport metalloprotein in the red blood cells of almost all vertebrates as well as the tissues of some invertebrates. Hemoglobin in

blood carries oxygen from the lungs or gills to the rest of the body.

- **Correct Option: Answer is C**

14. Inhaled air is made warm and moist in the

- **A.** epiglottis
- **B.** nasal cavity
- **C.** trachea
- **D.** mouth

Correct Option: Answer is B

15. Which of the following structures is not involve in respiration?

- **A.** Lung books
- **B.** Mouth
- **C.** Stomach
- **D.** Trachea

Correct Option: Answer is C

16. Filtrate in the Bowman's capsule contains vitamins because

- **A.** only all le amount is required by the body
- **B.** they can be reabsorbed into the blood
- **C.** they have low molecular weight
- **D.** most of them are fat soluble.

Correct Option: Answer is C

17. More sweat is produced during muscular exercise because

- **A.** the contracting muscles produce water
- **B.** fermentation occurs in muscles
- **C.** the temperature of the body rises
- **D.** the muscle fatigues

Short Explanation: During exercise, heart rate and blood pressure increase, which **in** turn cause the body to pump out **more sweat**.

- **Correct Option: Answer is C**

18. Which of the following neurons may not have myelin sheath?

- **A.** Effector cells
- **B.** Intermediate neurons
- **C.** Motor neurons
- **D.** Sensory neurons

Correct Option: Answer is A

19. Which of the following is an effector organ?

- **A.** Tongue
- **B.** Skin
- **C.** Nose bridge
- **D.** Ear glands

Short Explanation: The Somatic Nervous System (SNS) includes all nerves controlling the muscular system and external sensory receptors. External sense **organs** (including **skin**) are receptors. Muscle fibers and gland cells are **effectors**. The reflex arc is an automatic, involuntary reaction to a stimulus.

- **Correct Option: Answer is B**

20. During sexual reproduction in Paramecium, how many times does the zygote divide to produce eight nuclei?

- **A.** 1
- **B.** 2
- **C.** 3
- **D.** 4

Short Explanation: in each conjugant, the zygote undergoes mitosis division 3 times to produce 8 nuclei.

Correct Option: Answer is C

21. The testes in male mammals descend into the scrotal sac because

- **A.** there is congestion in the lower abdomen
- **B.** they run the risk of being destroyed
- **C.** they need special support
- **D.** they require a relatively low temperature

Short Explanation: The **testes** are located in the **scrotum** to maintain **testicular temperature** at several degrees below core body **temperature**, a necessity for normal spermatogenesis.

- **Correct Option: Answer is D**

22. Some animals return to water bodies to reproduce because

- **A.** natural enemies destroy their eggs on land
- **B.** water nourishes their embryos and their young ones
- **C.** they are close to their ancestors and imitate them
- **D.** the temperature on land is not suitable for the development of their embryos

Short Explanation:

- **Correct Option: Answer is D**

23. Which of the following methods is appropriate for the cultivation of cassava?

- **A.** Budding
- **B.** Fragmentation
- **C.** Root cutting
- **D.** Stem cutting

Correct Option: Answer is D

24. What will be the chromosome number of the hybrid of two varieties of a plant with 36 chromosomes in the endosperm cell?

- **A.** 12
- **B.** 24
- **C.** 35
- **D.** 48

Correct Option: Answer is C

25. Kreb's cycle occurs in the

- **A.** mitochondria
- **B.** cytoplasm
- **C.** nucleus
- **D.** ribosomes

Short Explanation: The citric acid **cycle** (also called as **Krebs cycle** or tricarboxylic acid **cycle**) **takes place** in the mitochondria and is an integral part for the generation of adenosine triphosphate (ATP).

- **Correct Option: Answer is A**

26. The process whereby electrons are emitted from chlorophyll and returned to it unchanged is known as

- **A.** non-cyclic phosphorylation
- **B.** photochemical reaction
- **C.** photophorylation
- **D.** cyclic photophorylation

Correct Option: Answer is A

27

27. Which of the following statements about photosynthesis is not true?

- **A.** Plants can photosynthesize without an increase in dry weight
- **B.** Carbon dioxide is absorbed by the leaves in bright light
- **C.** oxygen is produced during photosynthesis from the breakdown of water
- **D.** Photosynthesis occurs in green plants

Correct Option: Answer is A

28

28. The removal of all the phloem tissues of the stem of a plant close to the root system for a long period of time is likely to

- **A.** provide more energy to the roots
- **B.** accumulate more starch in the roots
- **C.** cause the underground roots to develop buds
- **D.** cause the plant to wither and die

Correct Option: Answer is D

29

29. Which of the following groups of substances are not six-carbon compounds?

- **A.** Glucose and lactose
- **B.** Lactose and cellulose
- **C.** Glucose, Lactose and Pyruvate
- **D.** Pyruvate and cellulose

• **Correct Option: Answer is D**

30

30. The conversion of glucose to starch in the leaf during the day principally

- **A.** enables photosynthesis
- **B.** prevents osmotic problems
- **C.** enables the leaf to store the starch
- **D.** enables glucose to be used up

Correct Option: Answer is C

31. A purple colour was obtained when sodium hydroxide solution and a drop of copper sulphate solution were added to a food substance. The food substance is likely to be a

- **A.** carbohydrate
- **B.** fat
- **C.** protein
- **D.** sugar

Correct Option: Answer is C

32. Which of the following enzymes is active in the duodenum?

- **A.** Pepsin
- **B.** Renin
- **C.** Trypsin
- **D.** Amylase

Short Explanation: **Trypsin** is an enzyme that helps us digest protein. In the small intestine, **trypsin** breaks down proteins, continuing the process of digestion that began in the stomach.

- **Correct Option: Answer is C**

33. The following organisms are producers except

- **A.** Hibiscus
- **B.** Mushroom
- **C.** Cactus
- **D.** Spirogyra

Correct Option: Answer is B

34. Which of the following structures produces the greatest variety of digestive enzymes?

- **A.** Salivary glands
- **B.** Pancreas
- **C.** Stomach
- **D.** Colon

Correct Option: Answer is B

35. The following are abiotic components of an ecosystem except

- **A.** temperature
- **B.** oxygen
- **C.** bacteria
- **D.** soil

Correct Option: Answer is C

Use the information on capture-recapture method to calculate the total number of organisms in a habitat.

First capture = 200

Second capture = 120

Number of organisms with mark in second capture = 40

36. The total number of organisms therefore is

- **A.** 200
- **B.** 360
- **C.** 600
- **D.** 800

Short Explanation: formula for estimated population = $\frac{\text{No of 1st capture}}{\text{No of specie}}$

$$\frac{200 \times 120}{40} = 600$$

- **Correct Option: Answer is C**

37. Ecological investigation in a habitat includes the following procedures except

- **A.** choosing a habitat
- **B.** identification of specimens
- **C.** determining the genetic makeup of specimens
- **D.** measuring abiotic and biotic factors

Correct Option: Answer is c

38. Which of the following substances when lost from the body of a mammal will not be returned to the ecosystem?

- **A.** Sweat
- **B.** Urea
- **C.** Heat energy
- **D.** Carbon dioxide

Correct Option: Answer is C

39. The rate of decomposition of organisms is faster in the tropical rain forest than in other biomes because

- **A.** of the relatively constant daylight
- **B.** there are more plants per square feet
- **C.** of abundance of water
- **D.** of constant cool temperature

Correct Option: Answer is B

40. A xerophyte conserves water by possession of the following features except

- **A.** thick cuticle
- **B.** sunken stomata
- **C.** broad leaves
- **D.** fleshy stem

Correct Option: Answer is C

41. The falling off of leaves of deciduous trees is helpful to the plant because it

- **A.** reduces the rate of transpiration

- **B.** enables the plant to conveniently eliminate its excretory products
- **C.** enables the plant to bear more fruits
- **D.** ensures that the limited mineral salts get to only growing regions²

Correct Option: Answer is A

42

42. A sample of wet garden soil of known weight was heated to constant weight. The loss of weight is due to loss of

- **A.** water
- **B.** organic matter
- **C.** water and organic matter
- **D.** water and inorganic matter²

Correct Option: Answer is C

43

43. Assuming that a period of stable population size is followed by a period when natality increases by ten (10) per cent and immigration increases by twenty (20) percent of population size. The population size will

- **A.** be at equilibrium
- **B.** increase by forty (40) per cent
- **C.** begin to increase
- **D.** begin to decrease

Correct Option: Answer is C

44

44. Which of the following methods does not make water fit for drinking?

- **A.** Addition of chlorine
- **B.** Boiling

- **C.** Distillation
- **D.** Addition of alum

• **Correct Option: Answer is C**

45

45. Vaccination results in

- **A.** aiding red blood cells to carry more oxygen
- **B.** production of antibodies which destroy toxins of germs
- **C.** arresting excessive bleeding
- **D.** production of white blood cells which engulf and digest bacteria.

Short Explanation: Vaccination is the administration of a **vaccine** to help the immune system develop protection from a disease. **Vaccines** contain a microorganism or virus in a weakened, live or killed state, or proteins or toxins from the organism.

Correct Option: Answer is B

46. Leguminous crops are incorporated into crop rotation in order to

- **A.** improve aeration of the soil.
- **B.** promote nitrogen fixation
- **C.** increase the rate of soil formation
- **D.** improve upon the water holding capacity of the soil

Short Explanation: Rotating crops adds nutrients to the soil. **Legumes**, plants of the family Fabaceae, for instance, have nodules on their roots which contain nitrogen-fixing bacteria called rhizobia.

• **Correct Option: Answer is B**

47. Which of the following practices improves crop yield in a clayey soil? Addition of

- **A.** more water and humus
- **B.** lime and humus
- **C.** fertilizers
- **D.** weedicides and fertilizers

Correct Option: Answer is B

48. Which of the following activities promotes forest conservation?

- **A.** Lumbering
- **B.** Use of firewood for cooking
- **C.** Afforestation
- **D.** Production of paper

Short Explanation: Afforestation is the process of introducing trees and tree seedlings to an area that has previously not been forested. **Afforestation** can be done through tree planting and seeding, naturally or artificially.

Correct Option: Answer is C

49. Which of the following statements is not true about continuous variation? It

- **A.** is usually controlled by several genes
- **B.** can be influenced by environmental factors
- **C.** follows a normal distribution curve
- **D.** is usually controlled by one or two pair(s) of genes

Correct Option: Answer is D

50. Acquired characters are

- **A.** received from parents
- **B.** passed to offspring
- **C.** caused by the environment
- **D.** caused by mutation

Correct Option: Answer is C

BIOLOGY PAST QUESTIONS 2011 THEORY WITH SOLUTION

1 (a) (i) What is the effect of gravity on plant growth?
(ii) Describe an experiment to demonstrate the effect of gravity on the roots of plants.

(b) Explain how bitter taste is detected in humans.

(c) Copy and complete the table below on endocrine glands and their functions.

S/ N	Endocrine glands	Hormone produced	functions
1	Thyroid		
2		Insulin	
3			Development of

			secondary sexual characters of females
--	--	--	--

Short Explanation:

(i) Effect of gravity on plant growth: The (main) root grows down in response to gravity/positively geotropic; the (main) stem/shoot/grows upwards away from gravity/negatively geotropic.

(ii) Experiment to demonstrate the effect of gravity on the roots of a plant: -

- Select three germinating bean seedlings/seeds on a moist/damp sawdust/wet blotting paper
- Pin/keep one seedling in a horizontal position
- Place another seedling in a vertical position with the radicle pointing upwards
- Place the third upwards vertical with the radicle pointing downwards, to serve as control
- After few/2-3 days, it will be observed that the radicle of the seedling placed horizontally bends/grows downwards
- the radicle of seedling placed vertically bends/grows downwards
- Indicating positive geotropism/ positive response to gravity.
- The seedling placed vertically with radicle point downwards did not bend. (N.B. observation must correspond with the experimental procedure to score).

ALTERNATIVE to 1 (a)

- (ii) Take a clinostat and two bean seedlings

- Pin one seedling horizontally on damp/moist/wet/cotton wool/blotting paper attached to the cork of the clinostat
- Pin the other seedling onto some damp/moist/wet cotton wool/blotting paper attached to the cotton tile
- Start the motor of the clinostat; Observe the seedlings after 8 —10 hrs.
- It will be observed that the radical of the seedling placed horizontally bends/grows down wards
- Indicating/showing positive geotropism.

(b) Detection of bitter taste in humans

- Taste buds are unevenly distributed on the surface of the tongue
- Taste buds for detecting bitterness are at the back/end of the tongue
- For a substance to be tasted it must be dissolved in a solution/saliva
- When bitter substance is dissolved in saliva, sensory cells at end/back of the tongue are stimulated
- Which pass impulses to the brain; Which interprets the taste as bitter.

(c) Hormones of endocrine glands and their functions

S/ N	Endocrine glands	Hormone produced	Function
1	Thyroid	Thyroxine	influences growth/development; Regulates/controls rates of metabolism
2	Pancreas	Insulin	Control blood sugar level/causes liver to convert to convert glucose
3	Ovary/	Oestrogen	

	ovaries	n	to glycogen Development of secondary sexual characters of females
--	---------	---	--

2. (a) Make a diagram 8 — 10 cm long to show the relative position of the following organs in the mammalian gut:

- (i) stomach
- (ii) pancreas
- (iii) liver
- (iv) gall bladder
- (v) duodenum and label fully.

(b) (i) List two diseases of the liver.
(ii) State five effects of the diseases in (b) (i) on the human body.

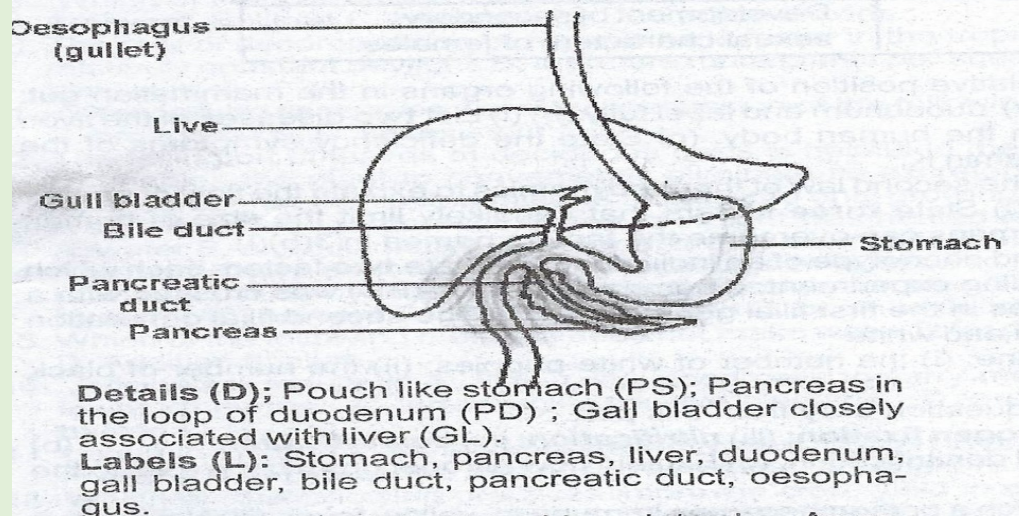
(c) State the deficiency symptoms of the following vitamins:

- (i) vitamin A
- (ii) vitamin C
- (ii) Vitamin K;

Short Explanation:

(a) Diagram showing some organs of mammalian gut
Title (T)

Diagram showing some organs of mammalian gut



(b)(i) Two diseases of the liver:

- Liver cirrhosis
- Amoebic liver abscess
- Hepatitis
- Cancer/tumor of the liver
- Gall stones.

(b)(ii) Effects of liver diseases on humans:

(i) Weakness and tiredness

(ii) Jaundice/yel-lowish eye/skin

(iii) (Slight) fever

(iv) Oedema/ swollen abdomen/accumulation of fluid in the abdomen

(v) High blood pressure

(vi) Loss of weight

(vii) Abdominal pain

(viii) Loss of appetite

(ix) Headache

(c) Deficiency Symptoms Vitamin A:

- Night blindness /poor vision
- Colds/bronchitis
- Dry scaly skin.

Vitamin C:

- Swollen joints/scurvy/bleeding gums/sore mouth
- low resistance to infection
- dental disorders.

Vitamin K: - Excessive bleeding/haemorrhage of wounds/inability of blood to clot.

3. (a) State the second law of thermodynamics.
(ii) Use the second law of thermodynamics to explain the flow of energy across different trophic levels in a food chain.

(b) (i) State three factors that can likely limit the size of human populations.

(ii) Outline two ways each by which humans can overcome the factors named in (b)(i).

Short Explanation:

(a)(i) Second law of thermodynamics: This states that when there is conversion of energy from one form to another; there is always a loss/ decreases; in the amount of useful energy.

(ii) Flow of energy across different levels in a food chain

- Transfer of energy between trophic levels
- Is not 100% completely efficient
- Successive levels have less energy
- And support fewer organisms
- Primary producers/green plants have the highest energy
- When herbivores/ primary consumers feed on primary producers
- The energy level is reduced
- When carnivores/secondary consumers feed on herbivores
- The energy level is further reduced
- Part of the energy is lost as heat at each trophic level
- Not all parts of the preceding organisms are eaten by the predator/organism at the next trophic level; Energy is lost in the process of respiration/feeding/movement/metabolic active

(b)(i) Factors that can limit human populations: -

- Famine / scarcity of food
- diseases/epidemics
- increased mortality/death rate by wars /natural disasters emigration
- increase mortality - reduced natality/birth rate.

(b)(ii) Ways of overcoming the factors:

1. Famine: - Availability of food (through improved farming techniques)
2. Mechanized farming
3. improved storage facilities
4. Use of fertilizers/agro chemicals
5. Genetic engineering
6. Improved planting materials/seeds/stems etc.
7. Improved finance

Diseases:

- Health care
- Immunization
- Preventive drugs
- Improved diet
- Good sanitation.

Wars:

- Improved/enhanced communication/dialogue
- Mediation.
- Prohibition of illegal firearms
- Good governance
- Natural disasters
- study of the environment
- promote environmental friendly activities
- preservation of natural habitats
- Accurate and prompt forecast of disaster

Increased Mortality/death rate:

- Provision of balanced/good diet
- Good Medicare
- Adequate security
- Peaceful resolution of conflicts/prevention of wars
- Effective management of disaster

Emigration: -

1. Provide adequate food/shelter
2. Improved health/environment/sanitation
3. Urbanization/development
4. Reduced natality/Birth rate
5. Apply birth control methods
6. Legislation against abortion
7. Education/ awareness against sexual transmitted diseases (STD's):

4. (a) (i) What is the difference between the genotype and phenotype of an individual?
(ii) State two factors each which determine each of the terms in (a)(i).

(b) In a breeding experiment, a homozygous dog (BB) was crossed with a homozygous white dog (bb) to produce black puppies in the first filial generation (F₁) The second filial generation produced a total of two hundred and forty (240) black and white puppies. With the aid of a genetic diagram, determine:

(i) the number of white puppies
(ii) the number of black puppies in the second filial generation.

Short Explanation:

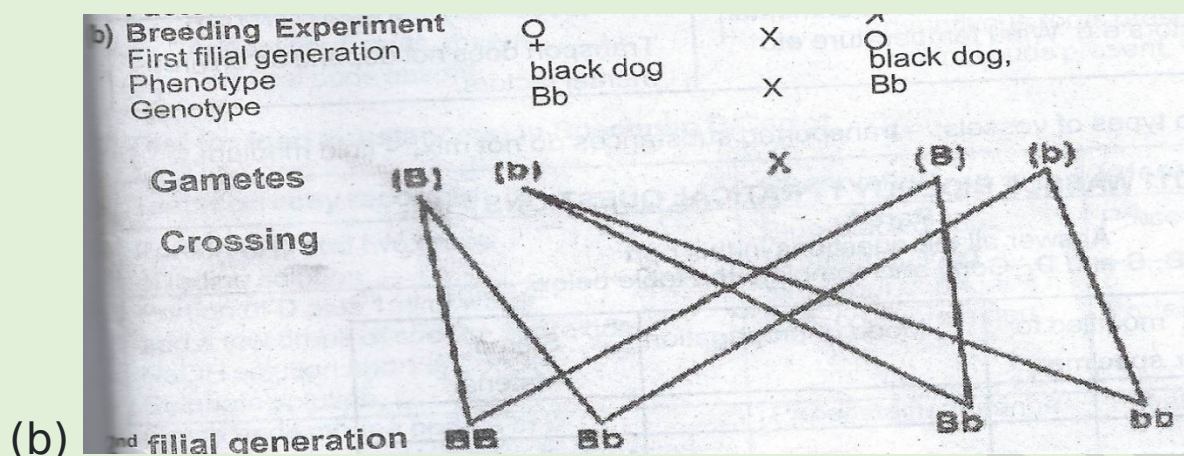
(a)(i) Difference between genotype and phenotype: A person's genotype is the sum total of the person's genetic constitution; while the phenotype is the observed trait/physical characteristics.

(ii) **Factors which determine genotype: -**

- Genotype of parents
- Mutation.

Factors which determine phenotype: -

- Genotype
- Environment;
- Diet
- Health.



Phenotypic ratio 3 black 1 white puppies

(b) **Calculation:** The number of white puppies using the phenotypic ratio

3 black: 1 white = 24

Total No of puppies produced = 240 .

No of white puppies produced = $\frac{1}{4} \times 240 = 60$ white puppies

(b)(ii) Number of black puppies = $\frac{3}{4} \times 240 = 180$ black puppies

OR calculation b(ii)

No of black puppies $\frac{3}{4} \times 240 = 180$

b(i) No of white puppies $\frac{1}{4} \times 240 = 60$

5. (a) Explain the following terms:

(i) putrefaction

(ii) nitrogen fixation

(iii) nitrification

(iv) denitrification

(b) State three differences between nitrogen fixation and denitrification.

(c) Explain how nitrogen from urine gets to the roots of plants.

Short Explanation:

(a) Explanation:

(i) **Putrefaction:** A process by which agents of decay/saprophytic/putrefying bacteria/fungi; decompose/breakdown; dead organic remains/waste products of other organisms; into ammonium compounds/nitrates.

(ii) **Nitrogen fixation:** The process by which free atmospheric nitrogen; is converted to nitrate in the soil; by thunder storm/bacteria in root noddles of legumes/soil.

(iii) **Nitrification:** The conversion of ammonia/ammonium compounds; to nitrites; then to nitrates; by denitrifying bacteria.

(iv) **Denitrification:** This is the conversion of nitrates; in the soil to atmospheric nitrogen; by nitrifying bacteria/fungi.

(b) Differences between Nitrogen fixation and Denitrification

Nitrogen Fixation	Denitrification
free atmosphere nitrogen is added to the soil	Nitrates in the soil is converted into atmospheric nitrogen
Nitrogen fixing bacteria are involved, it involves symbiotic and non symbiotic organism/bacteria	Denitrifying bacteria are involved; it involves purely non-symbiotic organism /bacteria.
Modern soil temperature is	High temperature is

required	required
----------	----------

(c) How Nitrogen from Urine gets to the roots of plants: -

- When a person passes urine in the soil
- Ammonium compound/urea present in urine; is converted into nitrite
- by nitrifying bacteria/nitroso monas
- The nitrites are oxidized to nitrates
- by other nitrifying bacteria /nitrobacter
- Roots of plants absorb the nitrates

BIOLOGY PAST QUESTIONS 2010 WITH SOLUTION

1. Which if the following organisms is not classified as an animal?

- **A.** Amoeba
- **B.** Paramecium
- **C.** Euglena
- **D.** Obelia

Short Explanation: Euglena are **single celled organisms** that belong to the genus **protist**. As such, they are not plants, animal or fungi. In particular, they share some characteristics of both plants and animals.

Correct Option: Answer is B

2. An organism that operates at the cellular level of organization, carries out its physiological activities by using its

- **A.** cell membrane
- **B.** organelles

- **C.** small size
- **D.** cytoplasm

Short Explanation: An **organelle** is a tiny cellular structure that performs specific functions within a cell. **Organelles** are embedded within the cytoplasm of eukaryotic and prokaryotic cells. In the more complex eukaryotic cells, **organelles** are often enclosed by their own membrane

- **Correct Option: Answer is B**

3. The organelle which eliminates water from the body of a protozoan is the

- **A.** plasma membrane
- **B.** contractile vacuole
- **C.** nucleus
- **D.** cell wall

Short Explanation: The **contractile vacuole** is a specialized type of **vacuole** that regulates the quantity of water inside a cell. In freshwater environments, the concentration of solutes is hypotonic, lesser outside than inside the cell. Under these conditions, osmosis causes water to accumulate in the cell from the external environment.

- **Correct Option: Answer is B**

4. Which of the following cell types has the least number of mitochondria?

- **A.** Cardiac cells of the heart
- **B.** Cells of the cornified layer
- **C.** Muscle cells of the bladder
- **D.** Muscle cells of the diaphragm

Correct Option: Answer is B

5. A typical plant cell is mainly distinguished from an animal cell by the possession of

- **A.** chloroplast and nucleus
- **B.** cell wall and cytoplasm
- **C.** chloroplast and cell wall
- **D.** cell wall and mitochondrion

Short Explanation:

- **Correct Option: Answer is C**

6. Which of the following characteristics do fungi share in common with animals?

- **A.** Presence of digestive tract
- **B.** Movement from one place to another
- **C.** Storage of carbohydrates as glycogen
- **D.** Movement of centrioles during cell division

Short Explanation: Fungi cells may **store carbohydrate** as glycogen (remember that plant cells **store carbohydrate** as starch).

Correct Option: Answer is C

7. Which of the following processes involves diffusion?

- **A.** Opening and closing of stomatal pores
- **B.** Turgidity of herbaceous plants
- **C.** Absorption of water through the root hairs
- **D.** Absorption of digested food into the villi

Correct Option: Answer is D

8. The mechanism of opening and closing of stomata in plants is based on

- **A.** turgidity and diffusion
- **B.** turgidity and flaccidity
- **C.** osmosis and diffusion
- **D.** diffusion and flaccidity

Short Explanation: **Turgidity** occurs by endosmosis when cells are placed in hypotonic solutions. On the other hand, **flaccidity** occurs when cells are placed in isotonic solutions. Water potential of the cytoplasm is equal to that of the surrounding solutions in **flaccidity**. Therefore, no net movement of water occurs

- **Correct Option: Answer is B**

9. In cellular respiration energy is made available to organisms by

- **A.** removal of a phosphate group from ADP
- **B.** breaking off a phosphate group from ATP
- **C.** adding a phosphate group to glucose
- **D.** breaking off a hydrogen ion from NADPH

Correct Option: Answer is B

10. Excretion in Paramecium sp. is by diffusion because

- **A.** its habitat is water and moist places
- **B.** it has simple, small and few internal organs
- **C.** it has a large surface area to volume ratio.
- **D.** it has a large efficient macronucleus.

Correct Option: Answer is C

11. The axial skeleton is composed of the

- **A.** skull and vertebral column
- **B.** limbs and girdles
- **C.** atlas and axis.
- **D.** radius and ulna

Short Explanation: The **axial skeleton** is the part of the **skeleton** that **consists** of the **bones** of the head and trunk of a vertebrate

- **Correct Option: Answer is A**

12. The inorganic components of bone consist of

- **A.** magnesium, sodium and calcium
- **B.** magnesium, phosphorus and calcium
- **C.** sodium, phosphorus and calcium,
- **D.** potassium, magnesium and calcium

Correct Option: Answer is B

13. A pulse is best described as

- **A.** contraction of arteries to let out blood.
- **B.** contraction of veins to allow blood into them
- **C.** dilation of arteries to accommodate blood rushing through.
- **D.** pumping action of the heart to move blood round the body.

Correct Option: Answer is D

14. The following characteristics are associated with the mammalian lungs except

- **A.** large surface area
- **B.** expandable lungs
- **C.** moist surface
- **D.** highly vascularized

Correct Option: Answer is B

15. The excretory system in mammals consists of the following parts except

- **A.** two kidneys
- **B.** two ureters
- **C.** two bladders
- **D.** one urethra

Correct Option: Answer is C

16. The organ that receives reduced flow of blood during vigorous exercise is the

- **A.** brain
- **B.** heart
- **C.** lung
- **D.** limb

Short Explanation

- **Correct Option: Answer is A**

17. The ear pinna is strengthened by

- **A.** blood pressure.
- **B.** cartilage
- **C.** bone
- **D.** turgor pressure

Short Explanation: Elastic **cartilage** is a structural **cartilaginous** tissue for non-load-bearing body parts, ... It is found mainly in the larynx, the external part of the ear (**pinna**)

- **Correct Option: Answer is B**

18. The relationship between the retina and the brain is similar to that between the

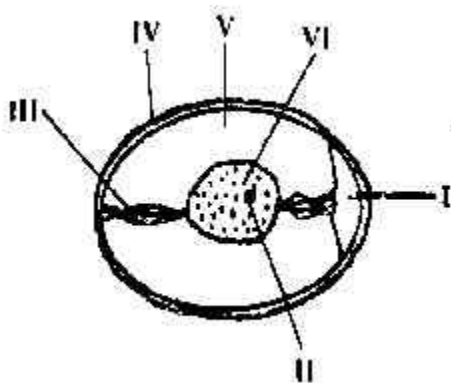
- **A.** cochlea and the auditory nerve
- **B.** cochlea and the brain.
- **C.** cochlea and the semi-circular canals
- **D.** eardrum and the brain

Correct Option: Answer is B

19. Which of the following statements about the development of the foetus is not correct?

- **A.** The circulatory system of the foetus is directly connected to the maternal blood vessels
- **B.** The foetus depends on the mothers blood for its food and oxygen
- **C.** The foetus is surrounded by a water sac
- **D.** Food and oxygen are carried across the placenta to the embryo's blood

Correct Option: Answer is A

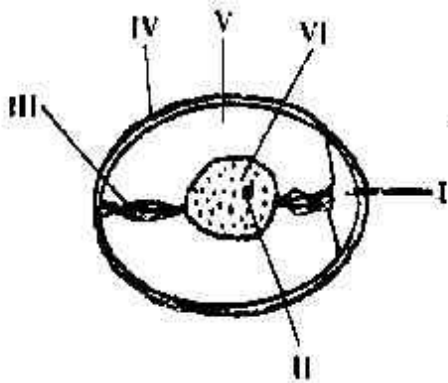


20. The parts that supply food and water to the developing embryo are labelled

- **A.** I and II
- **B.** III and V.
- **C.** V and VI
- **D.** I and III

Short Explanation

- **Correct Option: Answer is C**



21. The part labelled III is the

- **A.** albumen
- **B.** chalaza
- **C.** yolk
- **D.** air space

Correct Option: Answer is B

22. Which of the following statements is true about transpiration? It is the

- **A.** loss of water in the form of vapour from the surface of the leaf
- **B.** loss of water in the form of vapour from the surface of the root
- **C.** absorption of water in the form of vapour from the body of the plant
- **D.** movement of water through the body of the plant.

Short Explanation

- **Correct Option: Answer is A**

23. Transportation of water in the xylem tissue involves the following except

- **A.** root pressure
- **B.** capillary action
- **C.** transpiration pull
- **D.** translocation

Correct Option: Answer is D

24. The following are the major reasons why the butterfly lays eggs under the surface of a leaf except to

- **A.** shade the eggs from the direct rays of the sun.
- **B.** protect the eggs from predators
- **C.** protect the eggs from being washed away by rain drops
- **D.** camouflage the eggs

Correct Option: Answer is D

25. In which of the following organic compounds is the hydrogen-oxygen ratio equal to 2:1?

- **A.** Proteins
- **B.** Carbohydrates
- **C.** Lipids
- **D.** Vitamins

Correct Option: Answer is B

26. A man suffering from obesity must avoid meals containing

- **A.** margarine and butter
- **B.** rice and beans
- **C.** carrots and oranges
- **D.** beef and fish

Correct Option: Answer is A

27. In testing for starch in a leaf, the leaf is first boiled in water for about a minute so that the

- **A.** cell walls are hardened
- **B.** cells are killed
- **C.** chlorophyll is dissolved out
- **D.** iodine will penetrate

Correct Option: Answer is B

28. Which of the following food substances is incorrectly linked to its enzyme?

- **A.** Protein - trypsin
- **B.** Fat - lipase
- **C.** Sucrose - pepsin
- **D.** Starch - amylase

Correct Option: Answer is C

29. Which of the following pH values is the best for the action of the enzymes - renin and pepsin in the stomach?

- **A.** pH2
- **B.** pH7
- **C.** pH8
- **D.** pH9

Short Explanation

- **Correct Option: Answer is A**

30. The bacterium in a mutualistic association with legumes converts

- **A.** nitrates to ammonia
- **B.** ammonium compounds to nitrites
- **C.** nitrogen gas to ammonia
- **D.** nitrites to nitrates

Correct Option: Answer is D

31. An herbivore cannot feed on a piece of meat because it has no

- **A.** incisors to cut the meat
- **B.** canines to tear up the meat
- **C.** premolars to grind the meat
- **D.** molars to mash the meat

Short Explanation: **herbivores** feature ridged molars and jaws capable of moving sideways. Both of these traits help **herbivores** to grind their food more effectively. Most **herbivores** **are** missing **canines** entirely, and those that **do** possess them usually **have** very small or reduced **canines** that **are not** very important for chewing food.

- **Correct Option: Answer is B**

32. In a particular habitat the role of an organism is referred to as its

- **A.** level
- **B.** biomass
- **C.** niche
- **D.** space

Short Explanation: In ecology, a niche is the match of a species to a specific environmental condition

- **Correct Option: Answer is C**

33. The distribution of organisms in a fresh water habitat like a stream or pond is determined by the following factors except

- **A.** light penetration
- **B.** pH of the soil
- **C.** rainfall
- **D.** temperature

Correct Option: Answer is B

34. The speed of wind is measured using the

- **A.** wind vane
- **B.** anemometer
- **C.** photometer
- **D.** barometer

Short Explanation: An **anemometer** is a device used for measuring wind speed and direction. It is also a common weather station instrument. The term is derived from the Greek word anemos, which means wind, and is used to describe any wind speed instrument used in meteorology.

- **Correct Option: Answer is B**

35. Which of the following is a biotic component of an ecosystem?

- **A.** Soil
- **B.** Rainfall
- **C.** Light
- **D.** Fungi

Correct Option: Answer is D

36. A moss plant can withstand drought by means of its

- **A.** spores
- **B.** rhizoids
- **C.** antheridia
- **D.** achegonia

Correct Option: Answer is A

37. Which of the following factors is the main problem facing xerophytic plants?

- **A.** Competition for sunlight
- **B.** Conservation of excess water
- **C.** inadequate roots for water absorption
- **D.** Lack of adequate water

Correct Option: Answer is D

38. The following are greenhouse gases except

- **A.** carbon dioxide
- **B.** chlorofluorocarbon

- **C.** methane
- **D.** oxygen

Correct Option: Answer is D

39. Rural-urban migration in developing countries can be prevented by

- **A.** increasing the number of years in school
- **B.** putting up more factories in urban areas
- **C.** having boarding schools in rural areas
- **D.** even distribution of facilities in both regions

Correct Option: Answer is D

40. Muscles are attached to bones by means of

- **A.** tendons
- **B.** ligaments
- **C.** cartilage
- **D.** nerves

Short Explanation: A **tendon** or sinew is a tough band of fibrous connective tissue that connects muscle to bone and is capable of withstanding tension. **Tendons** are similar to ligaments; both are made of collagen.

- **Correct Option: Answer is A**

41. In a population, food shortage may lead to

- **A.** an increased survival rate
- **B.** high reproduction rate
- **C.** intra-specific competition
- **D.** low rate of migration

Short Explanation: Intraspecific competition is an interaction in population ecology, whereby members of the same species **compete** for limited resources. This leads to a reduction in fitness for both individuals, but the most fit individual survives and is able to reproduce

- **Correct Option: Answer is C**

42. The following effects are associated with deforestation except

- **A.** gully erosion
- **B.** extinction of plant species
- **C.** migration of wildlife
- **D.** increase in rainfall

Correct Option: Answer is D

43. The main reason for the conservation of wildlife is to

- **A.** create national parks for recreation
- **B.** maintain ecological balance in communities
- **C.** prevent hunters from being 'cruel to animals
- **D.** save some species from extinction.

Short Explanation: Wildlife conservation refers to the practice of protecting wild species and their habitats in order to prevent species from going extinct and to preserve natural Ecosystems.

- **Correct Option: Answer is D**

44. Which of the following is an example of variation?

- **A.** Blood

- **B.** Tongue rolling
- **C.** Reproduction
- **D.** Growth

Correct Option: Answer is B

45. The parameters of size, height, weight and colour in a population of living things are examples of

- **A.** environmental variation
- **B.** non-heritable variations
- **C.** continuous variations
- **D.** discontinuous variations

Short Explanation

Correct Option: Answer is C

46. Fingerprints are useful in crime detection because

- **A.** the police have sophisticated fingerprint machines
- **B.** thieves may leave their prints at the scene of a crime
- **C.** no two people have the same fingerprint
- **D.** fingerprints are easy to make

Short Explanation

Correct Option: Answer is C

47

47. A person with blood group, O can be given blood from persons who have blood belonging to

- **A.** group O only
- **B.** group A only
- **C.** groups A and O
- **D.** groups A, B and O

Short Explanation

Correct Option: Answer is A

48

48. A man heterozygous for albino gene marries a woman who is also heterozygous for the gene. Both have normal skin colour. The probability that they will have an albino child is

- **A.** 1/3
- **B.** 1/4
- **C.** 4/4
- **D.** 2/4

Short Explanation

- **Correct Option: Answer is B**

49

49. Which of the following statements about heredity is not true? In heredity the traits are

- **A.** carried by genes
- **B.** contained in the ovum and sperm
- **C.** always transmitted by one parent
- **D.** transmitted from parents to offspring

Short Explanation

- **Correct Option: Answer is C**

50

50. An example of a saprophytic relationship is a / an

- **A.** vulture feeding on decaying meat
- **B.** mushroom growing on decaying vegetation
- **C.** boy eating stale bread
- **D.** earthworm feeding on decaying vegetation.

Short Explanation: **saprophyte** or saprotroph is an organism which gets its energy from dead and decaying organic matter. This may be decaying pieces of plants or animals. This means

that **saprophytes** are heterotrophs. They are consumers in the food chain. This is the typical life-style of fungi.

- **Correct Option: Answer is B**

BIOLOGY PAST QUESTIONS 2010 THEORY WITH SOLUTION

1.(a) Write the chemical equation for each of the following processes:

- (i) aerobic respiration
- (ii) anaerobic respiration in plants
- (iii) anaerobic respiration in animals.

(b) State four functions which the liver of a patient suffering from cancer of the liver will not be able to perform.

(c) Make a diagram 8 -10 cm long of a flame cell and label fully.

(d) Name one appropriate hormone each responsible for the following conditions in plants:

- (i) ripening of fruits
- (ii) breaking dormancy in seeds
- (iii) weed control
- (iv) leaf fall.

Short Explanation

(a) Chemical Equations for processes:

(i) Aerobic respiration $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 38ATP$ Energy/690k cal.

(ii) Anaerobic respiration in plants $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + 2ATP$ Energy/50 k cal.

(iii) Anaerobic respiration in animals

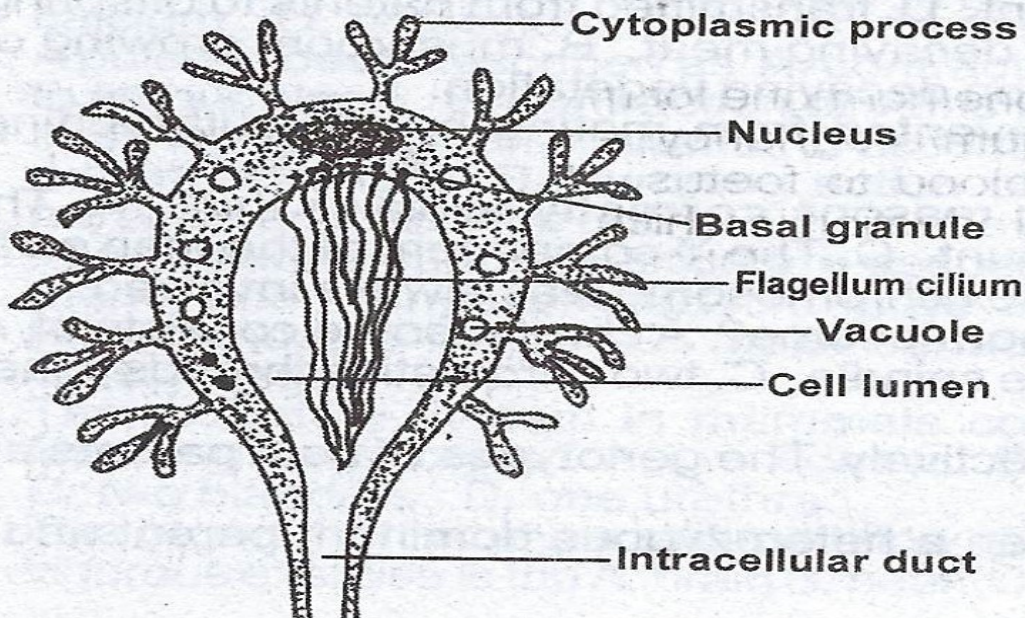
$C_6H_{12}O_6 \rightarrow 2CH_3COOH + 2ATP$ Energy/150 k cal. $C_6H_{12}O_6 \rightarrow 2C_3H_6O_3 + 2ATP$ Energy/150 k cal.

(b) A cancerous liver will not be able to: -

1. regulate blood sugar
2. carry out detoxication/deamination/regulation of blood proteins
3. produce bile
4. form red blood cells
5. produce heat
6. store vitamins
7. regulate lipids
8. manufacture protein
9. store iron

10. reserve blood.

(c) Diagram of flame cell:



Labels: Cell lumen, nucleus, basal granule, flagellum/cilium, excretory duct, excretory tubule, cytoplasmic process, vacuole

(d) Hormones responsible: -

- Ripening of fruit
- auxin/ethane/ ethylene/IAA
- Breaking of dormancy in seeds
- Auxin/ Gibberellins/cytokinins (kinta)
- Weed control
- Auxin/2,4
- D/ 2,4
- Dichlorophenoxyacetic acid
- Leaf fall
- Absciscic acid

2. (a)(i) Name two types of germination of seeds giving one example of each type.
(ii) Describe a simple experiment to demonstrate that oxygen, water and warmth are necessary for germination of seeds.

(b) State two distinguishing features each of
(i) epiphytes
(ii) parasites

(c) Give two examples of parasitic plants.

Short Explanation

(a)(i) Types of germination with examples: Epigeal germination e.g. cowpea/bean seed. Hypogeal germination e.g. maize grain/wheat/oil palm/guinea corn.

(ii) Experiment to demonstrate factors necessary for germination:

- Some dry cotton wool are placed at the bottom of each of conical flasks labeled A,B,C and D.
- In flask A, few/2 to 4 bean seeds are placed on dry cotton wool; the flask is placed in a warm place.
- In flask B the cotton wool is moistened together with the seeds in it
- Test tube containing pyrogallic acid and caustic soda/pyrogallol is suspended in flask B; to absorb oxygen; flask B is put in a warm place
- In flask C a little water is placed on the cotton wool and seeds and it is placed in a warm place; In flask D a little water is poured on the seeds to moisten them and it is placed in the refrigerator
- the four flasks are observed for four to five days/few days; only the seeds in flask C germinated. This shows that oxygen, water/moisture and warmth are necessary for seed germination.

(b)(i) Epiphytes: -

- A plant which grows on/supported by another plant/lives on trees
- Where it competes with other producers for sunlight
- It does not take manufactured food from the support plant
- It is capable of photosynthesis.

(ii) Parasites: -

- An organism that lives on or in another organism is called a host
- it obtains shelter/protection/food from the host At the expense of the host/leading to harm/death of the host.

(c) Examples of plant parasites: -

Dodder

Striga

Mistletoe

Cassytha

3. (a) Write short notes on each of the following:

- (i) ecological niche
- (ii) population density
- (iii) climax continually

(b) Describe briefly the interactions between abiotic and biotic components of an ecosystem.

Short Explanation

(a)

(i) **Niche:** Refers to a specific portion of a habitat/dwelling place which is occupied by a particular species of organism. It is also a functional position/role of an organism within the community. Each organism living in

the same habitat has its own living space, source of food and tolerance for the effect of the habitat/environment.

(ii) Population density: It is the number of individual organisms of the same species. Per unit area/volume of the habitat at a particular time. It determines the interaction between members of the same species and members of other species. It is mathematically expressed as $\text{Total } \frac{\text{population}}{\text{population size.}}$ It is estimated in humans/animals by counting. In plants by using quadrat.
Area of habitat

(iii) Climax community: Is established when a stable/unchanging community is attained in ecological succession. The community is in equilibrium with the environment. The vegetation reaches its highest development. The same species of animals, plants/organism re-occur from year to year.

(b) Interaction between biotic and abiotic components of an ecosystem: In an ecosystem the abiotic factors are non-living things (carbondioxide, oxygen, sunlight, soil and water, etc) while the biotic components are plants and animals. Green plants use carbondioxide in the presence of sunlight to produce carbohydrate/ starch/food. Animals/herbivores feed on carbohydrate/ plants and release carbondioxide from respiration. Decomposers break down dead plants/animals/organic matters to release nutrients to the soil. These nutrients are absorbed by plants for use in food production/ photosynthesis. Plants give out oxygen during photosynthesis, which is used by animals for respiration.

4. (a) List four characteristics which can be genetically transmitted in humans.

(b)(i) Name two sites each in plants and animals where meiosis occurs.

(ii) State two differences between meiosis and mitosis

(c) A woman with blood group 'A' (heterozygous) claimed that her son who has blood group 'O' was fathered by Mr. James who has blood group 'A' (homozygous). With the aid of suitable genetic diagrams

(i) prove or disprove the woman's claim

(ii) give reasons to support your answer.

Short Explanation

(a) Inheritable characters: -

Skin/hair/eye colour - Blood Group

- voice

Height/structure - Haemophilia

Albino/albinism

Weight Size - Sickle cell anaemia

Colour blindness

Size - Tongue rolling

Intelligence - baldness

(b)(i) Sites of Meiosis: -

In plants:

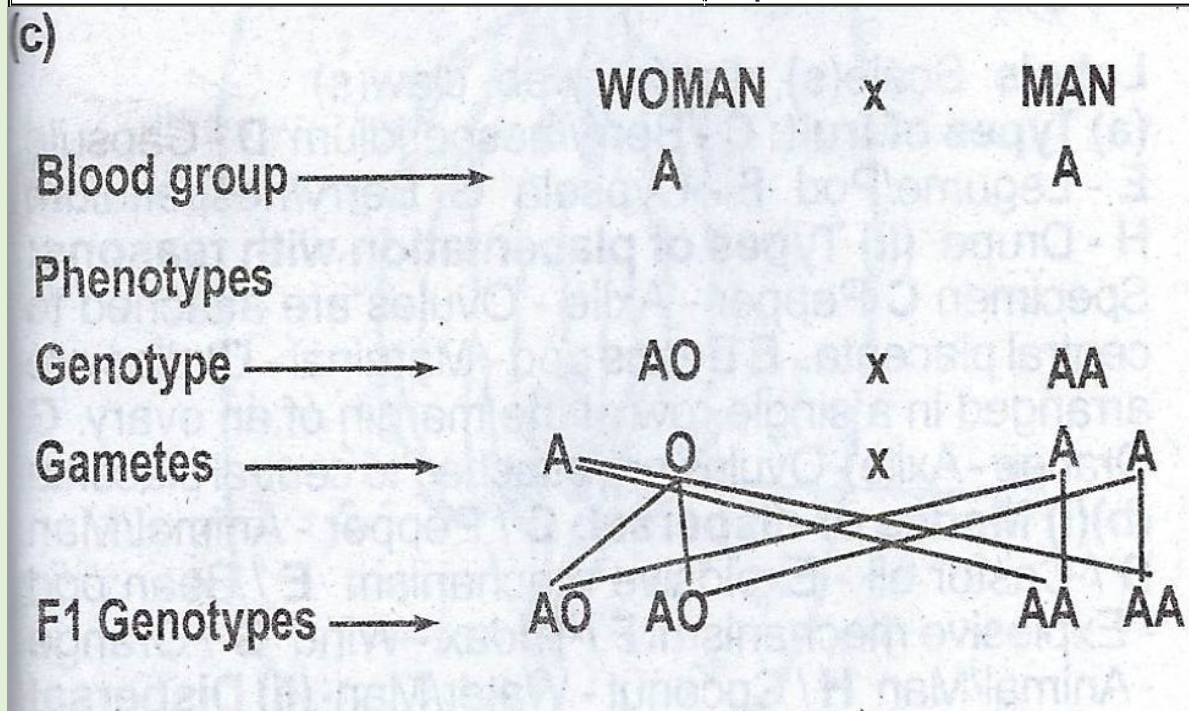
- antheridium/archegonium.
- Anthers
- Ovary
- sporangium/spore capsule.

In animals

- Testes
- Ovaries

Mitosis	Meiosis
Nucleus divide only once	Nucleus divides twice

Number of chromosomes remain the same after division. Diploid	Number of chromosomes is halved after division. Haploid
Homologous chromosomes do not associate	homologous chromosome associate to form bivalents
chiasmata are never formed	chiasmata are formed
crossing over never occurs / no exchange of genetic material	crossing over occurs/exchange of genetic material occurs
Daughter cells identical to parents cells	daughter cell generally different from parent cell
Two daughter cells are formed	four daughter cells are formed
chromosomes form single row at the equator	chromosomes form double row at the equator
Occurs in somatic cell	occurs in reproductive cell
occurs in asexual reproduction	occurs in sexual reproduction



The woman's claim is wrong.

Reasons

- (ii) Half their children are heterozygous blood group A and Half homozygous blood group A
(iii) No child is group O.

5. (a)(i) What is placentation?

(ii) Name two organs associated with the human placenta.

(b) State five functions of the human placenta.

(c) Write short notes on the following:

- (i) viviparity
- (ii) territoriality
- (iii) seasonal migration

Short Explanation

(a)(i) **Placentation:** This is the arrangement of ovules, within the ovary.

(b) (i) Organs Associated with human placenta: Umbilical cord, Uterus

(b)(ii) Functions of Placenta:

- Establishes connection between the embryo and the mother.
- Carries oxygen/water/food/ nutrients from mother's blood to foetus.
- Removes waste/urea/salt/carbon dioxide from foetus blood to mother's blood for elimination.
- Produces progesterone/hormone for maintenance of uterine wall/ endometrium/pregnancy.
- Carries antibodies from mother's blood to foetus.

(c) Viviparity: This is a type of reproduction in which the young develops inside the mother's body. Obtaining

nutrients directly from the parent. When fully developed, the young are born alive.

(ii) Territoriality: In many animal species individuals, usually males mark out areas/territories where they can live and breed.

- They guard these territories fiercely.
- Chasing away other members of the species.
- Only vigorous individuals are able to establish large territories, secure suitable mates and produce many offspring.
- Territoriality limits overcrowding.
- Animals that carry out this phenomenon are lizards, lions, etc.

(iii) Seasonal migration:

- In many habitats food supply is affected by seasonal changes. Food is abundant in rainy seasons but scarce in dry seasons.
- Animals overcome these seasonal food shortages by moving to other habitats during unfavourable seasons.
- Such large scale seasonal movements of animals between habitats are called migrations.
- Other causes of migrations are temperature changes and breeding seasons.

6.(a) Explain the term adaptation in relation to the mode of life of an organism.

(b) Explain how each of the following structures adapt the organisms that possess them to their modes of life:

(i) succulent leaves in Aloe sp

- (ii) succulent stems and reduced leaves in cactus plants
- (iii) short, strong beaks in fowls
- (iv) counter shading in fish
- (v) tiny scale leaves on needle-like branches of pine trees.

(c)(i) What is population?

(ii) State five effects of overpopulation on a community.

Short Explanation

(a) **Adaptation:** These are special features/characteristics that an organism possesses, which enable it to live/ survive successfully in an environment

(b)(i) **Aloe:** -

- The succulent leaves are adaptations for water storage.
- They are also adapted for surviving high temperatures and drought.

(ii) **Cactus:** Stems are thick and succulent for water storage, while leaves are reduced to conserve water, by reducing rate of transpiration.

(iii) **Fowl:** The beak is short/ strong/straight for picking insects/seeds, earthworms/small pieces of food, pecking of feathers.

(iv) **Fish:** The dorsal side is dark while the ventral side is silvery/counter shading. - This dual coloration protects the fish from predators, looking at them from above and below the water.

(v) **Pines:** The needle like leaves reduce water loss by transpiration. - Also conserves water for use in the long dry season.

(c)(i) Population: This is the total number of individuals of the same species, living together in a habitat, at a given time.

(ii) Effects of over population: -

- Competition for limited resources.
- Predation/cannibalism.
- Spread of diseases.
- Food shortage.
- build up toxic waste in the environment.
- Limited space for individual dwellings.
- High mortality/mortality rate.
- Increase of social vices/behaviour.